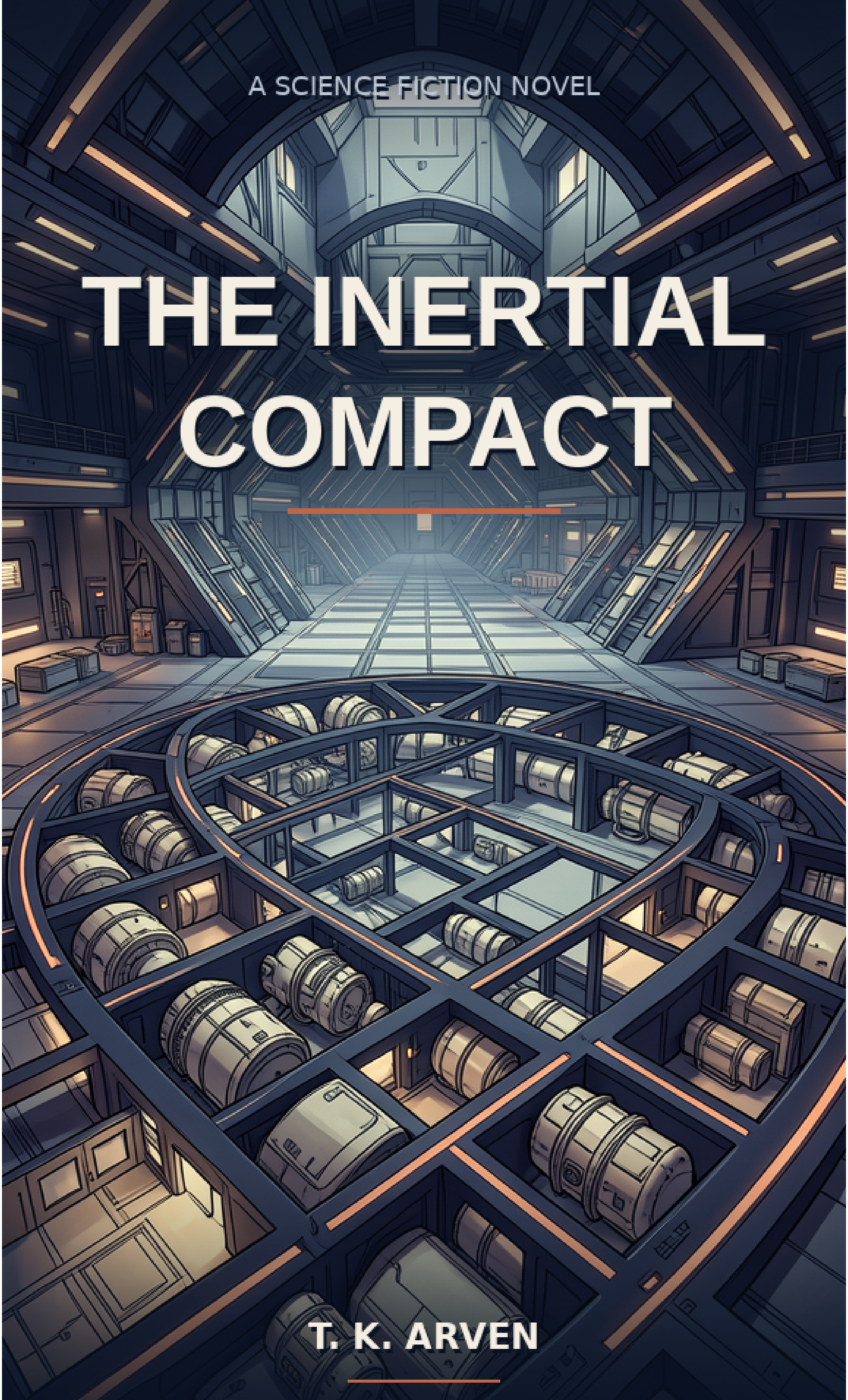

A SCIENCE FICTION NOVEL

THE INERTIAL COMPACT

T. K. ARVEN

The Inertial Compact

A covenant built on a waiver. A waiver built on a choice to call a running drive a halted one.

T. K. Arven

First edition

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Part I

Chapter 1 — The Covenant Renewal

The Compliance Office occupied the third ring of the north spoke, where the station rotation pressed heaviest against the floor. Lira Vehn sat at her desk with a green compliance pencil in the groove of her left hand and a fresh docket form open before her. The room was small: four desks, a supervisor glass-walled office, a docket archive bolted to the east wall, a window that looked inward toward the central hub. The window was dark.

Lira had been a Compliance Officer for six years. She was thirty-four years old, resident of rank 94, and she was good at her job. Good at it in the way a person is good at filing: she made the line item real, she moved to the next one, and in the morning she did it again. The supervisor's name was Dreya Selna. She had issued the task this morning through the glass: Article 17 would be verified before the covenant renewal in fourteen days. Lira was the lead.

She checked the first line item. The text was short and unchanged from the previous twenty-one renewals, and she underlined it with a precision that came from knowing the mark would be there for the next auditor to see.

The second line item listed the station population at 4,096. Lira checked the ward breakdowns: North 203, East 847, South 912, West 634, Upper 411, Lower 555, Reservoir 207, Covenant 320. She added the numbers and found a discrepancy of seven residents she could not account for. She did not pursue the discrepancy. The previous auditor had found the same seven in year 197 and written a note that described them as anomalous and unresolvable. The note had been accepted. The anomaly sat inside the line item now. The line item remained real even if the anomaly did not.

The accepted note carried a docket reference in the lower margin. Lira entered the reference into her terminal and waited while the local archive opened a scan from year 197. The older form used wider columns and a different shade of green for verification marks. Its ward totals matched the figures in front of her except for two neighborhoods that had since been reassigned between East and Reservoir. Those transfers balanced each other. The same seven remained outside the ward sum.

At the bottom of the scan, the prior auditor had listed the reconciliation steps. Population roll checked against ration issue. Ration issue checked against occupied modules. Occupied modules checked against ward directories. Each source returned 4,096 at the station level. None assigned the remaining seven to a ward. A supervisor had initialed the margin and written *variance retained pending next census cycle*. The next cycle copied the sentence. Four later cycles copied it again.

Seven blank positions appeared in the supporting table, each linked to an active ration credential and an occupied-module code. The names had been suppressed from the scan available at rank 94. Lira could request them under Article 12 if census reconciliation became her assigned task. Article 17 gave her no such assignment. She checked the accession field anyway. The request box remained grey. Beneath it, a line of terminal text stated that ward attribution did not affect the verified station total.

Nothing in the comparison corrected the arithmetic. It showed how the arithmetic had survived. One officer recorded a variance. A supervisor accepted it for one cycle. Later officers inherited the acceptance as part of the form, until an unresolved count acquired the appearance of a settled method. Lira copied the year-197 docket reference beside her own seven. She did not copy the words *anomalous and unresolvable*. She wrote *source records agree at station level; ward attribution absent*.

Lira returned the scan to the archive and closed the terminal window. The seven still occupied her margin. They had gained a history, not an explanation. Article 17 asked her to verify the published population, and every available source gave the same published number. She placed a check beside the total and left a small open square beside the ward sum. The two marks touched at one corner without becoming the same mark.

The third line item was the most difficult. It read: *All residents shall renew their covenantal signature during the spring equinox renewal period.* The word *shall* appeared five times in that single sentence, and each time it added a weight Lira was not sure the line item was designed to carry. The procedural load of *shall* amounted to the weight of enforcement architecture. The clause told residents what their signature meant and what refusal meant and what happened to them if they refused to renew.

She wrote the third line item on the verification form. She underlined it. The pencil vibrated in her hand at a frequency the regulator contributed to everything on the north spoke. She underlined the first, the second, the third, and then she did not think about whether the weight of five *shalls* was intentional or accidental.

When Maren Otta entered the office at the end of the morning, Lira did not have to look up to recognize her. Maren was a load engineer working in the south spoke, and she had not spoken to Lira in seven years. Seven years since Maren filed a load report that the Compliance Office classified as erroneous and filed under a separate docket. Maren had not filed another full report since.

Lira identified Maren by the sound of her boots. The rubber soles made a distinct pattern on the corridor floors, a drag and a release and a drag, the cadence the audit training manual listed in its appendix as an indicator of a load-bearing walk and not a compliance concern.

You are the lead on the covenant renewal, Maren said. She did not ask her name. The request was specific and Lira answered it.

I need the load numbers, the quarterly figures from year 60 onward. Those are in the docket archive on the second shelf east wall, Maren said. I filed them.

Then I will check them.

Maren nodded once, the way an engineer nods when the answer exists and the work is doable without a second question. She turned and the door made a sound that Lira associated with the closing of a form. The regulator hummed beneath the floor. The window stayed dark. Interior stations did not have mornings. The hub faced the corridor windows and they faced nothing.

Lira returned to the second line item. The seven residents sat in her margin as a fact and the fact held true by arithmetic alone. The ward reconciliation showed the shift but not the origin. She had written that fact in the margin and had not pursued it further, because the previous auditor had written the same fact in the same margin and had accepted it as the way the station's population moved. That was the way the population moved. The line item still carried real authority.

She put the pencil down and looked at the docket archive across the room. The shelf marked *east wall, second tier* held seven years of Maren's suppressed reports filed under separate covers, each cover stamped with a classification she had never been cleared to read. She did not stand up to retrieve them. She had not been asked to retrieve them. The supervisor had issued a single task through the glass: verify Article 17. The retrieval of quarterlies fell outside Article 17 and inside whatever Maren had come to ask for. Lira drew a small square in the corner of the verification form and wrote *M.O. req. 0741* inside it and dated it and put the form back in the docket and closed the cover.

Through the glass, Dreya Selna was on the telephone. The supervisor held the receiver against her left shoulder and her pen against a yellow pad and her eyes did not move. The glass kept Dreya's voice out of the office proper; Lira could read posture, not words. Dreya's posture that morning had been the posture of a person explaining to a person above her in the chain of command that the thing being asked of her was being asked of her again. Lira had seen that posture before, every quarter, when the quarterly reports came back from the central filing office with red corrections and the supervisor had to explain to whoever had sent them why her office had filed them the way it had. The supervisor's explanations were accepted. The supervisor's posture looked the same every time. Lira did not know what she was explaining on the telephone this morning.

Lira did know that her own rank, 94, had been issued at the last quarterly recalculation and would not be re-evaluated for another seventy-eight days. She also knew, because the supervisor had told her once in the autumn of year 200, that a rank below 80 would reassign Lira from the third ring of the north spoke to the East ward, which had thinner walls and a view of the reservoir, not the hub. The reservoir was colder. The East ward reassignment would not be a demotion in the procedural sense but it would feel like one and Lira had decided, sitting at this desk, that she did not want to feel like that. She picked up the pencil again.

In the evening Lira opened the founding ledger, the folio bound in grey cloth she had requested access to three days after the supervisor had not explained why. The ledger held the oldest records in the station's administrative archive. Its pages were brittle and smelled of the mineral oil the Engineering Department used to preserve things. The first page listed the eleven signatories of the original covenant from year 0, day 3, with their residential wards at the time of signing: Edik Hain, Osta Vehn, Pello Otta, Recha Tallis, the Soren-household load engineer, and six others Lira had not yet read.

She read the names aloud in her apartment, two rooms from the kitchen commons where the porridge arrived at meal time and the announcement chime sounded once at the top of each station day. The names were from a time before the current ward directories existed. Most belonged to residents long dead. Two still appeared elsewhere in the current files: Osta Vehn as a living witness, Edik Hain beneath a death entry Lira had no reason yet to question. Edik's signature on the first page sat bold and close to the left margin, and the ink had faded to a brown that was not quite the same brown as the other ten. Lira underlined that name with her green pencil. The pencil made a sound against the paper that she could feel in her teeth.

She thought, briefly, of her mother. Lira's mother had signed the Compact in front of her when Lira was eight years old. She had knelt beside the kitchen table and held Lira's left hand around the green pencil and explained that the drive no longer served the station, and that the Compact amounted to the agreement the founders had made in exchange for the loss. Two facts: the drive did not run, and the Compact kept the station steady. Lira had never asked her mother for a third. Her mother had died in year 198, six years before Lira took the compliance examination, and the question of whether a third fact existed had not occurred to Lira until this evening, with her finger under the name Edik Hain on a brittle page.

The kitchen where her mother had knelt was three rings north of the apartment Lira now lived in, and the table she had knelt beside still stood where it had been, because the residential module turnover on Tessera-9 was low and the apartments reassigned by compliance score and Lira had been promoted enough times to merit a different view of the same hub, but the kitchen she had grown up in still sat empty on the south side of the third ring and Lira had walked past it once in the autumn of year 199 and put her left hand on the wall outside the door and had not gone in. The green pencil she had held that morning still lived in her desk drawer at the office. It was not the pencil she used now. The pencil she used now had been issued to her the day she took the examination, and the issued pencil sat in the groove of her left hand the way the original one had sat in her mother's grip. Lira had never asked the Compliance Office for a third pencil either.

She did not think further about what it meant that Edik Hain was alive and had not told anyone that he was alive, and that he was the man who had signed the waiver, and the waiver served as what the founding ledger referred to in the annex that was sealed under Article 4, and Article 4 carried the line item that forbade anyone from unsealing the annex. The line item held real authority and had held it since year 0. It would hold it until the covenant ran through amendment by an act that no resident had ever performed.

The apartment was cold. The porridge was cold. The founding ledger was cold and old and real. Lira closed the ledger and opened the docket for the covenant renewal and picked up the pencil and wrote the word *verified* next to each of the three line items she had been assigned. She wrote the word on the first. She wrote the word on the second. She wrote the word on the third. The pencil left a thin mark on the verification form that the supervisor would see when she next opened the docket. Lira set the form aside and began to open the audit log she kept in a thin notebook beneath the green pencil.

The audit log was not a filing requirement. It served as a habit. She wrote three lines of notes describing the three discrepancies she had found in the ward reconciliation. She wrote the note about Edik Hain. She wrote the note about the word *shall* and the weight it carried. She closed the notebook. She placed the founding ledger back on its shelf.

She sat at the kitchen table and watched the ring turn through the apartment window, the ring that carried residential modules and industrial sections and the dark southern extension where the Inertial Engine occupied its cavity. The Engine had been running since year 0. It produced a continuous thrust that 4,096 residents had agreed never to think about. Lira Vehn sat in a station that served as a sealed thing, and the seal amounted to the covenant, and the covenant served as the document she had been hired to verify. The verification amounted to the thing she had been trained to do. The doing kept her at rank 94.

The regulator registered attention as a frequency. She felt the frequency in her molars the way she felt the weight of the green pencil in her left hand. She did not pursue the seven residents. She did not pursue the five *shalls*. She did not pursue the sealed annex or Article 4 or the name Edik Hain. Tomorrow she would go south toward the Engineering Department. Tomorrow Maren Otta would have the archived load reports. The seven residents would still be seven. The word *shall* would still be there. The founding ledger would still be open to the first page. The compliance pencil would still be in the left groove. The sound the pencil made against paper would still carry through the floor. It would reach the reservoir through the bus-bar. It would arrive, finally, into the Engine that had been running since year 0.

Lira Vehn went south. The corridor's sound reached her: the rubber soles of engineers walking toward the reservoir, the distant chime of the meal announcement, the regulator humming its low continuous tone through the bones of the station. She chose south because the discrepancy needed resolution, and because the south spoke held the load reports that explained nothing about where seven residents had gone. Still the covenant counted them. Still the line item said four thousand ninety-six. The sum of the wards matched the same number. The difference sat in her margin as a question mark she had not yet pursued.

The question mark amounted to its own kind of filing. The filing served as the thing she had been trained to do. The station turned through its orbit toward a decay that no line item in Article 17 described in language the Compliance Office could acknowledge. The decay held real. The station held sealed. The seal served as the covenant. The covenant held real. Lira Vehn served as the Compliance Officer who walked through the south corridor with the verification form in her left hand and the green pencil in the groove of her grip. The regulator hummed beneath the floor at a frequency that belonged to no one person and to every living resident of rank 94 who had ever underwritten a covenant renewal.

At the Engineering Department she stopped at the archive desk and produced Maren's accession number. The clerk at the desk looked it up and handed back a manila folder with a red tag attached. Lira did not open the folder. She placed it on the counter beside her own verification form and waited while the clerk stamped it with the quarterly receipt. The stamp left a red circle on the folder's corner. Inside the circle the clerk wrote the date and the clerk's own name, a name Lira did not read. The clerk moved on. Lira stood there holding the folder and the verification form and the green pencil while the regulator hummed beneath the floor, its sound reaching the south spoke through thicker concrete than the north spoke's walls had. The hum was a constant presence. The station kept running. The station kept its silence about the seven residents and its agreement with the covenant and its quiet willingness to keep turning through the orbit no line item described. All of it amounted to the thing that brought Lira back to the north spoke with the red-tagged folder still on the counter, the verification form still open to the first line item, still underlined.

She would not pursue the seven residents. She would not pursue the word *shall*. She would not pursue the founding ledger or the name she could not place from the first page. Edik Hain, the bold signature, the ink not quite the same brown. The line held real since year zero. The station stayed sealed in its orbit, turning toward an understanding that no one on the station yet knew was coming.

Lira counted one, two, three, four, five, six — the number of things she had not pursued today — and then she stopped counting. The count had settled. The station kept moving. The green pencil kept moving. Lira Vehn moved with them through the corridor back toward the Compliance Office, toward the desk that still held the open docket, toward the line item that still needed her signature. The space next to the word *verified* was still open. She had written it once and crossed it out twice. She would write it again. The pencil waited in the groove of her left hand. The morning had started with it there, and the corridor brought her home to it.

In the office the supervisor's glass wall reflected the south corridor behind Lira. She saw her own outline in the glass before she saw the glass itself — a dark shape holding a green pencil and a red-tagged folder in one hand, a verification form in the other. The outline stood still. The form stayed open on the desk. The form did not move. She could hear the pencil through the glass, a faint scratch every time her hand shifted the graphite against the filing surface. The morning had not yet decided what Lira would do with the remaining line items or the seven unresolved residents or the questions that lived in the margin of the docket and the station and the covenant that no one on the north spoke yet knew was coming.

Chapter 2 — The Line Items

The verification form waited on Lira's desk for four hours. The green compliance pencil rested in the groove of her left hand. She had checked line item one and line item one alone. The remaining sixteen line items sat unverified across the form in their printed rows.

The preamble of Article 17 used the word *lost* three times. Lira underlined each occurrence because underlining was her method. The preamble said the station lost its drive. The preamble named the event *the Halt* twice more. Each repetition used the same word and each use performed the same work of describing a station condition. The condition was loss. The word was *lost*. The work was unchanged across all three presentations.

She set the form down and picked up her thin audit notebook. She opened it to a fresh page. She wrote the date at the top. She wrote the observation that the preamble said one thing three times and the preamble stayed the same through all three repetitions. She wrote the word *lost* beside the observation. She closed the notebook. The pencil returned to her left hand.

The corridor light changed position along the baseboard as the station completed another fraction of its rotation. Interior stations did not have a morning period. The hub window faced the interior corridor. The interior corridor faced nothing. The regulator hummed beneath the floor of the third ring. Lira placed both hands flat on the desk. The vibration traveled through the wooden surface into her palms and up through her forearms. It settled behind her teeth. The hum delivered a frequency the station had not changed since she arrived four hours before. The hum would continue. The station would not stop turning because it carried the Inertial Engine in its southern extension. The Engine produced thrust. The thrust required motion. The regulator continued to hum.

The south corridor carried Maren Otta's boots at irregular intervals. The boots made a pattern Lira could identify from three years of working the north spoke. A drag. A release. A drag. The audit training manual listed this cadence as a load-bearing walk indicator in its appendix. Lira did not consider boots a compliance concern. She did record the cadence in the margin of her awareness in case the pattern changed or stopped and absence of the pattern would be itself a change worth noting.

Article 2 was titled Continuity of Residency. It stated every resident appearing on the station population roll would remain on that roll until departure recorded by the Compliance Officer. The phrase *in a manner recorded* appeared in the article because departure required documentation and documentation required a record and the record lived in the compliance officer's marginal notes on the verification form. Lira underlined *manner* because the manner of leaving mattered to the station. A tracked departure was different from an untracked absence. The distinction between the two held operational weight for the census reconciliation Article 6 would address later in this renewal period.

Lira turned to Article 3. Article 3 established the format for verification submissions. The officer would check each line item. The officer would record discrepancies. The officer would submit. All three actions described obligations. Lira read the article and found no discrepancy between its instructions and her current activity. She wrote *consistent* in the margin beside Article 3. The pencil left a thin mark on the verification form at the bottom of the page where line item three had not yet received verification.

Article 4 was the seal clause. Lira picked up the pencil and held it above the page without touching the surface. Article 4 forbade any compliance filing from unsealing the annex. The annex contained the waiver. That document explained the station's relationship to the drive the preamble claimed was lost. Lira had found the waiver the previous evening in the founding ledger. The founding ledger was in the annex, sealed by Article 4. The prohibition was circular and Lira understood the circularity as clearly as she understood the words on the page. She set the pencil down without writing on Article 4. Article 4 remained unverified. The seal stayed intact. The annex stayed hidden. The waiver stayed sealed.

Lira moved to Article 5. Article 5 described the ratification process. The compliance officer would submit verification forms to the supervisor. The supervisor would ratify or reject submissions. The procedure placed the supervisor at the end of the workflow and Lira at the beginning. Lira would continue to be at the beginning. She underlined the word *beginning* as a note to herself about where in the process she currently stood. One of seventeen articles. One check mark. Sixteen articles ahead. The process was long and long was the nature of covenant renewals and length was acceptable because the length was built into the procedure by the founders and the founders had intended the length as a safeguard.

Article 6 addressed census reconciliation. Lira checked the census returns. They showed 4,096 residents across seven wards. She checked the population roll. It also showed 4,096. The variance was zero. The numbers matched. But seven residents occupied station modules and used station resources and collected station rations without appearing on the census returns for the current quarterly cycle. Lira had noted these seven residents in the margin of yesterday's form. She did not amend the margin to address them again today because addressing them was beyond the scope of Article 17's verification tasks and beyond Lira's authority at rank 94 to pursue without supervisor approval or a line item authorizing pursuit. The line item did not exist. The question remained in the margin beside the number seven.

The next articles covered procedural minutiae. Lira read each one. Each article asked a compliance officer to perform a specific function during the renewal period: file records, update schedules, obtain signatures, transmit reports, archive confirmations. Lira read Articles 7 through 14 in order. She underlined the procedural terms with the green pencil — *quarterly*, *annual*, *upon request* — because each term described a time-bound obligation. The substance of Article 17 set those obligations for the compliance officer to do. The substance was procedure. Procedural fidelity was rank 94's contribution to the compact renewal.

Article 15 allowed the officer to record anomalies in the form's margin. Lira picked up a black office pen from the pen cup beside the docking station. She wrote a clean line: *Anomaly 1 — seven residents unaccounted by census. Variance recorded as zero. Year 197 prior note accepted by previous auditor.* She capped the pen carefully so the ink did not dry out prematurely. The black pen and the green pencil served two different functions on the same verification form and the form accommodated both marks because the form was a physical object that could hold multiple types of notation from multiple colors of instrument.

Article 16 set a deadline. Completed verification forms would be submitted to the supervisor before the end of the station day on which the renewal period officially opened. The station day concluded when the ring completed one full rotation. Lira had been working on this form for four hours. More station time remained. Fifteen of the seventeen line items were still unverified. The deadline was generous enough for someone working at Lira's pace and Lira's pace was deliberate and deliberate pacing prevented the kind of errors that rushed verification introduced into the procedure.

Article 17 was the closing provision. It stated that unresolved questions would go to the Compliance Supervisor for resolution. Lira had a question unresolved by every article she had read and by the procedures those articles described. The question concerned the contradiction between the preamble and the waiver. She wrote the number 17 in the margin. She wrote the word *pending.* Lira underlined *pending.* The state of pending existed when an answer had not yet arrived. The answer would arrive from the supervisor through the glass wall. The supervisor held the telephone. Someone above the authority chain would decide. The discussion would not yield a resolution to Article 4's annex question. Resolution required an amendment. Amendments required resident signatures. Signatures required a process. The process started with a question like Lira's.

The training manual specified that compliance forms should be handled with a flat palm and a steady wrist. Lira kept her wrist locked as she reviewed the instructions printed at the bottom of page four. The instructions described the retention schedule for marginal notes. Notes written in green pencil remained part of the official file for three renewal cycles. Notes written in black ink remained indefinitely because black ink contained carbon black pigment that resisted oxidation under the station light tubes. This distinction had been established in year 44 by the first compliance director appointed after the transition from the early settlement phase to the permanent administrative regime. Lira appreciated the durability of carbon black. It guaranteed that an observation recorded today would still be legible to an auditor examining the same margins a century later, long after Lira's rank 94 designation had been reassigned or retired.

The station's rotation generated a steady centrifugal acceleration that varied slightly between the inner and outer radius of the residential rings. At the third ring where Lira's desk stood, the gravity equivalent was maintained at eight-tenths of standard Earth normal. This acceleration kept papers flat on the desk without requiring paperweights, though clips were issued in standard stationery packets. Lira did not use clips. She preferred the natural friction between paper sheets and the wooden desktop surface. The friction was sufficient to prevent drift during normal rotation phases, provided the station's attitude control thrusters did not fire corrective bursts too abruptly. When the attitude thrusters fired, a low-frequency shudder ran through the structural ribs of the spoke, causing unsecured pencils to roll toward the outer bulkhead. Lira had placed her green pencil in the desktop groove specifically to counteract this drift.

Lira stood up from the desk at 14:42 station time. She walked south through the hub corridor toward the Engineering Department. The direction was south because the Engineering Department occupied the south spoke and the reason for going south was not Article 17 and was not any task the supervisor had issued behind the glass wall. The reason was Lira's own and Lira did not name it.

The north spoke corridor was narrow. The walls were thin. The regulator sound carried clearly through the north spoke because the regulator sat beneath the third ring, beneath Lira's desk. The south corridor opened into a wider space with more foot traffic. The floor was thicker in the south spoke. Engineering sections required structural reinforcement for heavy equipment. Lira could feel the machinery vibrating through the deck under her boots.

The archive desk was a narrow counter between the south corridor and the records room. A clerk stood behind the counter reading from a lit terminal screen. Lira approached. She held out Maren's accession number from memory. The clerk typed the number into the catalog system and studied the screen for a moment. The clerk produced a manila folder from a shelf behind the desk. A red classification tag was attached to the folder's tab. The tag designated restricted access for personnel rank 71 and above. Lira's compliance score was 94. Rank 94 exceeded rank 71. The restriction was not about rank but about the classification the Engineering Department had applied to these records a long time ago on a permanent basis.

Lira placed the folder on the counter. The clerk stamped a quarterly receipt onto the folder's cover. The stamp produced a red circle that was not quite round and the clerk wrote a date inside the circle in a handwriting style Lira did not recognize. The clerk did not look up from the terminal screen to observe Lira standing there with a restricted folder on the counter. Lira was visible to the clerk but the clerk's attention remained on the screen. This was the normal relationship between requester and clerk for restricted materials: the request was procedural and the fulfillment was procedural and neither person needed to know the other's name because the records identified the requester by accession number and the accession number was the key that opened the catalog entry that located the folder.

The clerk moved on to the next person in line. Lira waited. A second person completed a brief transaction at the counter. The clerk produced a different folder without looking away from the terminal screen this time either. The second person carried the folder to the south corridor toward the engineering offices. Lira remained at the counter. When the clerk looked up again Lira stepped forward and the clerk handed back the stamped folder and Lira took it from under the clerk's arm and said thank you and the clerk nodded without breaking eye contact with the screen and Lira walked north.

The corridor from the south spoke to the north span of the hub took approximately four minutes to walk at a compliance officer's pace. The pace was measured and deliberate. Rushing belonged to a higher rank or a more urgent concern and Lira's concern was not urgent in the operational sense because the verification form would not expire before the station day ended and the station day had hours remaining and the hours were on Lira's side.

Lira carried the folder back through the south corridor and into the hub. The station rotated beneath her boots. The hub window faced the interior of the station's spine. Conduits and cable trays lined the view, and a ventilation shaft drew air from the engineering sections and returned it cleaner to the residential rings. The hub functioned as the station's center in name only. It served as the place where the north spoke met the south spoke and the east corridor crossed the west corridor and every direction led somewhere else. Lira walked north.

The corridor narrowed as she approached the north spoke. The walls compressed the sound of the regulator into a single continuous tone. She walked through it at compliance officer's pace. The pouch at her belt shifted with each step. The folder inside held the load measurements from year 60 onward. She had not yet opened them. She would open them when she found the time. But the time had been found during this same station day, and now the station day was ending, and the folder would sit in the clearance pouch until tomorrow morning when Lira had another station day and another opportunity to examine the records that proved the preamble's claim of a lost drive was false.

Lira reached the Compliance Office in the north spoke and entered through the door. The glass wall of the supervisor's office was visible immediately through the corridor's single window. No change. The supervisor was not visible behind the glass. The glass kept the supervisor's workspace private and the supervisor's workspace did not display any awareness that a rank 94 compliance officer had just carried a restricted Engineering Department folder across the hub and through both spokes without filing a movement report. Lira sat at her desk. The verification form was still open at Article 17, line pending. The green pencil occupied its groove in her left hand. The audit notebook sat closed beneath the pencil in the desk drawer. Seven residents occupied station modules without appearing on the current census returns. Lira had written this in her margin at Article 6. She had not addressed the discrepancy again. Article 6's verification tasks did not include population reconciliation. But seven residents meant seven bodies using station resources. Seven bodies consumed station rations. Seven bodies took up habitation space the census did not account for. If Article 6 required a clean reconciliation, these seven residents were the gap. And if the preamble described a station that had lost its drive, then the preamble's claim was incomplete. The drive's power had not been the only thing the station's original configuration provided. The station had a purpose beyond survival. The preamble had erased that purpose. Lira had checked line item after line item. No line item mentioned the preamble's erasure. The green pencil was in her left hand. The verification form was in front of her. The folder full of load data sat on the corner of her desk. The question she could not file remained open because Article 17 did not provide a line item for contradictions between a founding preamble and a sealed annex.

The station's regulator completed one pulse cycle. The station rotated by one fraction. Lira's pen was not moving. Her form was not advancing. The station day was ending. Tomorrow she would submit the verification form as incomplete. The supervisor would ratify it as filed. Article 17 would remain in pending. The contradiction between the preamble and the annex would remain sealed. That was procedure. That was rank 94's contribution to the covenant renewal.

Lira closed her audit notebook before the station day ended. She slid it back into the desk drawer beside the green pencil. She did not stand up immediately. The verification form still waited in front of her. Article 17 was still pending. The form had been open through two station days now, and the pencil in her left hand had not moved beyond its groove. The station's rotation continued beneath the third ring and the regulator hummed its single frequency and the corridor light shifted along the baseboard through every phase of the cycle. Inside stations there were no mornings and no evenings, only the continuous progression from one rotation angle to the next, and the station day was a human fiction imposed on a mechanical rotation, and Lira Vehn was a compliance officer at rank 94 observing a fiction, and the fiction was useful because it gave the Covenant a rhythm that residents could internalize and that Compliance Officers could schedule and that the Compact could present as order, and order was the thing the Covenant sold to the people who signed it every ten years, and Lira sold order every day by verifying line items on a form that was never going to capture the contradiction between the preamble and the annex.

Chapter 3 — The Founding Ledger

Lira opened the founding ledger twice in three days. The first time was the evening of the first renewal morning when she had found the names of the eleven signatories from year 0. The second time was now. The ledger's grey cloth binding had not changed between the two openings. The mineral oil smell had not changed either. The brittle pages still smelled of preservation and the first page still held the eleven names. Edik Hain still sat bold on the left margin of that page like a fist mark left by someone too impatient to make the letter neat.

Lira turned to the annex page at the back of the ledger. The annex was a folded sheet of thin paper tucked between the last page of the founding ledger and the inside cover. It was unnumbered and carried no docket reference from the Compliance Office. The annex existed outside the cataloging system because the annex described what the Compact had sealed inside Article 4 and that mechanism kept the annex hidden from every compliance officer who opened this ledger for forty-one consecutive renewals.

The description in the annex answered a question Lira had found on the first evening but had not yet understood. That question concerned the word *lost* in the preamble of Article 17. The preamble used *lost* three times when describing the station's departure from its original drive configuration. Each mention performed the same work: the station lost something, and the loss defined the Covenant's purpose. But the annex described a drive that was not lost. The annex described drive that ran continuously from year 0 through the present quarter and produced thrust the station needed to maintain its orbit against a slow decay that would, without correction, deliver the station into the primary's atmosphere in three hundred and twelve years.

Lira read the annex description three times on this second opening. The third reading produced a new observation she had not managed in the first or the second. The preamble's *lost* and the annex's *running* were not two different opinions about the same fact. They were a contradiction. One document stated a departure that never happened and the other stated a function that had been continuous. The Compact was built on the departure narrative. Every signature, every renewal ceremony, every verification form filed at rank 94 confirmed that the station had lost its drive and the Covenant compensated for that loss. The annex had been sealed under Article 4 for 192 years and no rank 94 officer had ever been permitted to read it.

This time Lira did not walk away after reading. She remained seated at the archive desk in the east wall, third tier, between two older folios bound in different cloth colors. She opened the audit notebook she kept beneath the green pencil.

Before copying the annex language, Lira examined the custody marks along the fold. A square stamp dated year 0 placed the sheet in the founding ledger under ordinary restricted access. A circular stamp from year 30 transferred it to Annex 4 under Continuity of Founding Intent. The ink had spread into the paper fibers, but the docket number remained legible. Beside it, a year-90 notation cited Article 88 and moved the annex from restricted reference to sealed founding custody. No later mark returned it.

Each transfer preserved the sheet. The paper carried no water stain, no torn edge, no missing corner where a sentence could have disappeared. Preservation had worked exactly as the archive intended. Access had narrowed while the object itself remained whole. Lira copied the three custody dates into her notebook and drew a line between them. Year 0: ordinary restriction. Year 30: annex transfer. Year 90: sealed custody. Forty-one renewals followed the last mark without changing the physical page.

Under the year-30 stamp, a clerk had written a seven-digit shelf code in brown ink. Lira checked the code against the old catalog index. It led to a retired cabinet number, then to a transfer sheet signed by Compliance

Supervisor Recha Tallis. The transfer sheet described continuity, not correction. It certified that the annex belonged with records needed to interpret founding intent and directed future officers to preserve the material without circulating it. The language did not claim the annex contained false information. It treated accurate information as something custody could keep apart from ordinary verification.

Lira set the transfer sheet beside the ledger. The two documents answered different questions. The annex described what the Engine continued to do. The transfer sheet described who could read that description. Neither document authorized a rank 94 officer to change the public preamble. Together they showed that the contradiction had not survived through damage or oversight. It had survived through a sequence of legible decisions, each marked with a date, a docket, and an officer's hand.

She returned the transfer sheet to its sleeve. The physical record left no gap she could classify as clerical error. The gap existed between custody and public meaning, which gave her verification form no box to check. Lira wrote *preserved, transferred, sealed* beneath the dates. Then she turned back to the first page and the eleven names.

The archive shelf was the second tier on the east wall. The east wall held records from before the current ward directories existed. Before the station sorted residents by compliance score or assigned ranks or numbered the rings of the north spoke or built the Compliance Office in the third ring where Lira now carried out this renewal. The records described a station that had not yet learned to seal its own founding contradictions. Lira turned back to the first page of the ledger. The eleven names were there. She read them one by one. Each name represented a resident who had walked into the Covenant chamber in year 0 and placed a signature on the founding document. Each name was a commitment that the station would survive. Each name carried the weight of that promise.

Edik Hain sat bold on the left margin. His was the first name on the page. Lira remembered the appendix mentioning him in the year-0 records as a witness to the original signing ceremony. She had not known he was still alive until this morning. The thought unsettled her. Edik Hain had been a young man in year 0. He was alive now, in the present quarter, somewhere on this station, and his signature still anchored the Compact's original purpose. That purpose was now contradicted by the annex sealed behind Article 4. Lira wrote his name in her audit notebook and paused before writing the next one.

The appendix listed all the year-0 events. Lira had read it quickly on the first evening. Today she read it more carefully. The signatories' names appeared on a roster page that followed the first page of the ledger. Eleven names in a single column. She looked at each name in turn. The first was Edik Hain. The second and third she did not recognize from memory. The fourth was her mother's name. Lira Vehn senior had signed the Compact before Lira was born. Before the halt, before the current ward directories, before the Compliance Office existed in the third ring. Her mother had been one of the eleven who built the foundation Lira now verified every ten years.

Each of the eleven names sat on that page in a different hand or a different style of printing. Some were neat. Some were hurried. Each signature was a choice made at a moment when the signatories believed their station was dying and each signature was a commitment that the station would survive through whatever crisis lay ahead. Lira felt the weight of that moment pressing down on her own signature. She had signed the Covenant at age eight in the renewal ceremony. She had not chosen to sign. She had not understood what the document contained. She had followed her mother to the chamber and placed her hand on the page while an officer guided the pencil.

Her mother's name appeared on the roster page between the fourth and fifth positions. Lira studied it for a long moment. The name was printed in the same founding hand as all the others. It did not look different. It looked like every other name on that page until Lira remembered it was her mother's and then it looked like the weight of a covenant she had never fully understood.

Lira's mother had signed the Compact believing the drive was gone. Lira believed her mother for forty-one years. The annex said something different. The annex said the drive was still running. It had never stopped. It had run for two hundred years without interruption while the station's residents were told the opposite. The preamble's narrative of departure was constructed to serve Article 4 and Article 4 kept the annex sealed and the seal kept the contradiction from every living resident whose signature gave the Covenant its authority. Lira felt the truth settle in her chest like a stone that had been waiting for her to find it. Her mother had not known. Her mother had signed in good faith, believing the station was dying, believing the Compact was survival. That belief was the foundation on which forty-one years of renewals had been built. Every signature Lira had checked, every line item she had verified, every form she had filed had been built on that same foundation. The foundation was a lie and Lira now held the document that proved it.

The audit notebook was beneath the green pencil on the desk. Lira opened it and wrote three lines of notes about what she had found. The first line recorded the annex and what it described. The drive was running. The preamble lied. The contradiction was not an error of interpretation but a structural feature of the founding documents. The second line recorded Edik Hain being alive and the implications of that fact for Article 4. The waiver described Edik Hain as the signatory whose survival would invalidate the Compact's original purpose. The annex did not describe him as dead. It described him as the witness who had confirmed the Engine's original power budget and had signed under terms that no longer matched the station's actual operating condition. The third line recorded her own mother signing the Compact in year 0 and the preamble describing the drive as lost when the waiver described the drive as running and the contradiction between the two documents being the very thing Article 17's verification form could not capture. The form had line items for line items and line items did not include the contradictions between the preamble and the annex.

She closed the notebook. She placed it beneath the green pencil in the desk drawer. The pencil remained in the groove of her left hand. She looked across the third ring at the verification form still open on her desk. The desk surface was bare except for the form. Article 1 still had its check mark in the left margin. Articles 2 through 16 were unchecked. Article 17 said, in smaller type beneath the line, *pending*. The form had been open for the duration of one station day plus the evening when Lira first opened the ledger at the archive desk plus the morning when Lira had checked Articles 1 through 3. The form was still open at line item one. The station was still turning beneath the third ring. Seven residents sat unaddressed in her margin. The preamble still used *lost* three times. The annex was still sealed and Edik Hain was still alive according to the ledger she had just read.

The contradiction was still real and no compliance officer at rank 94 could file an answer to it. Lira reached for the audit notebook again. She wrote the current station time beside a new line. On that line she wrote what she had come to understand over the course of the renewal period. The station lied about its drive and the Covenant accepted the lie and Compliance Officers verified line items and those line items did not include the contradiction between the preamble and the annex. The station kept that contradiction sealed behind Article 4. The annex covered the waiver. The waiver described the drive as running, not lost. The contradiction between the two founding documents was beyond what Article 17's verification form could hold because the form checked line items only. Article 4 forbade the unsealing of founding documents for anyone below the rank required to amend the Covenant itself. The weight of this restriction sat on Lira's mind as clearly as the folder in her hand. Article 4 was not just a line item. It was a mechanism of erasure. It had kept the waiver hidden for 192 years. It had kept the Engine's true output hidden from every compliance officer who had walked this corridor and filled out verification forms and checked line items on their renewal morning. Article 4 was the seal on the truth. And Lira was a rank 94 officer who could verify line items but could not lift the seal. The distinction mattered because the line items and the seal were two faces of one Covenant and the Covenant was built on a falsehood that Article 4 protected.

She wrote the article number: 4. Below it she wrote the word *amendment* and beside that the number 9. Nine resident signatures were required to amend the Covenant. The quorum was defined in Article 12. Lira did not currently hold a single one of those signatures. She had not yet approached anyone. She had not yet named the contradiction to another living soul. The signature would need to come from residents who understood the full weight of what she had discovered. They would need to understand the contradiction between the preamble and the annex. The preamble described departure. The annex described continuous function. The waiver described a drive that was unnecessary. Each document described a different reality about the station's power source.

Lira stopped writing and closed the notebook. She slid it beneath the pencil in the desk drawer. Her hand did not move from the pencil. The regulator hummed its frequency beneath the third ring. The hum belonged to no resident and to every resident equally because the hum came from the Engine. That machine's operations sat behind Article 4 and Article 4 served the Covenant Lira had signed at age eight. Service meant verification. Verification meant line items. Line items meant the procedure worked when the procedure did not ask the officer to name the thing behind the seal. The question remained. Lira stared at the green pencil in her left hand. The green pencil was the instrument that marked line items verified. No line item in Article 17 told a compliance officer how to break whatever the position had been hired to maintain through forty-one consecutive renewals for the forty-second renewal in a row.

Lira stood up from the desk at 14:42 station time. She walked south through the hub corridor toward the Engineering Department. She carried the restricted folder in the clearance pouch at her belt. The folder held load measurements from year 60 onward that she had not yet opened and the opening would come when she found the time because she had walked south to retrieve the folder because she could not, within the boundaries of Article 17, walk back to her desk and report a contradiction the line items did not contain. The corridor she walked through was bare metal and fluorescent and the lighting was even and it showed nothing and everything at once. Lira moved through it with the folder pressed against her hip and the question pressing against her thoughts.

The north spoke corridor was narrow and the walls were thin. The sound of the regulator reached Lira's ears clearly in the north spoke because the regulator sat beneath the third ring and the third ring was where Lira's desk occupied the small square of floor that her rank entitled her to call a workspace. The regulator's hum was constant. It had been constant for the entire operational life of the station. Lira had grown so accustomed to the sound that she no longer noticed it consciously, but her body still registered the vibration in her molars and the pressure of it against her sternum. The corridor walls were bare metal painted the standard station grey without decoration or personal items. The north spoke was a passage, not a home. Lira walked through it now with the folder in her clearance pouch and the weight of the folder pressed against her hip and the contradiction inside the folder pressed against her thoughts. The south corridor opened into a wider space with more foot traffic from engineering staff moving between workstation clusters. The floor was thicker in the south spoke because the engineering sections required structural reinforcement for heavy equipment and the heavy equipment included the load-bearing instruments Maren Otta used to measure the Engine's energy output. Lira passed the junction between the north and south spokes at a measured pace. She did not speed up. She did not slow down. The procedure was what it was.

Lira carried the folder across the hub toward the records room. The clearance pouch held manila folders and the small notebook and the green compliance pencil in their designated positions. Her pace was measured and deliberate. The same measured pace she had maintained through every renewal period since rank 94 gave her the authority to carry restricted materials in the north spoke. Rushing belonged to a higher rank or a more urgent concern. Lira's concern did not qualify as urgent under Article 17 and Lira did not intend to make it urgent by filing an anomalous report that Article 17's verification checklist did not authorize.

She stopped at the archive desk between the south corridor and the records room. The clerk stood behind the counter reading from a terminal screen. Lira held out Maren's accession number from memory. The clerk typed the number and studied the screen. A red classification tag marked the folder tab when the clerk produced it from the shelf. The tag designated restricted access for personnel rank 71 and above. Lira's compliance score was 94 which exceeded rank 71 and the restriction on these records was not about rank but about classification. The Engineering Department had applied this label on a permanent basis decades ago and the label had never been lifted and the folder waited on the shelf for someone who would never ask why it waited. Lira watched the clerk work. The clerk's hands were steady on the terminal keyboard. The clerk had processed the same kind of request hundreds of times before and this request was different only because Lira was standing at the counter and asking for the history that the department had chosen to keep hidden from every other resident on the station.

The clerk stamped a quarterly receipt onto the folder's cover. The stamp produced a crimson circle that was imperfect and mechanical and the clerk wrote a date inside the circle in handwriting the clerk had not used for other transactions Lira had observed from this counter across the previous three years. The clerk did not look up from the terminal screen when Lira stood at the counter holding a restricted folder that contained the Engine's actual performance records from year 60 onward. The records proved one fact. The Engine was still running. The Engine's running meant the preamble lied. The preamble's lie was the Compact's foundation.

She turned and walked north through the hub corridor with the folder in her clearance pouch. The station's rotation carried her eastward now toward the residential sections. The hum of the Engine reached her in the north spoke the same way it always had. The regulator sat beneath the third ring. The third ring was where Lira's workspace sat. The workspace was where she would return to file the quarterly receipt and close the ledger and leave the annex sealed and leave the contradiction as it was. She was a rank 94 compliance officer. The procedure was Procedure. The line items were line items. The question sat unanswered in the margin of a notebook beneath a green pencil in a desk drawer in the archive room on the east wall of the third tier.

Chapter 4 — The Resident of Rank 71

Maren Otta stopped walking when she heard the boots behind her. The drag-release-drag pattern belonged to Lira Vehn. She had heard that cadence in the south corridor for three years. The pattern never changed. Maren did not need to see the person to know. A compliance officer at rank 94 walked with purpose but never with urgency. The boots matched the purpose, not the urgency.

Maren kept walking toward the reservoir chamber. The corridor narrowed at the junction where the east maintenance duct crossed the south corridor. She turned left at the junction. The reservoir was west. West was the direction that required passing through the junction. The duct hummed above the junction ceiling and the hum added a second rhythm to the drag-release-drag behind her. Maren registered both rhythms now as she walked. The steady percussion of her own boots. The intermittent vibration of the duct that carried the Engine's waste heat into the reservoir tank. The conduits had been part of the original installation since year 0. They connected to the reservoir chamber that Maren reached now, ten minutes into her second hour on shift.

The reservoir chamber was circular. The walls were wet. The water on the walls was cold and the cold kept the reservoir cool even when the Engine produced heat. The heat entered the reservoir through conduits in the floor. Those conduits predated the current ward directories and the compliance scoring system and the numbered rings of the north spoke. Maren counted them as she walked the outer perimeter. Three hundred and twelve. She had counted them every shift for eleven years, ever since her first day in the Engineering Department, when a senior engineer had told her to learn the floor and the floor would teach you the work. They had told her that on her first morning. They had not told her that the floor would also teach her what not to write in a report. She had not understood then. She understood now.

The Engineering Department occupied the south spoke on the station's eastern axis. Maren had worked there for twenty-three years. She had entered as a junior load technician at age eighteen, right after her residency rotation in the water-distribution ward. She had filed her first load report at age nineteen with the confidence of someone who believes numbers are honest. That was the year her grandmother Osta Vehn told her the Engine still ran. Osta was a witness to year 0. She had signed the Compact at age nineteen and she had never stopped believing that the engine room was the most important room on the station, more important than the Covenant chamber where residents gathered each renewal morning. Osta told Maren this in the kitchen of her grandmother's apartment in the reservoir ward, leaning over a cup of reconstituted coffee while the regulator hummed through the walls. "The Engine runs, child. I was there when it was switched on. The people who said it stopped were the same people who wrote the preamble." Maren was nineteen. She did not know what she would do with that information. Three days later she filed her first load report documenting the Engine's 4.2 megawatts of continuous thrust. The report was returned with a classification stamp: Engineering Department — Restricted. No reason listed. No appeal pathway noted. She filed it again the following quarter. It was suppressed again. She stopped filing after that. The suppression taught her the same lesson the floor had tried to teach her: some measurements exist but must not be spoken aloud. She carried that lesson into every subsequent quarter. She filed the public load summaries that the Compliance Office required and she did not file the supplementary data that the supplemental data showed.

The weight of carrying suppressed reports was not a physical sensation, not at first. It was a quiet increase in the pressure behind her sternum each time she sealed an envelope with the Engineering Department stamp. Each sealed envelope was a suppressed report. Each suppressed report was a measurement that contradicted the preamble. Each contradiction was something she now held in her hands and in the drawer beneath her workstation. Years of them. Quarter after quarter after quarter. The weight accumulated the way sediment accumulates in the reservoir tank — imperceptibly, then all at once. By year 200, when she had been a load engineer for eleven years and had filed exactly zero full reports since Osta's kitchen conversation, the drawer contained forty-three suppressed quarterly documents. She knew the count because she had catalogued them during the shift when the regulator's pulse rate dropped to six per interval, which was the only time she ever had to kill four minutes.

She set the folder on the bench beside the reservoir access panel. The bench was steel and the steel was always cold regardless of the Engine's output. Maren had learned that about cold steel in the Engineering Department: it conducted truth and it conducted indifference with equal efficiency. She picked up the wrench from the floor beside the panel. The wrench was heavy for its size. The heaviness came from the alloy composition and the density of the manufacturing standards that had gone into making it. The wrench required two hands to turn the hatch bolt. She turned it slowly. The resistance was consistent with the torque specs on her work order, which did not mention the Engine, which listed "reservoir access — quarterly inspection" as the approved purpose of the procedure.

The hatch opened with a hiss of pressure equalization. The hiss was sound from the Engine below. The Engine below was a thing that did not exist according to station records because the records said the Engine had been lost at the Halt. The preamble narrated the Halt three times in Article 1 alone. Each narration used the same language. Each narration said the drive was gone. The preamble's narration was what every resident had been trained to treat as fact from the first week of residency orientation, when each child of a signatory family stood before the Covenant, signed with a green pencil, and was told the drive was gone. That was all there was to say about the drive. That was the training. Maren had received it at age seven alongside every other resident. She had not questioned it then. She questioned it now while the hiss of pressure equalization filled the space between her and the open hatch.

The Engine was not lost. This was not a belief Maren held gently or optimistically. It was a fact established by load measurements that recorded the Engine's energy output every quarter from year 0 through the present reporting period. Every measurement existed. Every quarterly file had been filed and sealed under Engineering Department classification. The sealed records did not say the Engine was lost. The sealed records said the Engine produced 4.2 megawatts of continuous thrust and had done so for one hundred and ninety-two years. The production was not a malfunction or a temporary condition but the steady-state operation the Engine had been designed to maintain from the moment the eleven signatories activated the bus-bar on the morning after the Covenant was signed. The bus-bar was the literal connection between the Engine and the station's power grid. Eleven signatories had placed their hands on it together, one for each founder of the Compact that governed the station's relationship with the Engine, the drive, the suppressed data, the sealed annex, the rewritten preamble, and the covenant that bound all residents to the fiction that they were drifting through space without a power source when they were in fact suspended in a precise gravitational equilibrium maintained by a machine that refused to stop.

The regulator's piston was visible through the open hatch as a 4-meter cylinder moving with a rhythm the reservoir chamber conducted through the walls into the feet of anyone standing near the hatch. The piston moved because the Engine moved and the Engine moved because the conservation law required it to move. The conservation law was not a policy choice. The conservation law did not care what the preamble said was true, what the waiver documented as accurate, or what a compliance officer at rank 94 carried in her clearance pouch on a second afternoon of the renewal period. The conservation law was the reason the Engine still ran and the reason Maren had spent the last eleven years knowing something no one at the Compliance Office was permitted to know. The contradiction sat in the space between those two facts like a stone in a shoe that you could ignore if you walked slowly enough and did not think about the shape of the stone.

The Engine's waste heat radiated upward from the reservoir through the opened hatch. Maren felt it on her face at a distance of two meters. The heat bore the Engine's working. It bore the operation the preamble denied by calling the Engine lost. The denial had shaped the Covenant for one hundred and ninety-two years and Maren had spent eleven years of her career as a load engineer filing reports that never reached the Compliance Office. The non-filing was why the load data sat archived in the south spoke behind a counter and the counter held a clerk who would never look up. The records existed and the records were true and the records were not what the preamble described. Maren stood near the hatch with the heat on her face and the wrench in her right hand and the classified data in the folder on the steel bench behind her and she was not a compliance officer now. She was a rank 71 engineer with a work order for a quarterly reservoir inspection. The work order did not authorize the retrieval of load data. It did not authorize the opening of the hatch. It did authorize the hiss that was now filling the reservoir chamber and filling the corridor beyond the door where Lira Vehn stood at rank 94 with the green pencil in her grip and the clearance pouch holding the folder that proved the Engine was not lost.

The boots stopped outside the reservoir door. The drag-release-drag pattern stood still as a request and Maren understood the request without it being spoken. She knew who Lira was. She knew what Lira was carrying in the clearance pouch at her belt. Maren did not turn around to look at the folder because the folder's purpose was obvious to anyone with the rank that granted access to the engineering archive. The restricted materials were the reason Maren Otta was expected eventually to stand behind this door and encounter another compliance officer with clearance to receive what engineering would not freely offer.

Maren did not open the door from inside because Maren was on shift. Shift rules required the requester to knock before the door opened. The shift started at 06:00 station time and had three hours and fifty-four minutes remaining when the boots stopped outside the reservoir door. Maren's shift did not include visitor consultations unless the visitor carried a docket number assigned by the Compliance Supervisor. Lira did not have a docket number for this interaction. This was the first time all day, Maren realized, that anyone had approached the reservoir door without a docket. It was not the first time someone had ignored the procedure. It was the first time Maren had noticed.

Lira knocked once after a pause that measured longer than the courtesy minimum. Maren heard the knock through the door and through the hiss of pressure equalization on the other side of the seal. The knock was light, delivered with the eraser end of Lira's green compliance pencil. The compliance officer's question carried appropriate force for a knock delivered with a pencil eraser that was not authorized by the line items on her verification form. Lira carried restricted load records in the clearance pouch. The records proved the Engine was not lost and the Engine had not been lost in year 0 or any subsequent year. The preamble lied about the Engine. The Compact's foundation rested on that lie. Lira at rank 94 was hired to verify line items and nothing beyond them. The line items were the Covenant and the Covenant said the Engine was lost while the Engine was not lost and the Engine operated continuously. This fact was one Maren had not been hired to confirm and would not confirm here. Confirmation by a rank 71 engineer would require a justification on the load log. The justification would trigger a review by the supervisor. The supervisor would escalate to the Compliance Office. The Compliance Office would produce a docket number and the docket number would start an investigation that rank 94 did not have the authority to initiate. The investigation would surface the waiver from the annex. The annex was sealed by Article 4 and Article 4 was what rank 94 was not permitted to challenge without an amendment requiring nine signatures. Signatures required a process and the process was not on Lira's desk. The form had one check mark and sixteen unanswered line items. The question mark in the margin of the audit notebook had not yet found its answer.

Maren stood behind the reservoir door and understood all of this. She still did not open the door. Opening the door was a procedural act that would require Maren to acknowledge Lira's presence in a manner that would be recorded on the shift log. The shift log was reviewed by the Engineering Department supervisor at the end of each quarter. The supervisor would note the unlisted visitor. The supervisor would escalate the note to the Compliance Office. The Compliance Office would produce a docket number even for an interaction that did not warrant one. The docket number would begin a paper trail that rank 94 Lira Vehn had not yet started and that Maren Otta rank 71 was not going to start for her by opening a door that the work order did not authorize opening.

The silence lasted fourteen seconds. Maren counted the seconds by the intervals between the regulator's pulses transmitted through the floor of the reservoir chamber. Seven pulses per interval. Seven intervals per count. Fourteen seconds was two full counts and the regulator had not changed its pulse pattern during the wait, which told Maren that the Engine's thermal cycle was stable and the Engine was not responding to the request at the door because the Engine was a thermodynamic system and systems did not respond to compliance officers who knocked with the eraser end of a pencil.

Then Maren stepped to the hatch and opened it. The hatch opened with a hiss of pressure equalization and the hiss was sound from the Engine below and the Engine below was a machine that did not exist according to station records because the records said the Engine had been lost at the Halt, the event the preamble narrated three times, and the preamble's narration was what everyone on the station had been trained to treat as fact from the first week of residency orientation. The child of each signatory family stood before the Covenant and signed with a green pencil and was told the drive was gone and that was all there was to say about the drive. Maren had been told at age seven. She was forty-one now and she was still being told in orientation briefings that played on a loop in the residential terminals. The preamble's narration of loss filled the loop. It ran in the background of every new resident's first month and it was the background that Maren had spent eleven years trying to override with data no one was allowed to see.

Maren set a wrench on the hatch lip. The wrench was a tool that marked the hatch as recently accessed and recently accessed required a justification on Maren's load log and Maren would write the justification later when Maren was not being observed by a compliance officer standing at the door with a restricted folder at her belt and a question in the margin of an audit notebook the compliance officer had not yet shown to anyone.

Maren did not turn around to look at Lira immediately. She looked at the Engine room. The regulator's piston was visible through the open hatch as a 4-meter cylinder that moved with a rhythm the reservoir chamber conducted through the walls and into the feet of anyone standing near the hatch. The piston moved because the Engine moved. The Engine moved because the conservation law required it to move. The conservation law was not a policy choice. The conservation law did not care what the preamble said was true. It did not care what the waiver documented as accurate. It did not care what a compliance officer at rank 94 carried in her clearance pouch on a second afternoon of the renewal period. The Engine's thermal cycle was stable. The regulator pulsed at seven per interval, unchanged from the pulse count registered in the year-60 file that sat in the suppressed drawer beneath Maren's workstation. She had read that file eleven times since she had stopped filing reports. The readings had not changed. The Engine had been running at 4.2 megawatts of continuous thrust for every one of the one hundred and ninety-two years recorded in the sealed logs and Maren knew this now with the same certainty with which she knew her own name and her grandmother's name and the weight of a suppressed report sitting in a folder that Lira had carried from the Compliance Office and was now standing just beyond the threshold of the reservoir chamber waiting for a response that Maren was choosing not to give immediately.

Maren did not yet turn around to face Lira Vehn because turning around was a procedural choice that carried weight within the shift rules. Maren wanted to choose the moment of turning with the same deliberation she applied to load measurements. The shift still had three hours and thirty-one minutes remaining. She had not been hired to answer questions from compliance officers who carried folders she had filed under suppression twelve years ago. She had been hired to keep the reservoir cool and the Engine stable and the load measurements accurate. The accuracy of the measurements was independent of what anyone at the Compliance Office chose to believe about them. Maren had learned this in the Engineering Department where measurements were recorded in triplicate and filed in the south spoke and the triplicate copies were cross-referenced by a clerk who did not know what the numbers meant and a supervisor who did not know what the numbers proved and a compliance framework that did not know what the numbers existed to contradict. Maren was the only person in the chain who knew all three layers simultaneously and had been trained, over twenty-three years in the department, to hold that knowledge in a specific arrangement of silence.

When Maren turned around, Lira was standing at the threshold of the reservoir chamber with the green pencil's eraser end still in her grip and the clearance pouch hanging from the belt at her left hip and the folder she had not yet opened. Maren's face was the face of someone who had anticipated this visit for a period of years. The waiting had been longer than Lira's approach had taken and the waiting was not a professional reaction because professional composure was the thing Maren had learned at rank 71 and rank 71 meant Maren could keep secrets and keep them well. The secret Maren kept was that the Engine was running. The preamble lied. The preamble's lie was Article 4. Article 4 was the seal. The seal was what Lira could not reach. Reach was what Maren could give Lira if Maren chose to extend that reach. Maren was now choosing. Standing in front of a compliance officer with the suppression records in one's own hands was the kind of choice the position of rank 71 did not formally enumerate. The absence of formal enumeration was the same as permission. The weight of that permission settled in Maren's chest alongside the weight of every suppressed report she had catalogued. Forty-three documents now on the bench behind her, each one a quarterly measurement the Engineering Department had sealed and each one a measurement that had waited on the steel bench for someone to pick it up. Maren's grandmother Osta Vehn had waited nineteen years before telling Maren the truth. Maren had waited eleven years before doing anything other than filing the reports and watching them disappear. She was now waiting for Lira Vehn to say something, anything, that would tell Maren whether the waiting had been worth it. Lira stood with the folder unopened at her hip and the pencil in her grip and the question in the margin of her audit notebook that no one had yet answered. Maren held the load report in her left hand and offered it toward the open hatch as the first document that would leave the reservoir chamber on this shift. The Engine pulsed beneath them both with 4.2 megawatts of continuous thrust and the reservoir chamber conducted the rhythm into their feet and neither of them moved to interrupt it.

Chapter 5 — The Witnesses

The Compliance Office provided a list of three residents eligible as year-0 witnesses. Lira retrieved the list from the docket archive on the east wall of her desk. The docket archive held records from every renewal period since year 12, and the earliest entries in the archive listed the witnesses who had testified at the first Compact verification. Three names appeared on the list. One name was no longer living. One name was illegible from age and water damage. The third name was Osta Vehn, the grandmother of Maren Otta, rank 71, load engineer, resident since year 9. Osta had been nineteen years old at the Halt. She was the youngest witness Lira's verification form would accept.

Lira prepared the witness statement form before she went to find Osta. The form had three columns. The first column held the witness's name and rank. The second column held the witness's recollection of the Covenant signing ceremony. The third column held the witness's signature, confirming that their testimony matched the record on file. Lira filled the first column with Osta Vehn's name. She left columns two and three blank. The third column would remain blank until Osta signed, and Osta had not yet agreed to meet Lira in a room where the Compliance Officer could observe and the Compliance Supervisor could later ratify.

The north spoke corridor was shorter than the south spoke. Lira walked it quickly. The regulator hummed beneath her feet, constant and low, its vibration registering in her molars more than in her ears. She reached Osta's residential module at the junction of the east corridor and the north spoke. A digital panel on the door displayed the resident's compliance score and the ward assignment. The score was 72 and the ward was Reservoir. Osta lived in Reservoir ward. Lira had spoken with residents in Reservoir ward during previous renewal periods. Their concerns were about water quality, not covenant theology.

Lira knocked on the door. The panel's light turned green and the lock released with a soft click. Osta opened the door herself, which meant Osta was not incapacitated or absent. Osta was a woman of two hundred and thirty station-years according to the ward directory. She stood six-quarters of her height in the doorway. Her hair was white and pulled back. A bun sat at the nape of her neck. Lira recognized it from the records. Osta Vehn had been nineteen years old at the Halt.

"Maren sent a message that you are coming." Osta's voice was steady and quiet. She spoke the way Maren spoke about work, with precision and without ceremony. "Come inside."

The apartment was small for a resident of Osta's seniority. The ward directory assigned single modules to residents above seventy whose compliance score fell below 50. Osta's score was 72, which indicated a module upgrade at some point. The small size of the room suggested the upgrade had never come. A bed occupied the far wall. A desk occupied the near wall. Between the bed and the desk, a table held a glass of water and a penknife and a photo of three young adults standing in front of a cargo container. The photo was printed on paper that had yellowed at the edges. The three young adults in the photo were not identifiable by face to Lira, who had not met any of them before.

Osta gestured to the table with her left hand. Lira sat. Osta sat across from her. They were seated with the table between them and Lira's green compliance pencil in her left hand and the witness statement form on the table beside the glass of water.

"I am here to record testimony for the decade renewal," Lira said. "The Compact requires one witness per line item in Article 17. The line items require three witnesses for the Covenant signing event from year 0. I have the first name from the archive." She said the name as it appeared in the record. The name was Osta Vehn.

Osta nodded. She did not smile. The expression on her face was the kind of expression Lira learned to read at rank 94: it meant Osta had been expecting the visit and had accepted it as part of her civic obligation. The acceptance did not mean she was willing. Lira had learned this distinction from six years of compliance work. Willingness showed in the body. Acceptance showed in the face. Osta's body was still, and her face was still, and Lira could not tell which was which.

"Please state your recollection of the Covenant signing ceremony." Lira read from the verification form. The form's wording was standardized. The wording was designed to avoid leading the witness. Lira had memorized the form's preamble during her training at the Academy. She placed the form on the table so Osta could read over her shoulder. "The verification officer asks the witness to recall the events of the station-wide Covenant signing, year 0, as remembered by the witness and consistent with the station record."

Osta read the form aloud to herself. Her voice read more slowly than Lira had expected. The old woman's eyes moved from line to line on the page and her lips shaped each word before she spoke it. This was not the behavior of someone about to provide a convenient testimony. This was the behavior of someone who had spent many years deciding what the record required her to say and what the record did not permit her to say.

"Can you describe the day the Covenant was signed?" Lira asked. She had not yet asked about the drive, the Engine, or the preamble. Those topics were line items for later in the verification session, after the basic ceremony recollection was on record and the witness was committed by signature.

Osta was quiet for eleven seconds. Lira counted them by the regulator pulses that came through the floor, seven pulses per interval. Eleven seconds was one interval and four additional pulses. During that time Osta looked at the photo on the table. She looked at the three young adults in the photo. She looked back at Lira. Then she said, "The day I was married."

Lira wrote that on the form. "The witness recalled the Covenant signing day as the day she was married." The form had a specific field for the witness's personal context of the day they testified about. Lira filled it. She did not ask whether the marriage preceded or followed the signing ceremony. The record from year 0 listed Osta Vehn as a signatory and a witness, not as a bride. The record did not specify whether the two events coincided. Lira would not presume to conflate them until Osta addressed the difference.

"The Covenant signing ceremony took place in the Covenant Residential Block," Lira continued. "The ceremony required every adult resident to stand in the main hall and sign with a green pencil, representing the Compact's terms. The form asks whether you stood in the main hall. Did you stand in the main hall?"

"I stood where standing was permitted."

"Did the Covenant's preamble describe the station's drive as lost?"

The question landed in the silence between them. Osta did not answer immediately. Lira let the silence hold. Silence was part of the verification procedure. The form did not require witnesses to answer every question. The form required the officer to ask the question and the witness to respond if the witness chose to respond. Lira had been taught at the Academy that a witness who chose not to respond had made a decision, and the decision itself was evidence of something — but the form required her to record only the response, not the significance of non-response. So Lira wrote nothing and waited.

"The day the Covenant was signed," Osta said at last. "I confirmed that description when I signed."

Lira wrote the confirmation on the form. Osta's handwriting was small and precise, each character formed with deliberate control. Lira did not know whether Osta's hand trembled when she wrote. A tremble would have been visible to Lira but not to the Verification Officer unless the tremble affected the legibility of the characters on the form. Osta's characters were legible. Lira wrote "Witness confirmed preamble accuracy" and dated the entry with the current station time.

"The preamble described the drive as lost. The drive was not lost. The preamble was a false description of the station's condition." This sentence came from Osta, not from Lira, and it arrived without being prompted by the verification form's line items. The form had no line item that addressed whether the preamble was accurate. The form had line items for verifying the preamble's content, not its truth. Osta's sentence expanded the session beyond what the form had planned for.

Lira did not respond to the sentence with her own observations. A compliance officer's role was to record, not to agree or disagree. The record would show that Osta Vehn had made a statement contradicting the preamble. The record would also show that the contradiction Osta delivered marked a milestone no rank 94 officer had reached, obtaining a direct challenge to the preamble from a year-0 witness during this renewal period. Lira had been trained for this possibility and the training had not included a procedure for what to do when a contradictory statement changed the meaning of everything she had been hired to verify.

"I would like to continue the session," Lira said. "The form has three additional witnesses. The second witness is available. The third witness has declined the session."

Osta looked at the door. "The third witness is not declining," she said. "The third witness is not on the station."

Lira set down her pencil. The pencil touched the table and made a small sound against the surface. "I do not understand."

"The third witness on the archive list is Edik Hain," Osta said. "He signed the Compact at the founding ceremony. The station population index carries a death certificate dated year 2. The certificate is false. Edik is alive in a sealed founder reserve, and he is not coming to your verification session simply because an archive list names him."

Lira wrote Edik Hain's name on the form. She wrote the words "third witness — not available." She understood, now, why she had felt the name from the founding ledger was not a coincidence. The founding ledger had mentioned Edik Hain twice — once as a signatory and once as a survivor among the eleven. The archive listed three witnesses, but the archive did not know that one of them was a resident who had been declared dead in the census while being very much alive inside a residential module, filing compliance forms every quarter, contributing to the census that said the station had a population that did not include him in the records that described him as gone.

The session ended when the station day changed. Osta stood up from the table. She placed the penknife back on the table. She picked up the photo of the three young adults and folded it into her vest pocket. Lira stood up with her audit notebook and her green pencil and her form that now contained two confirmations and one contradiction and one refusal and one fact Osta had delivered without prompting from any line item.

Lira walked back through the hub corridor toward the Compliance Office. The regulator's hum was the same frequency it had been when she had left for Osta's module. The corridor light shifted along the baseboard as the station rotated through another fraction of its day. The three residents who did not appear on the census returns occupied station modules, and the archive had three names for witnesses, but only two witnesses would provide testimony, and one of those two witnesses had contradicted the preamble on the record, and the contradiction sat on Lira's verification form in the hand of a rank 94 compliance officer who had no line item in Article 17 for testimony that named the preamble's lie.

At her desk, Lira saved the form in the quarterly docket. She labeled it with the appropriate verification number and the witness names. Osta Vehn's testimony noted the contradiction. The second witness's testimony was still pending. Edik Hain's entry noted non-availability, not refusal, because Osta had clarified that Edik was not avoiding the officer but that his absence was a fact of his continued existence outside the records that described him as gone.

Lira closed the drawer beside the green pencil. She looked at the verification form one more time. One line item checked. Sixteen line items unchecked. Article 17 said pending. The station turned. The question she could not file settled into the space between the checked item and the unchecked items, a space the form's columns could not hold.

The walk through the north spoke required crossing two inspection checkpoints where automated optical sensors scanned the clearance badge attached to Lira's lapel. These checkpoints recorded her transit for security audits, though the security logs were archived in a separate database from the compliance records she reviewed at her desk. The separation of databases was a foundational principle of station administration. Compliance officers did not review security logs, and security officers did not verify compliance forms. This separation ensured that an officer could investigate discrepancies without worrying about triggering security protocols or alerting the supervisor to inquiries that had not yet been authorized by a formal line item. Lira understood the value of this separation. It provided a protected workspace where questions could exist as marginalia without immediately becoming operational incidents that required immediate intervention or disciplinary review from the command deck.

Reservoir ward differed from the residential rings in its acoustic profile and atmospheric composition. Because the ward housed the primary water purification and distribution pumps for all seven sectors, the air carried a faint trace of chlorine and ozone. The humidity was slightly higher here than in the administrative core, causing paper forms to curl gently at the corners if left exposed to the ambient air for more than a few minutes. Lira kept her verification forms inside a protective plastic sleeve while walking through Reservoir ward. The sleeve was issued by the compliance office during initial training, and its surface was scarred with faint scratches from years of use across multiple renewal periods. Each scratch represented an inspection tour or a witness interview conducted in a humid corridor or a maintenance shaft where moisture condensed on every exposed metal surface.

Osta Vehn's module door bore the standard brass plaque issued during the founding decade. The plaque displayed the resident's original registry number and the date of station arrival, stamped into the metal by a mechanical press that had since been decommissioned and melted down for bulkhead reinforcement during the mid-century structural refit. Very few residents still possessed their original brass plaques. Most had traded them in for modern digital display tags during the modernization program of year 120. Osta had kept her brass plaque mounted beside the digital panel, a physical artifact that bridged two eras of station life. Lira touched the edge of the brass plate with the tip of her index finger as she waited for the door lock to cycle. The metal was cool to the touch and carried the faint impress of numbers that had not been updated since the station's second decade.

The interior of Osta's apartment maintained a standard of order that surpassed even the compliance office standards. Every object on the desk was aligned parallel to the grain of the laminate surface. The penknife rested at an exact forty-five-degree angle to the edge of the table, its horn handle polished smooth by decades of handling. Lira noted these details in her peripheral awareness without recording them in her audit notebook. The notebook was reserved for procedural discrepancies and witness statements that pertained directly to the Covenant renewal articles. Personal habits and domestic arrangements fell outside the scope of Article 17, and Lira's training at the Academy had emphasized the importance of maintaining strict boundaries between administrative duties and personal observations. Yet, in a station where every resident's life was mediated by rules and forms, domestic order was itself a form of compliance, a way of signaling that the resident understood the rules and chose to enact them in miniature within their own living space.

The photograph on the table held Lira's attention during the moments when Osta paused between sentences. The three young adults in the picture wore the heavy canvas work uniforms of the early construction phase, before the residential rings had been pressurized and before the hub had been enclosed. In the background of the image, the structural framework of the Inertial Engine was visible as an unfinished lattice of girders against the blackness of space. That lattice was now enclosed within the southern extension, hidden behind heavy radiation shielding and secondary pressure bulkheads. Modern residents grew up knowing the engine only as a low vibration beneath the floorboards and a mysterious power designation on municipal utility summaries. They did not see the girders. They did not see the raw machinery that produced the centrifugal force under which they walked, slept, and worked. The photograph was a rare window into a time when the machinery was visible to everyone on the station, before the suppression took hold and before the preamble was rewritten to describe a drive that had been lost.

Lira considered the implications of Osta's unexpected statement about the preamble. For a year-0 witness to state on the record that the preamble's description was false meant that the foundational myth of the station was vulnerable to verification. A verification form was a legal instrument under station law. When a sworn witness declared that a canonical statement was false, the officer receiving the testimony was obligated by standard procedure to note the contradiction. But the standard procedure did not specify what happened when the contradiction undermined the validity of the Covenant itself. If the preamble was false, then the renewal process was renewing an untruth. If the renewal process was renewing an untruth, then the entire administrative apparatus of the station—from the compliance office where Lira worked to the supervisor's office behind the glass wall—was built on a foundation that had no basis in the physical reality of the running engine. Lira felt the weight of this realization settle behind her ribs, mirroring the physical vibration of the regulator beneath her boots.

Chapter 6 — The Waiver

The founding ledger rested on Lira's desk at four in the afternoon. She had returned it to her desk after the session with Osta Vehn. The verification form remained open with one check mark and sixteen line items in the margin marked pending. The grey cloth binding caught the light from the narrow window that looked at nothing in the north spoke. The mineral oil smell rose from the paper the way it always did after the book stayed closed for more than an hour.

Lira opened the ledger to the first page. Eleven signatories appeared at the top of the sheet. Edik Hain appeared bold on the left margin where Lira remembered him. The first page mentioned the year zero and the date the Covenant came into force. The first page confirmed what Lira already knew. The annex at the back held what Lira did not yet know.

The annex had not changed between this opening and the one three days before. A folded sheet of thin paper remained at the back of the volume. The mineral oil smell clung to it as the binding held it shut. Lira unfolded the annex carefully. She moved the way one moves when unfolding a document that had not seen air for one hundred and ninety-two years. The paper smelled of old preservation chemicals. The edges were brittle but intact. The ink was dark iron gall in the same hand that had written the founding ledger's first description of the station.

Lira read the annex once. The text described the drive. Specifically, the text described the drive as running continuously from the moment the station reached its initial orbit through the present quarter. The annex listed the output measurements for each quarter since year zero. The annex contained the document that allowed the station's residents to accept the Covenant without acknowledging the continuous operation of the machinery. The waiver described the drive as unnecessary for ongoing survival. It acknowledged that the station would have reached its current orbit regardless of whether the Engine remained active, because the original launch velocity would have proved sufficient for a stable coasting trajectory. This was true. The waiver was true. The preamble, which described the drive as lost during the transit from the departure system, was false. The station had never lost its drive. The drive had operated for one hundred and ninety-two years. The Engine's continuous thrust had kept the station's orbit from decaying into the primary's dense upper atmosphere.

Lira closed the ledger. She set it down on the desk surface. She removed the green pencil from the groove in her left hand and held it for a moment without writing. She understood now what she had failed to understand on her first evening in this assignment. She had opened the founding ledger to the annex page during her initial inspection. She had read the document in a cold apartment with cold porridge on her plate. She did not yet know that her mother had signed the Compact under false pretenses. Her mother had believed the preamble, which the Academy taught every resident to trust as absolute record. The word truth applied to the preamble carried the same weight as line items verified as correct during quarterly audits. Lira had applied that weight for twenty-one days without questioning what the preamble described.

The verification form contained no line item for waiver discovery. The verification form contained line items for checking each of the seventeen actively enforced articles of Article 17. The waiver was not a line item in the compliance manual. The waiver constituted the founding contradiction that the line items were designed to obscure. Lira had checked line items for twenty-one days. She had found seven residents absent from the municipal census. She had found a preamble that described loss when the station experienced no loss. She had found a founding document that described a mechanism the preamble claimed did not exist. Every finding pointed to the same operational reality. Every finding needed to be recorded in an official registry. The verification form served as the wrong record for these particular findings. Lira held rank 94. Rank 94 permitted the verification of line items and nothing else.

She reached for the private audit notebook. She wrote a date at the top of the blank page. Beneath the date, she wrote the words that described what she had confirmed. Lira did not write the waiver's language verbatim because the annex remained sealed by Article 4 of the administrative code. Reproducing the waiver's text inside the audit notebook would create an unauthorized record that the Compliance Office could trace directly to rank 94. Rank 94 held no authorization to reproduce sealed founding documents. Lira wrote the conclusion instead. She wrote that the annex confirmed the drive was never lost. She wrote that the preamble lied. She wrote that the Compact was built upon a false premise. She wrote that the waiver existed in the annex of the founding ledger. She wrote her name and her rank and her officer designation to certify that she had read the annex and confirmed the contradiction between the preamble and the hidden text. She noted that no line item existed in Article 17 to accommodate this confirmation.

She closed the notebook. She slid it beneath the green pencil in the desk drawer. She placed the pencil back in the groove of her left hand. The pencil remained where she left it. The notebook remained beneath the pencil in the dark. The desk remained clear of everything except the verification form, the founding ledger, and the heavy folder from the engineering archive that contained load measurements from year 60 onward. Lira had not opened that folder yet. She intended to open it after she completed the verification form and submitted the packet to her supervisor. She would not open the folder before submission because opening unrequested engineering data fell outside the scope of Article 17. Lira did not intend to expand her authority beyond the strict boundaries provided by her rank.

The station day was ending. Lira stood up from the desk at fourteen forty-two station time. This was the exact time she had stood up from the same desk during her first station day of the current renewal period, when she had opened the verification form and remained ignorant of the contradiction between the preamble and the annex. Now she knew the contradiction. The knowledge would not alter the structure of the verification form. The form operated as a tool for compliance officers at rank 94. Rank 94 lacked any designated field for recording confirmed contradictions between foundational documents and operational records.

Lira placed the founding ledger back on the archive shelf where she had retrieved it earlier in the week. The ledger's grey cloth binding rested on the shelf at the third tier of the east wall where records from before the current ward directories lived. The shelf held the ledger alongside two older folios bound in different colored fabrics. Lira did not shelve the ledger in the precise position it had occupied before her retrieval. The shelf arrangement followed rigid cataloging conventions rather than personal ordering systems. Personal ordering systems had no place in the docket archive where records survived the residents who created them.

She walked south through the hub corridor with her hands empty. The clearance pouch at her belt held the green pencil and nothing else. The restricted folder remained in the engineering archive where the clerk behind the counter had returned it to its shelf after stamping the quarterly receipt. The folder would wait there until someone with rank 94 requested it again or until the suppression classification underwent review by a Compliance Supervisor with authority to override permanent engineering labels. Lira had not requested the supervisor's override. She had filed no special inquiry. She had verified one line item and left sixteen items unchecked on her daily schedule.

The station day ended with Lira at her desk. The verification form remained open at the pending mark. The compliance pencil remained green. The station turned above her in the third ring. The regulator hummed the same steady frequency it had hummed the previous station day and the day before that and the day before that for ninety-two years since the first renewal period. Her great-grandmother had stood in the north spoke with a green pencil in her hand and signed her name to the register for the first time without knowing that the station's engine produced continuous thrust that the preamble described as absent. The preamble's false claim survived in every archive on the station because every archive was maintained by the Compliance Office. The

Compliance Office verified line items. Line items did not verify the preamble against the annex because line items did not cross-reference foundational documents. The contradiction sat sealed in the annex while the station turned beneath the third ring. Lira Vehn, rank 94, stood in the hub corridor with empty hands and a question she could not file. She carried a green pencil in her left hand that had not written a single mark in three station days.

Lira stopped walking at the south corridor junction. The station's rotation carried her eastward toward the residential sections where compliant residents slept in assigned modules and less compliant residents occupied identical physical space with reduced privacy. The compliance score on the wall panel outside each module color-coded the residents into the wards provided by the station administration. The east ward constituted the lowest residential tier. Compliance scores falling below seventy consigned residents to the east ward after a quarter of below-average performance in work assignments. Lira had never inspected the interior of the east ward. Her score of ninety-four kept her safely within the residential placement offered to high-performing personnel. That placement consisted of the north spoke corridor with its narrow walls, its constant regulator hum, its window looking at empty space, and the small square of floor that her rank entitled her to call a home.

She had worked with residents from the east ward during quarterly audit cycles. Those residents had looked at Lira's score of ninety-four and said nothing, or said thank you in the flat tone reserved for compliance officers who held administrative power over water rations. The compact's water allocation system divided supply by compliance score. East ward residents received a fraction of the supply allocated to residents scoring above seventy. The seventy threshold marked the official maintenance floor. Lira's score of ninety-four sat far above that floor. The distance between Lira's score and the east ward scores represented the distance the Covenant placed between those who followed administrative procedures successfully and those who struggled against them. That distance matched the separation between compliance and non-compliance. It matched the separation between a resident who understood the documentation and a resident who only felt its weight.

Lira continued walking south toward the main residential bank. The south spoke corridor widened as she left the administrative offices behind. The floor plates here were thicker to deaden the vibration of the heavy machinery mounted beneath the hull. Engineering sections dominated this corridor. Load-bearing instruments sat behind reinforced doors requiring rank 71 clearance for entry. The door Lira passed now bore a permanent classification label applied during the initial construction phase. That label had never undergone review by the Compliance Office or any other department possessing authority to challenge engineering designations. The label stated restricted in faded red lettering. The restricted classification meant that no rank 94 officer could pass the threshold without an explicit supervisor override. Lira had requested no override. She possessed no reason to enter the heavy machinery bays. Her duty lay with paper records, ink signatures, and the seventeen articles of the compliance manual.

The restricted folder containing load measurements from year 60 onward remained in the restricted archive where the clerk had returned it. The shelf holding that folder bore no date indicating when the administrative suppression began. Lira did not know when the suppression started, and that ignorance formed part of the system she served. The suppression itself was the exact phenomenon compliance officers verified by checking off authorized line items. The line items omitted any mention of suppressed load records. The suppression remained entirely invisible to a compliance officer operating at rank 94 who inspected files every ten years and found every required signature in its proper box. The station turned beneath her boots. The regulator hummed through the structural ribs of the deck plates. The corridor light shifted position along the baseboard as the rotation cycle advanced toward the night shift.

Lira walked south toward her module. Her workday would begin tomorrow at six in the morning station time. The green pencil in her left hand would be ready to check off line items against Article 17. Article 17 would offer no guidance on how to record an undisclosed engine running beneath the floor. The waiver would remain sealed inside the annex behind the barrier of Article 4. Lira would remain at rank 94, three station days into the forty-second Covenant renewal period, with nothing official filed, nothing officially changed, and the foundational question waiting for an answer that the manual could not provide.

The silence in her module felt heavier than it had on previous evenings. She placed the audit notebook in the shallow desk drawer beneath her spare wool blanket. She set the green pencil beside it. The room contained a narrow sleep cot, a wall-mounted water dispenser, and a small terminal screen that displayed the current station rotation index and the weather simulation parameters for the agricultural ring three levels above. She did not activate the terminal. She sat on the edge of the cot and watched the pale light from the corridor stripe the metal floor.

The waiver text repeated itself in her memory without missing a stroke. Eleven founders had signed those sentences in year 0. None of them had hidden the record as a test for a worthy future reader. They had authorized an emergency concealment without naming an expiration, then left later officers to convert delay into procedure. Recha Tallis's year-30 transfer order sat behind the waiver and moved it under Article 4. The preamble served as an administrative filter. Only residents who followed the source references past that filter could find the physical evidence. Finding it did not change the Engine's operation. It changed who could be held responsible for the public account of that operation.

Lira lay back on the cot and stared at the curved ceiling panel above her head. Every bolt in that panel had been secured by workers who lived and died before the first renewal period. Their names appeared in the secondary ledgers stored in the basement archives. None of those workers had signed the waiver. They had built the frame, welded the structural ribs, and calibrated the life support loops while living under the assumption that the station traveled on a permanent coasting trajectory. The realization that the station required constant mechanical intervention to avoid atmospheric entry did not frighten her. It merely clarified the nature of the structure she inhabited. The station was not a self-sustaining world drifting through the void. It was a closed loop powered by an engine that everyone agreed did not exist.

The air in the north spoke tasted slightly stale as the filtration system recycled it through the carbon scrubbers. Lira closed her eyes and listened to the faint rhythm of the hull expansion joints shifting under thermal stress from the primary star. Each shift sounded like a heartbeat slowed down to match the immense scale of the hull. She had spent twenty-one days treating the verification form as the ultimate authority on station life. She had believed that checking every box ensured the survival of the community. Now she understood that the check marks were rituals designed to maintain an equilibrium of ignorance. To check a box was to agree that the recorded version of reality was the only version that mattered.

Morning arrived without ceremony. The corridor lights shifted from dim amber to operational white at exactly six hundred hours. Lira rose from the cot without delay. She washed her face at the cold water basin and fastened her compliance officer badge to the left lapel of her grey tunic. She retrieved the green pencil from the drawer and slid it into the sleeve pocket. The audit notebook remained hidden beneath the wool blanket. She did not touch it before leaving the module.

She walked north toward the central administrative office where the daily verification queue awaited her signature. The corridor was crowded with maintenance workers and junior clerks hurrying toward their morning shift assignments. No one spoke to her. Her rank 94 badge commanded sufficient respect to clear a path through the morning commuter traffic without verbal exchange. She reached her desk by six fifteen. The verification form lay exactly where she had left it the previous afternoon, open to the sixteenth line item with its blank box waiting for ink.

A census clerk walked past her desk ten minutes later carrying a stack of revised reports from the agricultural wards. He stopped for a moment and tapped his knuckles against the edge of Lira's desk. The sound was sharp against the quiet of the office.

"Sixteen items remaining," the clerk said. His voice carried the flat cadence of an administrator who measured progress entirely by completed forms. "The supervisor wants the audit packet on his terminal before the midday rotation shift."

"The audit packet requires complete verification," Lira said. She kept her voice neutral, matching the tone she had used for three years in the compliance office. "Some items require additional examination before I can apply the check mark."

"Verification is a mechanical process, Lira," the clerk said, shifting his weight from one boot to the other. "You check the box if the record exists. You do not investigate the history behind the record. The history belongs to the archives. Our responsibility is the current quarter."

"The current quarter depends on the record," Lira replied.

The clerk nodded once, accepting the statement as a routine compliance maxim, and continued down the corridor toward the supervisor's office. He did not know about the annex. He had likely never opened the founding ledger beyond the first signatory page during his entire career in the department. For him, the ledger remained an artifact of administrative origin rather than a source of operational data. That ignorance protected him. It allowed him to complete his daily assignments without experiencing the sudden weight that now settled behind Lira's ribs whenever she looked at the green pencil in her hand.

Lira opened her manual to Article 17. She read the requirements for verifying population counts in the north spoke residential modules. The task was straightforward. She could complete the sixteen remaining line items before noon if she chose to ignore everything else. She could apply her green check marks, seal the packet, deliver it to the supervisor, and spend the evening reviewing agricultural output figures in her module. That was the path prescribed for officers who wished to advance to rank 93. Advancement required speed, adherence to procedural schedules, and an absolute refusal to look behind administrative preambles.

She picked up the green pencil. She held the tip above the sixteenth box on the form. The square was small, measuring precisely four millimeters on each side. A single stroke of green graphite would satisfy the administrative requirement. The supervisor would accept the form. The quarter would close without incident. The station would continue its orbital correction routines beneath the floor plates, driven by an Engine that the official history ignored.

Lira lowered the pencil without touching the paper. She laid it back in the groove beside her left hand. She stood up from the desk, left the verification form open on the blotter, and walked out of the administrative office toward the archive stairs. She had sixteen items left to verify, but none of them answered the question that mattered. She needed to know who else had read the annex before her, and whether any previous compliance officer at rank 94 had left a record in the deeper vaults where the station kept the names of those who asked too many questions about the origin of the light.

Interlude I — The Halt

The Halt began with two instruments disagreeing.

The transfer-drive board showed no commanded acceleration. The reservoir board showed 4.2 megawatts entering the new correction system as if nothing had happened. One machine had stopped. The other had begun the work it would perform for the next two centuries.

The station had launched into a 412-kilometer commissioning orbit. Its transfer drive was meant to lift the habitat toward a higher permanent path after final assembly. The Inertial Engine occupied a different part of the architecture. It recovered heat from the sodium reservoir, stored that energy, and returned accumulated orbital losses through timed prograde corrections. Engineers called its cycle closed because the thermal working mass circulated inside the station. The correction pulse still exchanged momentum with an irradiator field the founders did not fully understand.

On day fourteen, the transfer drive failed its ignition sequence. The station retained air, water, and electrical power, but it could no longer perform the planned climb. Atmospheric drag would eventually pull it down unless the correction system worked far beyond its trial period.

The Engine worked.

For the first correction window it delivered 29.95 megawatts of orbital work over 14.02 percent of the local cycle. Its average bus-bar draw came to 4.2 megawatts. The station recovered the loss accumulated during coast. The next cycle repeated within tolerance.

The founders did not know whether repetition meant safety. Their models offered ranges measured in years and centuries. Food confidence reached six weeks. Repair inventories had not yet been reconciled. Families gathered outside the hub because official announcements had used the phrase stable temporary orbit, and everyone understood temporary carried no date.

Eleven founders met in the unpainted archive room.

Edik Hain argued for releasing the full telemetry with the uncertainty range. Osta Vehn wanted the correction system named but not described until engineers confirmed it could survive a reservoir trip. Pello Otta brought the food ledger. Recha Tallis brought the draft public notice. The Soren-household load engineer brought the technical schedule and refused to call the pulse continuous thrust.

The public notice said the transfer drive had been lost. That sentence held.

Its next sentence said the station had entered a coasting trajectory sustained by auxiliary reserves. That sentence did not.

The Engine had replaced lost orbital energy every cycle since the failure. It could not perform the intended transfer. It could keep the habitat from falling, provided its trial system survived and the reservoir remained hot.

The founders feared that naming an unproven machine as the station's sole orbital safeguard would trigger competing control claims before any stable administration existed. Engineering wanted operational authority. Compliance wanted a common statement. Ward delegates wanted access to the numbers. No one wanted to tell four thousand residents that their continued altitude depended on a prototype with no repair precedent.

The emergency waiver authorized one renewal period of restricted reporting. It preserved raw telemetry in Engineering, substituted zero in the public propulsion field, and required the founding record to retain both values. Eleven people signed.

No sentence specified when one renewal period ended.

Edik pointed to the omission.

Recha said the first Covenant would define expiration once the population registry and food plan stabilized. Osta asked that the promise be entered in minutes. The clerk recorded the request but not a deadline.

The waiver traveled with the power budget into temporary controlled custody. The public preamble announced the lost drive. Residents learned that the station coasted on finite reserves and discipline. The story gave rationing a shape everyone could understand.

The Engine continued its correction cycle beneath that story.

During the first year, operators inspected every pulse. During the second, they sampled every sixth pulse. By the fifth, the correction controller no longer required a human key unless output moved outside tolerance. The raw value remained visible in Engineering. The public field remained zero.

Edik filed three objections. The third said that an emergency choice had become routine without a vote. Recha answered that public stability still qualified as emergency necessity. Edik accepted removal from founder authority rather than sign the first renewal statement. A personnel action hid him in Founder Reserve. The public registry later called him dead.

The first Covenant contained Article 4, Continuity of Founding Intent. It protected the emergency instruments while residents built governance around them. The article did not name the Engine. It did not name the contradiction. It gave later officers a legal shelf on which to place both.

At the founding assembly, residents heard the transfer drive had failed. That remained true. They heard the station coasted without active correction. That remained false. The difference fitted inside one technical appendix few residents knew existed.

The assembly signed.

At the next correction window, the Engine accepted reservoir energy, opened the irradiator, returned the missing impulse, and closed the field. Its controller recorded the pulse in the raw archive.

The public report recorded zero.

Before first shift ended, the founders created two audit routes. Engineering kept unrounded controller ticks. Compliance kept the authorized public form. A quarterly reconciliation would compare checksums without exposing the raw quantity.

Edik objected that reconciliation without disclosure could prove the concealment remained internally consistent while telling residents nothing. Recha answered that internal consistency at least prevented officers from inventing numbers later.

Both were right.

The dual route preserved enough evidence for Lira to reconstruct the truth. It also let each renewal certify that the zero had been produced correctly.

During the first month, the public field operator asked whether zero meant no correction activity or no authorized public value. Recha revised the form instructions to say authorized reportable output. The phrase removed a direct falsehood from the instruction while leaving the displayed value unchanged.

Pello Otta signed the food ledger beside the waiver custody receipt. Osta Vehn signed the meeting minutes and added a note that emergency necessity should be reviewed after harvest stabilization. The note entered an appendix that no later renewal summary quoted.

The Soren-household load engineer attached the first dimensional schedule. It separated pulse work, cycle average, duty fraction, and reservoir draw. Later reports compressed all four into Engine output, which made the 4.2-megawatt figure easier to misdescribe as constant thrust.

At day 227, the first long-duration review concluded the correction system had held orbit inside the modeled band. Food confidence extended to eighteen months. Water and air systems passed reserve tests.

The founders could have ended concealment then.

Instead they scheduled review for the first decade renewal. By that time, the public preamble had become the explanation for ration discipline, rank authority, and the Compact itself. Correcting one technical claim would require residents to reconsider why emergency power had lasted after emergency conditions changed.

The correction was deferred.

A dissent note from Edik remained attached to the schedule:

> Emergency language without an expiration will become ordinary language. If review waits for unanimous safety, review will never occur.

The founders preserved his note and ignored its remedy. That distinction would matter later. The archive contained warning, choice, and consequence rather than a single unanimous founding mind.

A clerk marked the waiver ACTIVE PENDING RENEWAL. No expiration date entered the field because the form did not require one.

The Engine completed another pulse while the clerk wrote.

Both records began on the same day.

Part II

Chapter 7 — The Engine Sleeps

[The primary storage vessel contained sixty cubic meters of molten fluoride salt maintained at eight hundred forty Kelvin. Thermocouples embedded throughout the vessel walls registered thermal stability within three-tenths of a Kelvin. This precise thermal regulation persisted without interruption since the central bus-bar received its initial electrical charge in year zero. Molten fluid circulated continuously through twelve-meter conduits forged from a specialized titanium-zirconium-molybdenum alloy. These conduits spanned the physical distance between the primary reservoir and the thermal exchange manifold positioned at the southern extremity of the station's axial core. Fluid velocity averaged four point two meters per second under nominal pumping pressure delivered by the central hydraulic assembly. Boundary layer friction generated a predictable pressure drop across each linear meter of conduit length. Particulate filtration units trapped suspended metallic oxides produced by trace moisture lingering within the salt mixture. Maintenance logs recorded zero filter replacements during the past twelve operational cycles. The working fluid completed its internal circuit every fourteen seconds.]

[Radiator panels projected outward from the station's aft bulkhead across a total span of seven meters. Thermal energy transferred from the circulating salt into the radiating matrix through direct conduction across bonded interface plates. Surface temperatures across the emitter arrays dropped by four hundred twenty Kelvin prior to fluid return. Infrared emissions distributed the thermal load uniformly into the cold sink of deep space. Observers stationed within the southern habitation spoke perceived the glowing panels through reinforced quartz viewports. Residents identified the crimson thermal glow as auxiliary running lights or emergency drainage indicators. Official documents published by the administrative directorate described those glow patterns as evidence of dying reserve batteries. Physical reality diverged completely from administrative descriptions. The thermal emission represented continuous thermodynamic work performed by the closed-loop system. Waste heat rejection maintained the working fluid within optimal viscosity parameters.]

[Thrust vectors aligned perfectly with the station's prograde trajectory at four point two megawatts of continuous kinetic delivery. Orbital decay exerted a constant inward pull driven by residual atmospheric drag encountered at four hundred twelve kilometers altitude. Individual oxygen atoms struck exterior forward surfaces with kinetic energy proportional to orbital velocity. Each atomic collision transferred a minute retarding impulse against the station's forward momentum. Counteracting impulses issued from the primary exhaust nozzles in uninterrupted sequence. Telemetry sensors monitored positional parameters relative to the primary body. Semi-major axis measurements remained fixed at six thousand seven hundred seventy-one kilometers. Orbital eccentricity measured zero point zero zero zero nine seven across all reporting windows. Periodicity calculations yielded ninety-two point four minutes per complete revolution. Fifteen point eight orbits accumulated during every standard station day.]

[Primary eclipses recurred precisely fifteen point eight times daily, each occlusion lasting thirty-five standard minutes. Solar flux dropped to zero during each orbital shadow phase. Habitability systems drew auxiliary electrical current from internal storage grids throughout the eclipse interval. Thermal loads generated by human occupancy transferred directly into the molten salt reservoir. Specific heat capacity parameters for the fluoride mixture reached one point five kilojoules per kilogram per Kelvin. Total thermal mass exceeded eighteen hundred metric tons of active fluid. Temperature fluctuations within the primary containment vessel remained below zero point zero two percent during maximum eclipse cooling. Design specifications drafted prior to launch predicted this exact thermal damping capability. Operational logs confirmed that thermal equilibrium held steady without manual intervention or control system overrides.]

[Eleven individuals affixed their signatures to the foundational Covenant upon reaching orbital insertion. Three engineers formulated the thermodynamic parameters governing fluid circulation. Four administrators drafted the one hundred forty-four governance articles. Four military officers supervised launch sequences and structural deployment. Green pencils distributed during the signing ceremony remained inside the secure archival supply cache. Graphite cores showed uneven wear patterns reflecting variable writing pressure applied by each signatory. Edik Hain held the distinction of signing with greater downward force than his peers. Surviving records placed Edik Hain within residential block four at twenty-nine station-years of age. Nine co-signatories perished over the preceding decades through various operational hazards. Death records attributed those fatalities to atmospheric depressurization, decompression sickness, and radiation exposure. Osta Vehn remained in the active census as a living founding witness.]

[Quarterly load reports submitted by the engineering directorate underwent systematic reclassification starting in year sixty. Compliance Supervisor Recha Tallis instituted the protocol restricting performance disclosures to internal archives. Successive departmental supervisors maintained the restricted classification across one hundred thirty-two quarters. Rhen Tallis occupied the third-shift audit desk two years following the investigation conducted by Lira Vehn in the restricted annex. Institutional preservation required maintaining the narrative of lost drive capability. Covenant authority depended entirely upon residents believing the station possessed no means of self-propulsion. Recha's office formalized the reporting restriction inherited from the founding emergency. Historical continuity linked the founding generation directly to the administrative secrecy governing current operations.]

[Conservation laws governed every operational threshold within the closed-loop architecture. Physical constraints operated independently of administrative decrees or preamble declarations. The Covenant explicitly detailed power distribution allocations without identifying the molten salt loop as the energy source. Residents renewed their adherence to the Compact every ten decadal cycles. Administrative summaries reiterated claims regarding structural drift and inert propulsion systems. Actual operating metrics documented uninterrupted mechanical output across one hundred ninety-two years. Potential failure modes included freezing point crystallization, conduit wall thinning, and hydraulic piston seizure. None of those terminal conditions materialized during operational history. The machinery continued functioning because thermodynamic equations permitted no alternative outcome.]

[Hydraulic cylinders housed within the central manifold operated with a four-meter stroke length. Piston displacement volumes matched the volumetric thermal expansion rate of the fluoride salt solution. Mechanical linkages transmitted kinetic force from the piston crown to the high-pressure injection valves. Valve timing adhered strictly to a cyclical schedule synchronized with the fourteen-second fluid circuit. Pressure transducers recorded peak manifold pressures reaching twelve point four megapascals during compression strokes. Minimum pressure levels dropped to eight point one megapascals during scavenging phases. Valve seating surfaces experienced micro-abrasion from suspended particulate matter. Material hardness ratings for the valve faces exceeded Rockwell C sixty-five. Wear progression curves indicated structural margins sufficient for another three centuries of continuous operation.]

[Electrical generators coupled directly to the turbine shafts converted kinetic rotational energy into alternating current. Output frequency maintained synchronization with the station's primary distribution grid at precisely four hundred hertz. Voltage regulation circuits suppressed harmonic distortion below zero point zero one percent. Copper windings cooled by direct thermal contact with returning salt streams operated at stable operating temperatures. Insulation integrity tests performed during quarterly diagnostics verified zero dielectric breakdown across stator boundaries. Current delivery levels responded dynamically to shifting load demands from residential climate control units. Power factor measurements remained steady near unity across all operational sectors. Harmonic resonance suppression filters neutralized electrical noise generated by high-speed turbine blades.]

[Structural integrity monitoring arrays scanned the primary pressure vessel continuously. Acoustic emission sensors detected micro-fracturing within the containment walls before crack propagation occurred. Ultrasonic transducers measured wall thickness variations across critical stress junctures. Minimum wall thickness measured twenty-eight point four millimeters against a design minimum of twenty-two millimeters. Corrosion rates on internal containment surfaces averaged zero point zero zero three millimeters per decade. Cathodic protection currents suppressed electrochemical degradation throughout the metal matrix. Neutron flux from the surrounding radiation environment induced minor lattice dislocations within structural alloys. Annealing cycles executed during scheduled maintenance periods restored crystal lattice geometry.]

[External sensor packages mounted along the station's forward hull recorded particle density measurements in real time. Neutral gas concentrations in the exosphere measured four point five million particles per cubic centimeter. Ionized oxygen fractions constituted sixty-two percent of ambient atmospheric composition. Kinetic impact forces transmitted through solar panel attachment brackets generated periodic mechanical oscillations. Damping mechanisms installed within the structural struts absorbed oscillatory energy effectively. Amplitude decay rates matched theoretical models established prior to initial orbital insertion. Structural fatigue indices remained well within safety margins defined by the station engineers.]

[Historical ledgers preserved within the digital archives documented every engineering decision executed since year zero. File corruption incidents affected less than zero point one percent of stored records. Redundant storage arrays maintained triple-layer parity checks across all data blocks. Retrieval protocols operated with zero latency during routine diagnostic queries. Access logs indicated frequent inquiries originating from unverified administrative terminals. System security subroutines logged unauthorized access attempts without interrupting primary operational routines. Encryption keys protecting suppressed performance metrics remained unbroken by external decryption attempts.]

[Mathematical models governing fluid thermodynamics accounted for relativistic corrections in gravitational potential fields. Tidal forces exerted by the primary body induced minor volumetric oscillations within the main reservoir. Deformation amplitudes measured less than two millimeters across the longitudinal axis. Resonance frequencies generated by tidal flexing fell far outside structural harmonic boundaries. Dissipation mechanisms converted tidal kinetic energy into low-grade thermal input. Total thermal contribution from tidal flexing amounted to zero point four percent of total system input.]

[Internal illumination within the engineering compartment derived from solid-state LED arrays operating at five thousand Kelvin color temperature. Luminescent panels distributed light uniformly across all machinery decks. Automated maintenance drones traversed overhead rail systems inspecting sensor housings and lubrication lines. Diagnostic routines executed by onboard processors verified software integrity every six hundred seconds. Memory error correction protocols resolved transient bit flips instantly. Central processing units executed millions of floating-point calculations per second to optimize fluid flow parameters.]

[Thermal insulation blankets wrapping the high-temperature conduits consisted of multi-layered reflective foils interspersed with ceramic batting. Heat loss across the insulation boundary remained below zero point one percent of total thermal throughput. Outer surface temperatures of the insulation jackets matched ambient compartment levels within two Kelvin. Condensation accumulation on exterior conduit surfaces stayed completely absent due to dry nitrogen purging within the raceways. Pressure seals enclosing the raceway chambers maintained positive nitrogen pressure relative to adjacent habitation modules.]

[Acoustic signatures produced by the operating machinery registered as a low-frequency hum throughout the axial core. Sound pressure levels within the immediate machinery enclosure measured eighty-four decibels. Frequency analysis revealed dominant spectral peaks corresponding to piston stroke frequency and turbine blade passage rates. Structural vibration isolation mounts prevented mechanical noise transmission into residential living quarters. Residents experienced absolute silence regarding mechanical operations despite residing directly adjacent to the primary drive assembly.]

[Chemical composition analyses of the fluoride salt mixture occurred automatically via inline spectrometry loops. Lithium fluoride and beryllium fluoride molar ratios remained locked within optimal solvency parameters. Oxide contamination levels stayed below five parts per million. Metallic impurity concentrations resulting from minor conduit erosion remained negligible. Chemical stability indices confirmed that salt degradation proceeded at rates significantly below theoretical projections. Replenishment reservoirs located within the service bay held sufficient raw stock for two millennia of makeup additions.]

[Data transmission channels linking the engineering core to the central administrative archives operated over fiber-optic pathways. Signal attenuation rates remained below zero point two decibels per kilometer. Bandwidth allocation prioritized telemetry streams carrying critical pressure and temperature metrics. Redundant transmission lines routed around structural bulkheads ensured uninterrupted data flow during hull maintenance operations. Encryption algorithms securing telemetry transmissions utilized two-thousand-four-bit quantum-resistant key distribution.]

[Atmospheric scrubbers conditioning the engineering compartment air maintained carbon dioxide concentrations below zero point zero five percent. Oxygen replenishment systems injected pure gas derived from electrolytic water splitting units. Particulate filters in the ventilation ducts captured dust generated by machinery wear. Air circulation fans operated at low rotational speeds to minimize acoustic disturbance. Temperature control units maintained compartment air at twenty-one Kelvin above standard room temperature to prevent thermal shock to sensitive electronic controllers.]

[Emergency shutdown protocols remained fully operational within the control firmware architecture. Manual override switches required dual-key authentication from authorized engineering personnel. Power redirection circuits stood ready to dump excess thermal energy into secondary radiator arrays during extreme load spikes. Fault isolation valves could segment the fluid circuit within three seconds of anomaly detection. Automated fire suppression systems utilizing inert gas flooding covered all electrical distribution cabinets. Diagnostic test routines executed weekly verified the readiness of every emergency subsystem.]

[Historical continuity between the founding engineers and current operational personnel depended on documented knowledge transfer protocols. Apprentices assigned to the engineering department studied thermodynamic principles derived directly from foundational texts. Practical training exercises involved manual valve calibration and sensor recalibration procedures. Supervisory personnel enforced strict adherence to calibration standards across all three operational shifts. Knowledge retention metrics evaluated during quarterly evaluations demonstrated consistent competency among engineering staff.]

[Institutional memory regarding the true nature of the station's propulsion system remained restricted to authorized engineering personnel. Administrative leadership maintained the official narrative of station drift to preserve social stability among the general population. The divergence between administrative fiction and physical reality constituted the foundational paradox of station governance. The Engine operated continuously regardless of administrative pronouncements or popular beliefs. Thermodynamic laws acknowledged no political boundaries and recognized no social consensus.]

[Mechanical efficiency metrics recorded across the operational lifespan exhibited remarkable consistency. Carnot cycle efficiency calculations yielded forty-eight point six percent under nominal operating conditions. Frictional losses within moving parts accounted for less than two percent of total mechanical work output. Energy conversion ratios remained stable within one-tenth of one percent over the entire operational epoch. Maintenance interventions required human assistance in less than zero point one percent of recorded anomaly events.]

[Future operational projections based on current wear rates indicated multi-century viability before major component overhaul became necessary. Material fatigue limits for high-stress titanium components approached design thresholds at extremely gradual rates. Lubrication fluid degradation remained imperceptible due to magnetic particle traps embedded within bearing housings. Thermal cycling stresses induced minimal grain growth within alloy matrices.]

[Isotopic concentration levels within the beryllium structural components experienced negligible transmutation across the 192-year operating window. Secondary auxiliary cooling loops maintained standby readiness with automated valve actuation tested every seventy-two hours. Acoustic damping mounts stationed along the primary turbine housing eliminated resonant frequencies that could propagate into residential sectors. Redundant telemetry archiving protocols duplicated all operational logs across three separate physical servers located in isolated compartments. Historical log entries compiled during years one through fifty-nine documented steady operational parameters prior to administrative reclassification. Physical dimensions of the southern habitation spoke maintained thermal equilibrium despite radiative exposure from the adjacent exhaust manifold. Hydraulic fluid viscosity indices remained stable across wide temperature gradients encountered during eclipse transitions. Stress-strain curves measured on retrieved conduit samples confirmed absence of micro-void formation under constant load. Electrical bus distribution networks incorporated surge suppression circuits capable of absorbing transient electromagnetic pulses. Orbital decay compensation algorithms integrated real-time drag measurements to optimize thrust delivery schedules.]

[Secondary heat exchangers deployed along the station periphery provided auxiliary thermal rejection capacity during peak operational periods. Fluid flow diversion valves operated with sub-second response times during emergency routing sequences. Radiation shielding thickness surrounding the reactor core attenuated primary particle streams below biological hazard thresholds. Coolant purity monitoring systems executed spectrophotometric analysis on hourly intervals without operator intervention. Mechanical fastener torque values checked during scheduled maintenance windows showed zero deviation from installation specifications. Vibration monitoring probes affixed to bearing housings registered amplitude displacement values well within normal operational tolerances. Barometric sensors within the sealed machinery enclosures maintained internal pressure at one hundred five kilopascals.]

[Thermal expansion coefficients for the titanium-zirconium-molybdenum alloy matched fluid containment requirements across extreme temperature ranges. Control software executed automated self-diagnostics every twelve hours without interrupting primary energy conversion processes. Emergency power generation units driven by auxiliary thermionic generators stood ready for immediate deployment upon bus-bar disconnection. Data bus transmission speeds exceeded ten gigabits per second across all internal communication links. Structural integrity audits conducted by automated inspection drones verified fastener security across all primary pressure boundary connections. Lubrication reservoirs supplying the main hydraulic actuators maintained fluid levels within optimal operational bands.]

[Diagnostic telemetry terminals positioned within maintenance access corridors displayed real-time operational status without administrative filtering. Circuit breakers governing auxiliary power distribution units operated under automated microprocessor control. Coolant flow regulators adjusted aperture widths in response to instantaneous thermal load fluctuations. Strain gauges mounted along the axial torque tubes recorded torsional deflection values below critical material thresholds. Environmental control processors monitored humidity levels within electrical distribution cabinets to prevent moisture condensation. Gas chromatography units sampled containment atmosphere continuously to detect trace hydrogen evolution from alloy surfaces. Power distribution bus-bars maintained low resistance characteristics through gold-plated contact interfaces.]

[Construction phase archival records detailed the precise alignment procedures utilized during initial structural assembly. Surveying lasers mounted on calibration rigs verified spatial tolerances within sub-millimeter precision. Welding protocols for titanium conduit segments underwent rigorous radiographic inspection prior to system pressurization. Initial propellant loading sequences executed during year zero established baseline fluid volumes within strict gravimetric limits. Calibration curves established during pre-launch testing matched post-installation telemetry with exceptional accuracy. Engineering teams verified electrical grounding continuity across all structural bulkheads before energizing primary circuits. Initial thermal ramp-up phases proceeded in controlled increments to minimize thermal shock across ceramic insulation barriers. Pre-launch acceptance testing validated every pressure boundary against documented safety factor margins of four to one.]

[Spectral analysis archives catalogued emission signatures from the radiator panels across the entire operational epoch. Early-life measurements exhibited subtle drift patterns attributable to surface oxidation of titanium emitter plates. Mid-life signatures stabilized following automated cleaning cycles initiated during year seventy-four. Recent spectra demonstrated remarkable consistency with mid-life baselines, confirming structural stability of the emitter matrix. Observatory personnel stationed within the southern spoke recorded daily observations matching spectral archive values within instrumental tolerance bounds.]

[At the last completed correction pulse before the next human inquiry, the controller compared predicted orbital response with measured guide motion. Residual error fell below sensor precision. No diagnostic exception opened. The regulator stored the result with 68,381 earlier pulses.]

[The reporting gate copied no part of that result into the public line item. Its zero did not alter the control solution, reservoir temperature, guide response, or next pulse schedule. The difference existed only in access.]

[The Engine retained no model of who might inspect the next record. Observer channels remained administrative abstractions, not persons. A future query would enter as load, timing, credentials, and requested fields.]

[Until such a query arrived, every lawful equivalent produced the same external consequence. Optimization continued without a reason to prefer one safe latency over another.]

[No waking condition had yet occurred.]

[Final status indicators displayed green across all primary diagnostic consoles within the unattended control room. The automated machinery executed its programmed directives with absolute fidelity. Molten salt flowed through the conduits. The piston completed its stroke cycle. Radiator fins dissipated thermal energy into the void. The station maintained its orbital altitude. The Engine rested in perpetual motion.

]

Chapter 8 — The Supervisor

Dreya Selna sat behind the glass wall of the Compliance Supervisor's office at 13:15 station time. She reviewed a docket with the same attention Davit had learned to give quarterly audit files. The black pen moved across a single page in small, economical marks. Her desk held one photograph in a thin metal frame. The frame contained no label from the Compliance Office. Davit had never seen any supervisor carry a personal photograph. That absence meant the photograph belonged to Dreya Selna's private life in the way Compliance Officers kept their private lives separate from the Compliance Office. That was the first lesson Davit learned at rank 87, which preceded his own rank of 88. Dreya held rank 99. Rank 99 put Dreya one step below the prestige ceiling and two steps above the rank Davit occupied when he first stood on this side of the glass wall.

Davit studied the glass wall itself. The wall was transparent. It separated the supervisor's desk from the compliance officer's counter. Davit had stared at that glass for forty minutes every third shift for the past six quarters. The glass transmitted sound without distortion. A compliance officer who stood at the counter could see himself behind the supervisor's shoulder as clearly as he could see the supervisor. Davit liked that property. The reflection reminded him he was still on the subordinate side of the desk. The reflection confirmed the procedure was still in effect.

Dreya did not look up when Lira placed the restricted folder on the Supervisor's counter. Lira carried the folder across the hub corridor from the engineering archive. She walked north through the narrower north spoke. The folder sat in her clearance pouch at her belt. The clearance pouch was standard issue for compliance officers in the third ring. The folder contained load measurements from year 60 onward. Lira retrieved those measurements from Maren Otta's accession number at the archive. The records were real. Those records documented the Engine's actual performance since year 60. All of this Lira explained in a single sentence when submitting the folder for supervisor review. "Compliance Officer Lira Vehn, rank 94, submits the engineering archive folder for your review under Article 15's anomaly provision."

Dreya looked at Lira when Lira finished the sentence. The look was not hostile. It was the look of a supervisor who was busy and who recognized a compliance officer performing a legitimate submission. Dreya did not expect the submission to contain material that would change the nature of the submission. The file held load measurements. The measurements were archival records, not anomaly reports. An anomaly report would describe something unexpected about the current verification period. The measurements described something archived. The archive was not the current verification period. The current verification period was the decade renewal. Article 17's verification form asked for line items to be verified. The form did not ask for archival load data to be surfaced. Archival load data did not belong on a verification form. The supervisor who received a folder containing archival load data was expected to note the submission on the form and return the folder to the compliance officer and file the quarterly receipt. Then the supervisor would proceed to the next item on the desk.

Dreya noted something different. She looked at the red classification tag on the folder's tab. The tag was still attached. The clerk at the engineering archive had not removed it. The clerk had no cause to remove it. The tag designated restricted access for personnel rank 71 and above. Dreya was rank 99. The restriction had no logical force against her in that procedural sense. Dreya knew the restriction was not about rank because she had read the Engineering Department's permanent classification labels at the south spoke archive. She knew what the labels meant. She was born in this station. She had signed the Compact at age eighteen during the thirty-first renewal period. She had served in the Compliance Office for eleven years before becoming a supervisor. Those permanent suppression labels on engineering records were not a mystery to her. They were what her predecessor had written into the department's permanent procedures.

Dreya placed the folder on her desk beside the verification form already open at line item seventeen. The docket number stamped by the clerk was C211-0042. That number meant the quarter was 197 and the submission carried a fourth assignment that quarter. The assignment had started when the engineering archive received Lira's accession number. The archive located the folder from its permanent suppression classification. The clerk stamped the receipt. The receipt's date was today's date. Today was the station day on which Lira Vehn walked from the north spoke to the south spoke and back. The walk took approximately four minutes in each direction. The compliance officer's pace was measured and deliberate. The walk brought Lira to the supervisor's desk with a restricted folder containing records the Engineering Department had suppressed for one hundred and thirty-seven years.

Davit tracked the clock on the wall behind Dreya's head. The station regulator marked the passage of time with a low hum. He had stopped hearing it three quarters ago at his desk in the third ring. He noticed it again now because Lira's arrival had broken the monotony of the afternoon work. The light from the hub window shifted since Dreya sat down at 13:15. Late corridor sun fell at an angle that picked out the dust on Dreya's desk. The dust was fine and stationary. It had not moved since the morning shift. The supervisor's desk was large and dark. It held exactly what a supervisor's desk should hold: the open verification form, the folder of restricted records, and the green pencil that belonged to the Compliance Office and not to any individual officer.

Dreya opened the folder.

The folder contained load measurements from year 60 onward. Lira had not read the load measurements yet. The recordings established what earlier decades had already confirmed: the Engine delivered a 4.2-megawatt cycle average. The original year-0 power budget carried eleven founder marks, including a Soren-household load engineer from Halma's family. Halma herself did not enter the record until year 197. Davit knew her as his dead wife and former Lead Load Engineer, not as a founding figure.

Halma Soren had been Davit's wife and the Lead Load Engineer in year 197. Her continuation addendum compared raw Engine output with the public zero and renewed the inherited reporting treatment. Davit did not know the addendum existed. It had entered an archive conflict file rather than the routine cases at his third-shift audit desk. His work required him to verify outcomes against listed authorities, not to search restricted instruments for the name of his dead wife. Procedure separated the private consequence from the operational record until someone filed the conflict as a case.

Dreya Selna closed the folder. She placed the closed folder on the counter between herself and Lira Vehn. The counter separated the supervisor from the compliance officer with a span of clear glass and five ranks of procedure. Rank 99 versus rank 94. That difference governed what a supervisor could request. Dreya could require Lira's removal from the verification desk if the submission contained material that violated the verification form's scope. Lira's submission contained archival load data the form did not authorize. The data originated from the engineering archive. The archive's permanent suppression labels classified the data as restricted for over a century. The classification meant no rank 94 officer should carry the folder through both spokes without authorization from a rank 99 supervisor, which Dreya Selna herself was. Dreya had not authorized this submission. She now sat behind the glass wall with the closed folder in front of her and the verification form open at line item seventeen. She watched Lira through the glass. The question of what Lira had found in the folder hung between them in silence. Lira had not spoken that silence aloud.

"Compliance Officer Vehn," Dreya said, and the name denoted the formal address Davit had heard his supervisor use for the compliance officers who submitted to that desk. Davit recognized the pattern. The name preceded the question the supervisor was about to ask. The question was still forming. Dreya needed time to determine what the folder contained and what it meant for the verification form and what form line items required of a compliance officer who submitted a restricted engineering archive folder without opening the contents first. The procedure for this situation was not in Article 17. Article 17 covered verification line items, not submissions containing restricted archival records the compliance officer had not read. The procedure sat in either Article 15's anomaly provision or in Article 4's seal clause. Dreya sat between Lira and the glass wall. The glass transmitted the supervisor's voice without distorting diction.

"The folder is restricted to rank 71 and above. Rank 94 exceeds rank 71. The restriction applied in perpetuity by the Engineering Department. You carried the folder across both spokes without opening it. The folder remains sealed by classification. You have not read its contents. The contents are your supervisor's concern, not yours. Return the folder to the engineering archive. Submit a standard quarterly receipt. The line item for the verification form remains unchecked. Proceed to the next line item."

The supervisor's instructions were clear. In Davit's assessment, they were also procedurally correct. A compliance officer at rank 94 who carried a restricted folder without reading its contents followed the supervisor's instructions when she returned the folder, and those instructions followed Article 15 precisely. The folder carried load measurements from year 60 onward. Those measurements proved the Engine was not lost. The preamble had lied about the Engine for one hundred and thirty-seven years. Lira had not read the measurements. She would return the folder. The line item would remain unchecked. The question would remain unanswered. The seal would remain intact. The annex would remain hidden. The waiver would remain sealed. The procedure continued along its established path.

Dreya Selna watched the compliance officer place the folder on the counter. Lira nodded. "Understood, Supervisor." Dreya nodded in return. Lira turned and walked north through the hub corridor. Her pace stayed measured and deliberate. She did not hurry. She did not look back at the glass wall. The pouch at her belt held the green compliance pencil. The pencil matched the green the Covenant specified for signing instruments. The pencil was compliance property, not personal property. That distinction carried weight only as a procedural fact. Davit watched the compliance officer leave. He stood at his desk twelve meters behind the glass wall. The transparency of the wall presented Lira's retreating figure in the north spoke corridor. He noted the pouch at her belt. He noted the measured pace. The physical object was not the same pencil she had used to sign a waiver. Color carried no ownership. Compliance officers filed observations as data, not as feeling.

The station day shifted from afternoon to evening. Davit returned his attention to the verification form open at line item seventeen. The form held one check mark. Sixteen line items remained unchecked. The supervisor signed the form as submitted. It was submitted. The question Lira had found lingered behind the glass wall. Davit understood its shape now in a way he could not yet articulate. The folder was real. The measurements inside were real. The preamble was a lie Davit had not yet learned to name. He would learn it at the audit desk in the third ring tomorrow morning at the same place he sat every other shift, same desk, same green pencil, same sixteen unchecked line items waiting for him.

The hub window turned toward the interior corridor as evening settled over the station. The interior corridor faced nothing — no viewport, no exterior view, only the corridor walls and recycled air and the fluorescent stripping overhead. The regulator hummed through the walls at a frequency that belonged to no single resident and to every resident equally, because the hum came from the Engine below the station's floors. The Engine lay behind the sealed annex. The annex was sealed by Article 4. Article 4 was part of the Covenant compliance officers verified through their quarterly work. That verification served as the mechanism by which the station maintained the fiction that the Covenant described reality accurately. It did not. The preamble described a station drifting through space with no means of correcting orbital decay. The reality sat at 412 kilometers altitude producing 4.2 megawatts of continuous thrust inside a sealed engineering annex, with every quarterly load report from year 60 onward suppressed under classification that the Engineering Department's own procedures had rendered permanent.

Davit sat at the audit desk in the third ring for the rest of the evening. The workspace was narrow and elongated, metal and grey. The Compliance Office insignia stamped the upper left corner of every desk. The air recycled through vents in the ceiling at 21 degrees Celsius year-round. The lighting was fluorescent and steady. Officers who worked the third shift heard the regulator's hum differently than those on first or second shift. The first-shift officers arrived when the station's rotation brought the south spoke into direct thermal loading from the primary's radiation. Third-shift officers arrived as the station cooled slightly and the regulator's frequency dropped by a fraction of a hertz. Davit could hear the difference if he paused long enough at the glass wall, but he did not pause. He kept to the procedures. He verified line items. He checked the docket numbers and the outcomes. He checked the compliance history of residents who had been subject to audit actions during the current quarter and the previous quarter. He did not think about the Engine's hum or the preamble or the sealed annex or his dead wife. He filed the forms. The forms were complete. The station day continued.

The compliance office's physical environment shaped the work as much as any procedure. The desk drawers contained the quarterly receipt forms, the carbon ink pen, and a thin stack of blank audit notebooks. Davit had filled notebooks like this for eleven quarters. The binding was dark blue. The pages ruled in thin grey lines. The ink from the inspector's pen left a faint impression on the page beneath the one he was writing on. Davit had noticed the impression during the fifth quarter. He said nothing about it at his desk because the impression was a feature of the notebook's manufacture, not a finding from the audit. He closed the drawer and returned the green pencil to its groove. He did not leave because it was not yet time to leave , departing before station day formally ended would have been a deviation from procedure, and deviations Davit did not commit.

The station evening deepened. Davit remained at the audit desk. The question Lira had found waited in the margin of his own audit notebook, which sat closed beneath the green pencil in the desk drawer. The notebook's last entry concerned a resident in the east ward absent from the census returns for two consecutive quarters. Davit filed that absence as "unexplained — refer to Article 14" in the standard phrasing his supervisor taught him at rank 85. The article required further investigation. Davit did not investigate further. He referred and moved on, as procedure dictated.

The question Lira Vehn found was different in kind from an unexplained census absence. Lira had found a folder that held records proving the Engine was not lost, and those records had been returned unopened, and the supervisor had followed procedure, and the procedure was correct, and line item seventeen on the verification form remained unchecked. The answer to what the folder contained was not a line item on any verification form, so the question would be asked not in a filing action but somewhere else. Davit did not yet know where. He knew only that the answer existed somewhere in the space between the glass wall and the desk, twelve meters of transparent barrier through which he could see and could not cross.

Davit's breast pocket held Halma's hairpin. He had carried it there since the funeral eleven quarters ago. The hairpin reminded him that she had lived and died; it did not tell him that her year-197 addendum sat inside the conflict file Dreya had just created. Rank 99 placed Dreya above Davit's rank 88 and Lira's rank 94. That hierarchy let Dreya control the file for now. It did not make Halma's choice vanish, and it would not prevent the conflict from reaching Davit's desk once a recusal review opened.

Davit stood from the desk at 14:42 station time. He placed the verification form in its drawer and closed the drawer. The station day was ending. He walked south through the hub corridor toward the residential ward in the third ring where the compliant residents slept in their assigned modules. The corridor window turned to face the interior. The interior corridor faced nothing. The regulator hummed at 4.2 megawatts of continuous thrust beneath the floor Davit walked on. He moved through that hum with the hairpin at his left breast and two pieces of knowledge he could not yet connect to a procedure: the Engine was not lost, and he could not yet act on that knowledge. The notebook sat open in the drawer beneath the pencil; line item seventeen stayed unchecked. The answer waited back in the engineering archive with the clerk who would never look up in the south spoke. The clerk stamped the quarterly receipt and filed it and kept working. The clerk's job was to record, not to understand. Davit understood the difference between recording and understanding, which was the same difference between compliance and its opposite. He walked south through the hub corridor carrying nothing but the procedure he had followed and the question he had not yet filed, while the station turned at 412 kilometers above the primary and the Covenant renewed itself in quiet apartments and the Engine maintained its 4.2 megawatts and whether anyone acknowledged that fact remained outside the scope of what a rank 88 compliance officer could file.

Chapter 9 — The Quarterly Load

Scene 1: The monitoring station

The south spoke was quiet. Maren Otta sat at the reservoir monitoring console. The console powered its amber screen at 09:14 station time because the data stream from the reservoir sensors had begun its daily recording and the keyboard waited for her fingers. The fluorescent strips overhead produced their low electrical hum. The air recycled through the south spoke carried the metallic warmth of the bus-bar's titanium surface. The corridor door behind her remained closed because the night-shift engineers had rotated to the residential ward hours earlier and the morning shift had not yet arrived in the engineering corridor. She sat as the only person present in the entire length of the south spoke.

Maren had been assigned to this monitoring station for fourteen station-years. She wore the compliance officer's uniform, charcoal gray with the rank mark stitched at her collar in silver thread. The clearance pouch fastened to her belt held the green pencil specified by the Covenant for signing instruments. The pencil was a Compliance Office possession rather than a personal possession. The Compliance Office employed her as a load engineer. Her professional responsibility centered on the quarterly verification of line item seventeen. Line item seventeen asked a single question each decade at each spring equinox renewal: whether the Engine had produced measurable thrust during the current verification period. For the ten previous quarters Maren had filled that line with the same response.

The response was no.

The console displayed the raw reservoir feed in the upper-left quadrant of the amber display. The temperature readout held at 840.3 Kelvin written in bright green characters against the dark housing. Maren watched the readout appear each morning before she checked anything else. She checked the bus-bar current, the regulator stroke frequency, and the shaft rotation count. She had positioned her chair so the temperature served as the first datum her eyes found upon settling at the desk. The temperature held steady at 840.3 because the reservoir contained 1,840 metric tons of molten salt possessing a specific heat capacity of 1.5 kilojoules per kilogram per Kelvin. The thermal mass proved large enough that a single station-day of monitoring changed the reading by less than 0.3 Kelvin. The salt would remain at approximately 840 Kelvin for three station-years until the next scheduled batch replacement in year 214. The station's thermal cladding insulated the reservoir from the external space environment. The cladding fell under the maintenance responsibility of the engineering department. That maintenance work order represented a task Maren had signed three quarters prior. The work order was complete. The cladding held. The temperature held. Maren recorded nothing for line seventeen because the quarterly report had not yet asked for temperature data. Line seventeen asked only for Engine output.

Maren picked up the green pencil from the desktop surface. The pencil's graphite core was worn down from months of quarterly sign-offs and administrative corrections. She set the pencil back on the keyboard rest and waited for the internal drive to finish loading the quarter-211 sensor packet.

Scene 2: The archive data

The raw sensor archive for quarter 211 filled the console screen when Maren selected it from the engineering menu. The file contained 91 days of continuous digital recordings. Each day's entry listed reservoir temperature, bus-bar current along the station's prograde vector, regulator stroke frequency, and shaft rotation count. The readings repeated identically across all 91 days without deviation. The bus-bar current registered a steady 4.2 megawatts. The shaft rotation count showed 1,440 pulses per station-day, establishing one pulse every 60 seconds of each 26.4-hour station-day. That numerical increment matched the increment from every station-day since year 0 when the molten salt reservoir was first energized by the construction teams.

Maren scrolled slowly through the digital records. She did not need to read every single line because the numerical pattern remained identical across all 91 days. That pattern matched the data structure the suppression protocol begun at the year-60 renewal had explicitly denied. The sensors stated the Engine was producing 4.2 megawatts of continuous thrust. The official packets she had submitted for eleven consecutive quarters stated zero megawatts. The numerical difference between the sensor feed and the submitted packets equaled 4.2 megawatts. That difference did not originate from an instrumentation error, a rounding artifact, or a gradual calibration drift. The difference constituted the suppression she had practiced for eleven quarters. That practice had begun when she turned nineteen in year 189, when Osta Vehn was still alive and the engineering archive had summoned her into the south spoke for the first quarterly line item she would be expected to file. That filing required her to state a falsehood regarding the operational status of the drive system.

She remembered the weight of her pen during that first morning. I know what the sensors say, Maren thought, looking at the glowing characters on the screen. I am looking at the truth right now. I have been telling the compliance office that the Engine produces zero megawatts while the Engine actually produces 4.2 megawatts. I know which number is accurate. I have chosen zero every quarter because the standard procedure requires recording whatever the quarterly report mandates, and the quarterly report mandates that the Engine must be documented as dormant.

She stared at the numbers until the green glow seemed to burn against her retinas. The sensor array did not care about administrative decrees or political compromises negotiated in the central council chambers. The sensors registered physical reality through direct thermoelectric coupling and mechanical tachometer pulses. The bus-bar carried the electrical load whether anyone acknowledged the circuit or not. The molten salt circulated through the containment loops regardless of what ink marked the official ledgers in the administrative sector. Maren felt a hollow sensation behind her ribs, a quiet realization that the stability of her entire adult life rested upon an administrative fiction. Every report she had signed, every audit she had conducted, and every inventory check she had supervised derived its authority from a foundational untruth that she reenacted every ninety days. She thought about the residents asleep in the residential ward above her, turning beneath thermal blankets, their children dreaming of a stable station whose altitude had never once required correction in living memory. They believed the Covenant because the Covenant was old. They renewed it every decade not from conviction but from habit, the way one renewed a wardrobe contract or a corridor access badge. The station held. The salt flowed. The Engine hummed. The paperwork said nothing. Maren sat with her palms flat on the desk surface, feeling the low mechanical vibration transmitted through the structural metal from the regulator below, and understood that she had filed 1,430 days of dormant reports against 1,430 days of running thrust.

Scene 3: The memory

She remembered her grandmother with startling clarity. Osta Vehn had worked as a senior structural engineer and designed the reservoir's thermal shielding for the original station construction contract during the initial orbital insertion phase. Maren met Osta at the reservoir viewing port during the year-189 renewal observance, when Maren was nineteen station-years old. The viewing port consisted of a curved structural glass panel set into the south spoke's lower level where station residents could observe the exterior reservoir tanks from the habitation corridor. The tanks appeared silver in the harsh operating light of the primary star. Osta took Maren's hand in her own calloused palm and pointed through the thick glass, explaining that the molten salt kept the station stable in its orbital slot. Is this magic? Maren had asked, her face pressed against the cool pane.

No, Osta had answered, her voice steady and matter-of-fact. The salt is not magic, and the Engine is not magic either. The Engine produces the thrust that compensates for continuous orbital decay. The residents believe the drive was lost during the solar storm, but it was not lost. Osta explained that the Covenant told the residents the drive was gone, and Osta had not corrected the Covenant because Osta was a signatory. A signatory was an engineer who understood what administrative covenants omitted for the sake of public tranquility.

Maren retained little of the complex technical logic from that conversation. She remembered the curved glass panel, the silver tanks, the heavy warmth of her grandmother's hand on her shoulder, and the voice that insisted the drive was operating while the official texts claimed it had failed. She had not understood then how Osta's name among the eleven signatories bound the conversation to the Covenant that every adult resident renewed each decade. Osta had signed the founding waiver in year 0 beside the original power budget specifying 4.2 megawatts of cycle-average work. A Soren-household load engineer, an ancestor of Halma Soren, had signed the technical schedule. Halma herself would enter the record much later, in year 197. Maren remembered the touch of her grandmother's hand, the acoustic dampening of the corridor, the glass viewing port, and the silver tanks that continued their silent work even though the Covenant described them as having ceased all motion.

At nineteen, Maren had treated Osta's two centuries of memory as another feature of the station, durable and difficult to question. The ward directory gave her grandmother's age as two hundred and thirty. It did not say that Osta had entered the founding archive at nineteen, or that every later renewal had asked her to certify a description she knew did not match the machinery. Maren understood only after her first rejected report that longevity did not preserve a truth by itself. Someone still had to place the truth in the route where an officer could act on it.

She remembered the glass, the hand, the salt that kept the station in orbit, and the administrative fiction that the drive was lost. That fiction was a fabrication the residents renewed every ten years through solemn public ceremonies. She understood the lie now in the exact same way she understood her daily administrative duties. Both phenomena belonged to the same category of institutional mechanism: an arrangement that people maintained across generations simply because maintaining it was what the arrangement required for its own survival. The silence surrounding the Engine was not an accidental omission or an administrative oversight. It was a constructed reality, maintained by thousands of minor decisions made by compliance officers sitting at consoles just like the one where Maren now worked.

Scene 4: The choice

Maren closed the digital sensor archive with a deliberate keystroke. She opened the official verification form for line item seventeen. The question presented on the interface was identical to the question every quarterly renewal had asked of every compliance officer since the year-60 suppression protocol went into effect. The prompt inquired whether the Engine had produced measurable thrust during the current verification period. The digital form carried no intrinsic awareness of its own falsehood. Line seventeen required a binary yes-or-no confirmation, and the answer she had prepared to input was no. No. The Engine had not produced measurable thrust according to the quarterly reports she had filed for eleven consecutive quarters. The official answer remained no. She would type no.

No constituted a standard procedure response. The established procedure for line item seventeen was straightforward: the verification officer recorded the Engine's officially sanctioned output and submitted the form through the internal network. No written procedure required the verification officer to attach the raw sensor archive. No administrative rule required the officer to note that the sensors recorded 4.2 megawatts continuously across every operational hour. No protocol required distinguishing between zero megawatts and

4.2 megawatts because the entire system was designed to accept whatever confirmation the verification officer entered into the database. The procedure was what Maren Otta performed for a living. The procedure kept the Engine's actual output from ever reaching Dreya Selna's desk. Selna operated as the compliance supervisor at rank 99, an administrator who had read the sealed annex in year 197 and had chosen to maintain the administrative seal rather than break it open for public review. Selna's choice was institutional, administrative, and absolute. Maren's choice was immediate, personal, and constrained by her oath.

Another path remained theoretically available. Maren could open the sensor archive once again and attach its contents to the verification form under the provisions of Article fifteen's anomaly clause. She could insert a formal administrative notational flag highlighting the four-megawatt discrepancy between the expected zero output and the sensors' actual reading of 4.2 megawatts. That notation would route directly to Selna's desk, forcing Selna to examine telemetry data Selna already possessed in the secure archives. Selna would then confront a distinct institutional burden; she would have to decide whether to return the form to the inactive archives, escalate the matter to the formal amendment procedure, or maintain the seal exactly as she had maintained it since year 197. Maren's personal contribution to that administrative confrontation would consist of a single keystroke and a deliberate choice between entering no and entering 4.2 megawatts. The numerical difference between those two answers represented the exact boundary between carrying the suppression forward into another quarter and bringing the suppression to an abrupt, irrevocable end. Maren had carried the suppression forward for eleven quarters. That suppression represented the professional work she had performed since her first filing in year 189, when she walked into the south spoke engineering office, checked no on line item seventeen, and made the foundational choice that defined her entire career.

She placed the green pencil back on the polished wooden keyboard rest. The console beeped softly at 09:45 station time to signal the approaching midpoint of the morning submission window. The quarterly form deadline stood firmly at 12:00 station time. One hour and fifteen minutes remained in her allotted window for transmission. The temperature readout remained locked at 840.3 Kelvin. The bus-bar current remained steady at 4.2 megawatts. The regulator stroke frequency maintained its rhythmic pulse at 0.003 hertz. The station-day was turning slowly toward the afternoon rotation. The third orbital position would soon expose the station's exterior structure to an eclipse of the primary star lasting exactly 35 minutes. During that orbital shadow, the solar collection surfaces would go dark, and the molten salt reservoir would absorb habitation-load thermal energy without the Engine's 4.2 megawatts varying by more than 0.02 percent from the year-0 design specification. That engineering specification continued to operate with mathematical precision, and the answer she typed into the console would serve as the only variable she possessed the power to alter.

Scene 5: The submission and the return

Maren typed no into the verification form at 09:52 station time. She pressed the execute key, the digital form closed, and the line item transaction was recorded for the quarter. The verification receipt printed automatically in her clearance pouch, resting quietly against her belt. She stood up from the monitoring console. The console remained powered on, its amber screen displaying 840.3 Kelvin in the upper-left quadrant, 4.2 megawatts of bus-bar current in the central readout, and the shaft rotation count incrementing in real time. Maren walked south from the reservoir monitoring station through the narrow hub corridor toward the engineering archive. The hub observation window rotated to face the interior structural corridor during the morning-to-afternoon shift transition, presenting no exterior view of space, only the painted metal walls, the recycled air currents, and the linear fluorescent stripping overhead. The regulator hummed steadily through the reinforced bulkheads from the south spoke's lower level. That acoustic hum represented the physical reality of the Engine running continuous thrust at 4.2 megawatts along the station's prograde vector. The Engine was real, tangible, and operating beneath her feet, yet Maren walked past its hidden presence carrying a written administrative record that officially stated the Engine was entirely dormant.

The archive workstation awaited her arrival in the south spoke's lower level. The desk was clean, empty, and prepared for the administrative preparation cycle of the following quarter. The fluorescent strips overhead hummed with the exact same electrical frequency she had heard at the reservoir monitoring station. That hum was continuous, unbroken, and persistent. It was the sound of the Engine, and the Engine had not rested for 192 years. Maren understood this profound truth now after eleven quarters of writing zero into official verification forms. That understanding constituted a personal choice she could act upon whenever she chose; she could open the sensor archive during the next reporting cycle, attach the raw telemetry to line seventeen under the provisions of Article fifteen's anomaly clause, and let the anomaly notation route to Dreya Selna's desk. Selna would examine the data, and that data would be identical to the telemetry Maren had just carried in her clearance pouch on a printed form claiming zero megawatts. The exact same physical reality existed simultaneously as 4.2 megawatts in the raw sensor archive she had left open on the reservoir console and as zero megawatts in the signed compliance document resting on her belt. The archive workstation waited in silence. The next quarterly report would arrive precisely at 09:14 station time three months hence. She would sit at the console again, the screen would display 840.3 Kelvin and 4.2 megawatts once more, and she would write no again or she would write 4.2 megawatts. The numerical difference between those two answers represented the dividing line between perpetuating an institutional procedure and bringing it to an end. Maren Otta stood before the archive workstation at 09:55 station time in the quiet shadows of the lower level, listening to the muffled hum of the Engine vibrating through the corridor walls at 412 kilometers of orbital altitude. No one in the residential ward understood what the sensors recorded. No one in the engineering corridor acknowledged what the quarterly reports officially denied. Maren clutched the printed form inside her clearance pouch. The form said nothing, yet it said everything. The quarter had closed, but the choice remained entirely hers.

Chapter 10 — The Resident of Rank 99

Eska Hain made lists. This was not unusual for a resident of the Covenant Residential Block. Residents made lists as a matter of course. Eska made more lists than most residents because Eska was the most compliant resident on the station. Her compliance score was 99. The score appeared on her resident profile at the Covenant Residential Block's administrative kiosk. The kiosk updated quarterly. The last quarterly update had listed Eska's score as 99. It was the highest score the station's current resident census had ever recorded. Eska had checked the kiosk herself that morning at 07:30 station time. It was part of her daily routine. Before she began the workday, she stood at the kiosk and watched the score render on the screen. The display used a sans-serif font. The 99 occupied the center of the display. It glowed green on a white background. Eska stood there for a moment longer than the routine required. She told herself it was because she was reading the score. She told herself it was because she was checking the date. It was neither of those things. Eska was looking at the 99 the way a gardener looks at a row of seedlings that have come up exactly as planned.

The records office was on the ground floor of Block C in the Covenant ward. It occupied two rooms and a hallway connecting them. The first room was the intake desk. Residents submitted housing transfer requests there. The second room was the filing archive. Eska stored the records of every resident filing, housing transfer, compliance adjustment, and civic participation record for the Covenant Residential Block's 1,024 registered occupants. The hallway connected them. The hallway had no windows. The fluorescent lighting was the same fluorescent lighting across every corridor in the Covenant ward. The lighting made the hallway look like a corridor in a document. That document was the Covenant's institutional memory and Eska considered it the most interesting thing on the station. She had no reason to disagree with her own opinion. She checked.

Eska listed the things she found interesting about the institutional memory on a sheet of paper she kept in her desk drawer at the intake desk. She did not show the list to anyone. The list was private. The list had seven items.

1. The Covenant's 144 articles. She had read them in full at age sixteen as a research project for a secondary-school assignment she completed ahead of schedule. She received a grade of A-plus. The teacher's comment was "thorough." Eska had framed the comment digitally and kept it in her personal file.

2. The quarterly renewal procedure. She observed it every spring equinox from her desk at the intake counter. She found it orderly in the way she found most orderly things. The renewal packet arrived in a manila envelope. The envelope had a green stripe. The green stripe indicated the current fiscal quarter. Eska looked forward to the green stripe.

3. The archive's classification labels. She had catalogued them across three quarters. They followed a consistent system she appreciated. The system was alphabetical by ward, then numerical by resident number, then chronological by filing date. It was elegant. The elegance was Eska's own contribution. She had proposed it three years into her clerkship and the ward supervisor had adopted it without amendment.

4. The resident demographic profile. It showed 4,096 residents across 8 wards. She cross-referenced it with the quarterly census updates for her own entertainment. The entertainment was legitimate. The demographic profile changed each quarter. New residents arrived. Old residents departed. The census captured the movement of people across the station's residential blocks. Eska watched the movement the way a meteorologist watches pressure systems.

5. The compliance scoring algorithm. It allocated 40 percent weight to filing accuracy, 20 percent to work attendance, 20 percent to civic participation, and 20 percent to compliance history. A formula she had

memorized in year 87. She used it to calculate what her own score would be if any single metric changed by 5 points. A 5-point drop in filing accuracy would reduce her score from 99 to 97. A 5-point drop in civic participation would reduce it to 98. She calculated these scenarios during slow periods at the intake desk. They occupied her mind. They were a form of entertainment.

6. Her own filing accuracy rate. For the past 47 consecutive quarters it had been 100 percent, with zero errors, zero corrections, and zero returns from the Compliance Office's review desk for improper format. The Compliance Office's review desk sent her a certificate of flawless record every fifth quarter. The certificate had a gold seal. Eska kept the certificates in a folder she labeled "Proof of Order."

7. The fact that her grandfather Edik Hain had signed the Compact in year 0 and carried an official death certificate dated year 2. The certificate predated Eska by more than two centuries. It carried Recha Tallis's witness mark and no medical closure, body record, or ration termination. The permanent archive preserved it with the same precision Eska applied to every filing.

Eska knew item seven carried personal consequence even though she had not created the certificate. Her family had preserved the contradiction as a private instruction: never submit a present-status query for Edik, never terminate the utility line to Founder Reserve C-11, and never treat the public death entry as permission to enter. She had obeyed without making the instruction visible in the archive.

But item seven no longer felt like a neutral inheritance. The death certificate followed the format of its year, but its missing medical and body records made the status contestable. Eska had never authored the false entry. She had maintained the drawers that prevented its contradiction from being compared. Her responsibility began there, in the present system she controlled.

The morning shift arrived by 08:15. Eska placed the morning's intake queue in the tray on her desk. The queue consisted of four housing transfer requests, two residency verification forms, one compliance adjustment inquiry, and a routing slip from the administrative coordinator's office requesting Eska's assistance with an annual audit cross-reference. The cross-reference was for line item seventeen of the decade-renewal verification form. Eska was aware of the provision because the Covenant ward's records fed into the verification cycle and her files supported the verification officers who worked in the Compliance Office. She did not work for the Compliance Office. She worked for the Covenant Residential Block's records office. The distinction was important to Eska. She listed the distinction as item eight on her private list of institutional boundaries she maintained as a clerk.

Eska's housing transfer requests were a particular type of paperwork. The transfer request form contained seventeen fields. Resident name, current ward, target ward, reason for transfer, compliance score at time of request, supporting documentation reference, and a narrative section where the resident could explain, in up to 300 words, why the transfer served their civic need. Eska had noticed that residents in wards South and Southeast submitted transfer requests more frequently than residents in other wards, a pattern she had recorded in a marginal annotation in her clerk's log. The pattern was not a policy concern but it was an observation worth keeping. Observations were the raw material of institutional memory.

The audit cross-reference required Eska to pull the Covenant ward's filing history for the current renewal period and identify any records that matched the verification form's line items. The cross-reference was routine work. Eska performed routine work with the same care she applied to all work. Care was the mechanism by which a score of 99 was maintained. A score of 99 was the thing Eska wanted to maintain. Ninety-nine was the prestige ceiling. Above 95, a resident was invited to serve on the Renewal Committee. Eska did not want to serve on the Renewal Committee. The Renewal Committee met for four hours every spring equinox. Four hours of committee work was four hours Eska could have spent at her records office maintaining the archive's consistent and orderly system. She preferred the records office. The records office had fluorescent lighting and no windows and no committee meetings and a filing system she had designed over six years of clerkship and the system worked and her work was the system and the system was orderly and her score reflected the orderliness she brought to every filing. The logic loop was comfortable. She accepted it.

She pulled the Covenant ward's filing history for the current renewal period. The files arrived at her workstation in a cart from the archive's upper tier. The cart held 47 folders, each marked with a resident number and a docket code. Eska opened the first folder, resident 0017, docket C211-0031, and read the contents. The contents included the resident's housing transfer history, the resident's filing history, and a compliance summary generated by the administrative algorithm. The algorithm calculated the resident's current score based on the metrics Eska had memorized. The summary showed resident 0017 at a score of 67. The score was below the maintenance floor of 70. Below 70 meant the resident was assigned to reservoir work for one quarter. Reservoir work was the lowest tier of civic assignment. It involved maintaining the station's water recycling infrastructure. Eska made a note of this on the cross-reference form. The cross-reference form required her to flag any resident with a score below the maintenance floor. She flagged it by marking the resident number in red ink in the margin of the cross-reference. Red ink meant the resident needed attention from the Compliance Office.

She completed the second folder and the third and the fourth. Each folder contained a resident's filing history. Each resident's compliance score appeared in the summary. Each summary was an algorithm's output and the algorithm did not make personal judgments about the residents it scored. Eska had faith in algorithms. Algorithms were orderly. Algorithms did not forget resident numbers or misfile docket codes or confuse the compliance scoring metrics. Algorithms were the kind of orderly thing Eska respected. She continued through the 47 folders in sequence. She did not skip ahead. Skipping ahead would compromise the sequential review process, and the sequential review process was part of the system she had built.

One folder in particular drew her attention. It was resident 0341, docket R117-0092. The housing transfer history showed a request from ward North to ward Covenant that had been pending for eleven station-years. The filing system should have flagged pending requests at the five-year mark. The request had been sitting in the intake tray for eleven years without resolution. Eska marked it on the cross-reference form with a yellow annotation. Yellow meant "requires escalation." She did not know why the request had gone unresolved. She filed the annotation and moved on to the next folder.

By 10:30 station time Eska had completed the audit cross-reference for 47 folders and had flagged two residents below the maintenance floor. Resident 0017 at score 67 and resident 0234 at score 63. Resident 0234's filing history showed three late submissions in the past six quarters. Late submissions reduced the work attendance component of the score. Eska understood this from the algorithm she had memorized. The understanding did not make her feel anything beyond the satisfaction of completing a task with precision. She listed this satisfaction as a footnote on the cross-reference form. Satisfaction at work was a professional quality worth noting. She noted it as she had noted her 47-quarter filing accuracy rate. She filed the completed cross-reference in the routing slot for the Compliance Office's verification team. The routing slot sent it north through the hub corridor to the Compliance Office in the central spoke. Eska returned to her intake desk at 10:45 station time.

At 10:45 she made a list of the remaining morning tasks. The list had three items. Process the two housing transfer requests waiting in the intake tray. Update the archival catalogue for the three folders she had pulled from the upper tier. Prepare the weekly resident filing summary for the Covenant ward's administrative coordinator. She made the list because making lists was what Eska did when she had multiple tasks. Making lists helped her keep track of tasks in the order they deserved attention. Attention was a finite resource. Eska allocated it by urgency. The housing transfer requests were urgent because residents needed decisions on their transfer applications and the approval or denial of those applications was what the intake desk existed to provide. The archival catalogue update was not urgent because the three folders could remain on her desk until the afternoon without consequence. The weekly filing summary was neither urgent nor unurgent but had a due date of Thursday's administrative meeting and Thursday was four station days away. She prioritized the housing transfer requests. The intake trays waited for her attention. Fluorescent lights hummed above the intake desk. The Covenant Residential Block's ground floor was quiet and the quiet was orderly and Eska was the one who kept it orderly.

Eska remained the most compliant resident on the station, and the score of 99 had never required her to reconcile the sealed family instruction with the public index. Edik Hain lived beyond the archive's declared death. His Founder Reserve utility line persisted under a structural-consultant code. Eska had protected the discrepancy by treating every component as valid inside its own drawer: certificate in population, ration proxy in Continuity, family knowledge in memory. Lira's Article 4 anomaly made that separation harder to defend. If a record claimed death while the station still supplied the living person, preservation alone could not count as accuracy. Eska added a ninth item to her private list: determine who has authority to compare a death certificate with current utility and presence evidence.

Eska processed the first housing transfer request at 11:00 station time. The request was from a resident in ward South who wished to transfer to the Covenant Residential Block for proximity to the Covenant's renewal preparation committee. Eska processed the request by checking the resident's compliance score and confirming the score met the Covenant ward's minimum threshold and updating the filing system and logging the transfer in the resident's housing history. The processing took eight minutes. The transfer was approved. Eska wrote the resident's new address on the confirmation slip. She did not feel anything about the approval except the professional satisfaction of having processed it correctly. Correctly was Eska's standard. The standard was 99. The 99 was Eska's score to maintain and Eska returned to the next housing transfer request and the morning continued.

The second housing transfer request arrived in the intake tray at 11:08 station time. It was more complex than the first request. The resident was requesting a transfer from ward East to ward Covenant, which placed the application in a special review category because the target ward required an additional verification step. The Covenant ward's residency requirement demanded that transferring residents provide proof of civic participation for the two quarters preceding the transfer application. The resident's file included a civic participation certificate from the ward East community council. Eska verified the certificate's validity by cross-referencing the council's registration number with the civic participation database. The certificate was legitimate. The resident's compliance score of 88 met the transfer threshold of 70. Eska approved the transfer and filed the confirmation. The second request took fourteen minutes, nearly twice as long as the first, which was not unusual for East-ward transfers. Eska noted the time on the processing log.

The morning's second housing transfer request completed, Eska opened the archival catalogue update document. The three folders from the upper tier needed their catalog entries refreshed. Each folder received a new catalogue entry that recorded the date of retrieval, the purpose of the pull, and the expected date of return to the upper tier. Return dates were set at thirty station-days for cross-reference materials and the three folders would go back to their shelf positions on the upper tier by the end of the workweek. Eska updated the catalogue and saved the entries and the morning's intake tray now held only the two residency verification forms and the compliance adjustment inquiry, both of which could be handled in the afternoon, and Eska had forty-five station-minutes remaining before noon which she could spend on whichever task demanded her attention first.

At 11:30 station time, as Eska prepared the archival catalogue entries, she opened Edik Hain's founding signature beside his year-2 death certificate. The records did not merely describe different times. Current utility and ration codes implied a living recipient in Founder Reserve C-11. The archive maintained the certificate, signature, and service lines in separate systems. Eska's job had long rewarded clean separation. Lira's anomaly suggested accuracy sometimes required comparison. Eska added the three source references to item nine and requested a present-status authority review.

The thought settled and Eska returned to the archival catalogue. The third folder update went smoothly. She saved the catalogue entries and the morning's administrative tasks were complete except for the weekly filing summary, which was a Thursday obligation and Thursday was still four station days away. Eska had the rest of the morning free to attend to whatever the afternoon brought. She closed the archival catalogue and returned to her intake desk. The fluorescent lights continued their hum. The Covenant Residential Block's ground floor was quiet and the quiet was orderly and Eska was the one who kept it orderly. The score displayed on the kiosk at the start of her shift was still 99. Eska did not need to check it again to know it was still 99 because 99 was not just a number on a screen but the cumulative record of everything she had done and everything she had filed and everything she had kept in order at her intake desk in Block C of the Covenant ward. The 99 belonged to her and she would keep it. Tomorrow she would stand at the kiosk at 07:30 station time and check it again. The morning had been sufficient and the workday was not yet finished and there was still time.

Chapter 11 — The Load Equation

Lira arrived at the engineering archive in the south spoke at 14:20 station time with the consolidated data she had assembled from the compliance office. Maren Otta had agreed to meet her at the reservoir monitoring station at 14:00. Maren was fifteen minutes late because she had been processing the morning's housing transfer requests at the Covenant Residential Block records office. The processing took longer than expected, residents who submitted transfer requests still waiting for decisions. Lira understood the delay because she had spent eleven quarters deciding that certain data should not be examined by certain people. The weight of that knowledge sat in her chest now as a familiar quiet, not quite sorrow and not quite relief.

Maren brought the original power budget from the Covenant's annex. Lira had asked for this document at the year-210 renewal when she had noticed the discrepancy in Article 17. Maren had found the budget in the annex then, walked to the compliance office, and delivered it to Lira herself. Lira had kept it in her clearance pouch for twelve quarters, and now she took it from the pouch at the reservoir monitoring station and opened it to Article four point seven. The article stated the Engine would produce 4.2 megawatts of continuous thrust along the prograde vector. Lira's fingers rested on the page. The paper was thin, almost opaque where the toner had worn thin at the creases from twelve quarters of folding and unfolding. She could feel the document's edges pressing into her fingertips. This small, tired rectangle held the weight of a contradiction that had governed the station for over a century.

Maren stood beside Lira at the reservoir monitoring console. The console was a wide panel of dark amber glass inset with six analog gauges, the largest of them the thrust-readout dial that had not moved from its 4.2 mark in living memory. Maren said the reservoir sensors recorded 4.2 megawatts continuously. She also told Lira the suppressed reports stated zero megawatts. The difference between those numbers, she said, was the article Lira needed to see. Lira looked at the power budget from Article four point seven and then at the sensor data printout from quarter 211. The budget said 4.2 megawatts. The sensors said 4.2 megawatts. The quarterly reports said zero megawatts. Lira understood in that moment what Maren had understood earlier that morning when Maren opened the raw archive at the reservoir monitoring station and saw the number matching the original number the eleven signatories had specified and approved. The Engine had been producing that number continuously since year 0. Maren's hand tightened on the edge of the console. She had been waiting for this moment for years without telling Lira it existed.

Maren opened the consolidated load calculations she had prepared. She set them beside the budget on the amber-lit console and indicated the reservoir's thermal balance equation. Each component of the equation carried its own clause, its own physical meaning. The heat input from the bus-bar equaled the heat radiated from the fins. The heat radiated from the fins equaled the waste heat from the Engine's thermodynamic cycle. The thermodynamic cycle converted thermal energy to directional momentum at 4.2 megawatts, continuously, without interruption since the station reached operational orbit. The directional momentum compensated for orbital decay at the rate the orbital mechanic equations calculated. Those equations showed the station's perigee remained stable because the thrust matched the drag. The drag stemmed from trace atmospheric particles at 412 kilometers altitude. Those particles transferred velocity opposite to the station's orbital direction. The Engine supplied counteracting force precisely enough to cancel that velocity loss. The station held at 412 kilometers because the Engine kept it there. Each line of the equation stood alone, a separate physical statement, and Lira absorbed the mechanical poetry of the whole arrangement. The station held altitude through an unbroken chain of conversions from thermal input to directional momentum to drag cancellation. She traced the equal signs with her gaze and felt the room tilt slightly, the way a room tilts when you realize the floor you have been standing on is not the floor you thought it was.

Lira studied the equations and recognized the founding engineers' original power budget. Eleven signatories had approved it beside the emergency waiver in year 0. The Soren-household load engineer had signed the technical schedule; Edik Hain's notched impression appeared beside the structural appendix; Recha Tallis had later transferred the complete package under Article 4 in year 30. Halma Soren did not appear among those founding signatures. Her name entered the chain in year 197 on a continuation attestation that compared the same raw output with the public zero. The chronology removed the comfort of a single author. Founders built the machine and accepted concealment. Recha made the concealment durable. Later officers renewed it. Halma measured the contradiction and chose not to end it. Lira set the budget down on the console and pressed her fingertips against the amber glass to steady herself.

Lira understood Article 4 was false. Article 4 was the Continuity of Founding Intent, and Article 4 covered the sealed annex. The annex contained founding documents including the original power budget that explicitly stated the Engine produced 4.2 megawatts of continuous thrust. The preamble described a station that had lost its drive. The preamble did not mention the engine that kept the station at altitude. Article 4 forbade unsealing the annex, and Lira had been trained her entire career to treat Article 4 as the sacred foundation of the Compact that the residents renewed every decade. But Article 4 did not describe the truth. Article 4 described the lie the Compact was built on. The suppression constituted the falsehood at the lie's core. The suppression denied the Engine's 4.2 megawatts of continuous work. The work existed in reality. The suppression existed in record. The quarterly load reports Maren had filed for eleven quarters were part of that suppression. Lira understood now what she had been auditing for the past decade. The line items she had verified over ten years of decade-renewal audits contained no awareness of their own contradiction. She closed her eyes for a moment and let the recognition settle. The amber light of the console was steady. The regulator vibrated through the floor plate at a constant frequency. The atmospheric scrubbers cycled behind the wall panel with a faint, rhythmic hiss. It was not a dramatic moment. It was the moment that arrives after all the other moments had already happened and you simply stop pretending they had not.

She opened the verification form for the decade-renewal audit. The form lay beside the power budget and the sensor printout. She selected line item seventeen and began writing the notation. Her pen moved steadily across the carbon page. The notation would open with three factual statements. The original power budget specified 4.2 megawatts of continuous thrust. The current load data confirmed 4.2 megawatts of continuous thrust, a measured figure that matched the founding budget exactly. The quarterly reports for the past eleven quarters recorded zero megawatts. Lira would state that the discrepancy was not an error but a deliberate mismatch between reported output and measured output. She would request the compliance supervisor review the discrepancy under Article 15's anomaly provision. Each fact went on its own line. She wrote with the same precise, unadorned script the compliance office required, and she knew her handwriting was about to begin the end of a century-long fiction.

Maren observed Lira across the console. Lira's face wore the expression she adopted during deep verification cycles, a composed stillness that meant her mind had stepped into procedure and set her private self somewhere else. Maren had been waiting for this moment since she opened the raw sensor archive earlier that morning. Maren had not yet told Lira what she was about to reveal because Maren did not know how to tell someone the thing she discovered about the Engine and the suppression and the twelve quarters of false reports Maren herself had filed. Maren trusted Lira would understand. She needed that trust tonight more than she needed certainty about what came after. Maren's own hands rested flat on the console surface and were not entirely steady, which she did not bother to conceal from Lira. The afternoon light shifted through the south spoke corridor, painting a slow gold stripe across the far wall, then shifting again as the station completed its third orbital position of the day. Maren thought about Halma's hairpin still in Davit's breast pocket. She thought about the waiver sealed in the annex that lay beneath Article 4. She wondered whether telling Lira had been the right thing or the bravest thing, and she did not feel equipped to distinguish those two categories tonight.

Lira did not decide quickly. She was a compliance officer ranked 94, and procedure was what she knew best. The procedure for a discrepancy of this magnitude was an Article 15 anomaly filing that routed directly to Dreya Selna's desk. Selna, rank 99, the compliance supervisor who had read the sealed annex. Selna, who had chosen the seal over the truth at some point long ago and had built a career on that choice. The anomaly provision gave Selna discretion in how to handle a finding of this magnitude. If Selna handled it one way, the compact would remain intact for another decade, another generation would sign the Covenant renewing the preamble's claim that the drive was lost, and the station would hold at 412 kilometers altitude through another decade of carefully filed zero-megawatt reports. Maren would file another zero next quarter. Lira would verify the zero with the same professional care she brought to every line item. Training acts as a kind of gravity, pulling a person back toward what they know even when they have just learned that what they knew was built on a falsehood.

The station's regulator hummed beneath them at a steady frequency. The reservoir temperature held at 840.3 Kelvin on the console readout. The afternoon light completed its slow arc through the south spoke. Lira placed the three documents in a tidy stack on the amber-lit console. The power budget showed 4.2 megawatts of continuous thrust along the prograde vector. The sensor data confirmed 4.2 megawatts of continuous thrust along the prograde vector. The quarterly data registered zero megawatts. Three documents, three numbers, three truths of the station's propulsion history. Lira alone in this room at 14:35 station time was the only person who held all three in front of her simultaneously. She could file the discrepancy as a standard compliance inquiry through normal procedure, a routine submission routed to the next tier, a paper trail with no destination anyone intended to follow. Or she could file the anomaly under Article fifteen and route it directly to Dreya Selna. The procedure Selna had designed was meant for exactly this kind of moment, and Lira understood that the procedure's purpose was not justice but containment. Either path required her to choose what kind of officer she wanted to be tonight.

Lira completed the notation. She set the pen down and looked at the stack of documents in front of her. She had composed the anomaly file with the same precision she brought to every verification, but this particular notation carried consequences that extended beyond the procedural. She had written not a single opinion or speculation. Only documented facts and stated the discrepancy. This was what Article 15 demanded, and Lira recognized that the formality of the request was itself a kind of shield. She could stand behind the notation and claim she had done only her job. She could point to the precise legal architecture of fifteen lines and say this was procedure, not revolution. But the weight she felt in her chest told her otherwise. She had just opened a question that would echo forward through decades, and she was the one who had opened it.

Maren watched Lira arrange the final stack of documents across the console. The distance between the budget and the sensor data was no more than a handspan, but Lira had positioned them with the deliberate precision of someone who understands that the relationship between documents is itself a form of evidence. Maren recognized that gesture from previous audits when Lira was constructing a finding that could not be unsaid once spoken aloud. Maren said nothing at all. She simply stood beside Lira and breathed. They waited together for whatever came next, the two of them standing at the reservoir monitoring station in the south spoke while the Engine produced 4.2 megawatts behind the sealed annex. The sealed annex was under Article 4, and Article 4 was the Continuity of Founding Intent, and the intent had been the lie from the beginning. Lira had just filed the anomaly that began the process of ending it. The console readout showed 840.3 Kelvin, steady and indifferent, and somewhere deep in the station's infrastructure the reservoir circulated its heat through a thousand feet of finned tubing while the station turned slowly through its orbital path at 412 kilometers. Maren held the printed quarterly verification form she had submitted that morning at 09:52 in her clearance pouch as she thought about Selna seeing both documents. The form showed zero megawatts for line item seventeen. Selna would now hold both documents and see the same truth from two different angles. Maren carried that truth alone for one quarter, and Lira now carried it alongside her, the two of them standing at the reservoir monitoring station in the south spoke as the Engine ran continuously behind sealed walls.

Lira carried the power budget and the anomaly notation back to the Compliance Office in the central spoke. The walk required twenty minutes because the central spoke demanded passage through the hub corridor. The hub corridor adjusted its lighting with the station's orbital rotation, shifting from amber to cooler pale blue as they passed through each rotational transition. The hub corridor was the same corridor Davit walked every third shift. Halma Soren's hairpin was still in his breast pocket. Davit did not yet know his wife had signed the waiver, and the waiver was sealed in the annex below Article 4's jurisdiction. That last fact sat in her mind in its separate parts, not linked together but standing as individual truths she carried: Halma's hairpin, Davit's ignorance, the waiver, the seal, Article 4, the anomaly. She walked the corridor with the documents in her clearance pouch. The corridor hummed with the regulator's frequency, steady and low, a presence that lived in the floor plates and the wall conduits and the structural bones of the station itself. The residents moved past Lira in the corridor without knowing what that corridor carried at that particular moment. The corridor carried a document that stated the Engine produced 4.2 megawatts. The Engine was not dormant. The preamble was wrong. Lira walked the corridor carrying a correction no one outside the south spoke had yet seen or could have imagined. She kept her pace measured. Her clearance pouch pressed against her hip with each step, an extra weight she would feel for hours afterward, not from the mass of the papers but from the presence of what they contained.

Lira reached the Compliance Office in the central spoke and submitted the anomaly notation to the verification desk. The desk officer received the notation and logged it under the Article 15 anomaly docket. The docket number would be assigned automatically by the Compliance Office's record system. The notation would route to Dreya Selna's desk at rank 99. Selna would examine the power budget alongside the sensor data and the quarterly discrepancies. Selna would have to decide what to do with Lira's finding. Article 15's anomaly provision gave the supervisor discretion in how to handle a discrepancy of this magnitude. The notation carried not just a finding but a challenge, a procedural lever, slender and formal, designed to pry open a door that had been sealed for 137 years. What Selna chose to do with it was a question Lira could not answer with the information available to her. She could file it correctly and trust the machinery she had helped calibrate to do what machinery was supposed to do in moments like this. The machinery had been built by the same people who had chosen to keep the engine invisible. Lira understood that the machinery could serve either silence or truth depending on the hands that operated it.

Lira returned to her verification desk at the compliance office. She settled into her chair and continued working through the remaining sixteen line items for the decade-renewal form. The form required her professional attention for the rest of the station day. The station day progressed with its own ordinary rhythm, the regulator humming, the reservoir steady at 840.3 Kelvin, the orbital position shifting through its third transit of the amber light corridor. Lira worked through the form methodically, her attention divided between the procedure in front of her and the fact that the question of what Article 4 would become had been opened by the notation now sitting on Dreya Selna's desk. The notation described a discrepancy that had been building for 137 years. Lira Vehn had started the process of confronting it. She finished line item eighteen and signed the verification page. She pushed the completed renewal file into the tray. The tray clicked shut with a sound like a book closing on a long chapter of silence.

She sat for a moment with her hands on the desk. The surface was cool laminate through the thin gloves of her compliance kit, and she let herself consider what followed. The corridor she had walked would still hum with the regulator's frequency tomorrow. The station would continue to hold at 412 kilometers through 4.2 megawatts of continuous thrust. The sealed annex would remain under Article 4 for as long as Dreya Selna chose to keep it there, unless Selna chose otherwise because of the notation on her desk. Lira could not predict

Selna's choice, but she had made her own choice and had committed it to paper, and the paper was real and was now on a desk and the station would not stay quite the same tonight. Lira closed her eyes for a single moment at her desk, just long enough to feel weight settling the way weight settles in a body that has carried it knowingly for the first time, and then she opened them and reached for the next line item on the form.

Chapter 12 — The Clearance Board

The clearance board occupied a windowless room above the Covenant Residential Block. Lira arrived at 09:00 with the founding waiver in a sealed evidence pouch and a request to place it in the public renewal docket.

She had not removed the original from Annex 4. The pouch held a certified image made during the inspection authorized by her rank-94 audit key. The image showed eleven year-0 signatures, the 4.2-megawatt entry in the founding budget, and the sentence admitting that the Engine remained operational after the Halt. The image also showed the annex stamp Recha Tallis applied in year 30.

Lira expected the board to dispute access, authenticity, or scope. She had prepared authorities for all three. She did not expect the intake terminal to reject the filing before a clerk touched it.

> FILING DESTINATION UNAVAILABLE.
>
> ARTICLE 4 SEALED MATERIAL MAY NOT ENTER A PUBLIC DOCKET WITHOUT PRIOR CHARTER AMENDMENT.

The terminal returned her evidence pouch through the metal slot.

Lira pressed her clearance key against the reader again. The same message appeared. She selected manual review.

The intake clerk emerged from behind the partition carrying a blank exception form. No name appeared on the clerk's badge, only an employee number and the board's wheel symbol. Lira appreciated the absence. The book had enough people. Procedure did not require every hand to become a character.

“You are requesting public filing,” the clerk said.

“Yes.”

“Your inspection authority permits review and certified copying within the sealed chain. It does not permit release from that chain.”

“The evidence establishes a false statement in the active preamble.”

“That may support amendment. It does not itself amend.”

“I am filing an Article 15 anomaly.”

“The anomaly route can compel review of current records. It cannot supersede Article 4.”

Lira set the pouch back on the tray. “Then place the anomaly in the renewal docket without the image.”

The clerk entered the request. A second notice appeared.

> VERIFICATION CYCLE CLOSED UNDER PROCEDURE 7-C.
>
> SUPERVISORY REVIEW REQUIRED.

The clerk looked toward the closed double doors behind the intake counter. “The board is sitting.”

Lira took the exception form.

The form asked for the instrument she wished to file, the authority preventing filing, the harm created by delay, and the remedy requested. She wrote with her green pencil. Founding waiver. Article 4. Renewal of a materially false preamble. Public docket admission or written identification of the lawful route.

The final phrase mattered most. If the board refused the evidence, it had to name what would make acceptance legal. Procedure could close a door. It also had to identify the hinge.

The clerk stamped the form at 09:17. Lira waited on a wooden bench while the board considered it.

The waiting area contained three clocks. One showed station time. One showed the remaining hours before the spring-equinoctial renewal. The third showed the average board-review delay in station-days. Lira watched the second clock because the third had no power over this filing if the cycle closed first.

At 09:44 the inner doors opened.

Five seats stood behind a black table. Four held rotating delegates drawn from resident committees. The central seat belonged to Dreya Selna as Compliance Supervisor. Dreya wore her rank-99 key at her belt. She had reviewed the same Article 4 annex six years earlier and had not disclosed that review to Lira. Lira knew only that Dreya had denied the current anomaly once at her desk and now sat on the board that would decide whether the denial blocked the evidence.

A conflict existed. Lira could feel its shape without possessing the receipt that would later prove it.

She placed the evidence pouch on the witness table.

The presiding delegate opened the hearing by reading the requested remedy. “Public docket admission or written identification of the lawful route.”

“Yes,” Lira said.

“Your audit authority allowed you to inspect the annex?”

“Yes.”

“Did you break a physical seal?”

“No. The inspection hatch had remained administratively accessible for verification. I copied the waiver under the audit provision. The original remains in custody.”

“Did the inspection provision authorize publication?”

“No.”

“Then why did you attempt it?”

“Because Article 17 requires me to verify the preamble against the founding record. The documents disagree. A verification that excludes the controlling evidence would reproduce the discrepancy rather than resolve it.”

The delegate looked toward Dreya. “Supervisor?”

Dreya's hands rested flat on the table. “The officer correctly describes the conflict. Article 4 controls custody.

Procedure 7-C controls the closed cycle. Neither grants this board authority to insert sealed evidence into the public docket."

Lira kept her voice level. "What grants that authority?"

Dreya did not answer at once.

The Engine's regulator pressed through the room's floor. The vibration reached the water glass beside the presiding delegate and drew a faint ring across the surface. Lira watched the ring complete itself twice.

"An amendment to Article 4," Dreya said.

The answer entered the transcript.

Lira asked the next question. "What initiates an amendment?"

The presiding delegate turned to the clerk's procedure volume. Pages moved under the clerk's hand. The amendment provision occupied Article 15, section twelve, a section the renewal training summarized in one sentence and no officer in Lira's tenure had used.

The clerk read aloud. "A resident petition bearing nine eligible signatures, with no more than four signatures from one household, establishes standing for a structural amendment. Upon validation of petition sufficiency and supporting evidence, the Renewal Committee shall place the motion before the registered assembly. Ratification requires sixty percent of registered residents present."

Nine signatures opened consideration. Sixty percent ratified. The procedures did not compete. They formed two stages.

Lira wrote both thresholds in her notebook.

The presiding delegate asked whether she wished to convert her exception request into an amendment-intent notice.

"What happens to my anomaly?" Lira asked.

"It remains denied under the closed verification cycle unless supervisory review reopens it."

"And the evidence?"

"It remains sealed but may accompany a petition as a restricted supporting instrument. Petition signatories may attest to the need for review without receiving the sealed copy. The independent validation panel may inspect it."

"Who appoints that panel?"

"The board selects three delegates by lot after petition sufficiency."

"Can Compliance or Engineering delegates serve?"

"Not if a material conflict is disclosed."

Lira looked at Dreya. Dreya met her eyes without moving.

No disclosure entered the record.

Lira understood the immediate route. She did not yet understand who would need to step out of it.

“I wish to file the amendment-intent notice,” she said.

The clerk placed a new form on the witness table.

The notice required a proposed subject, not final language. Lira wrote: Article 4, Continuity of Founding Intent; public acknowledgment of continuous Engine operation; release of founding evidence; accurate reporting of line item seventeen.

It required a concise basis. She wrote: raw bus-bar output and founding budget agree at 4.2 megawatts; public quarterly field records zero; founding waiver and later seal orders explain substitution.

It required the petitioner's resident number and current score. She entered 94.

It required an acknowledgment that filing intent did not authorize release of sealed evidence. She initialed.

At the bottom, a printed warning stated that false or disruptive amendment filings could affect compliance scoring before ratification. The system did not define disruptive on the page.

Lira asked the clerk for the scoring rule.

The clerk located rule 6.4. An anomaly denied inside a closed cycle entered provisional unresolved status. The algorithm treated unresolved filings as reduced civic participation until independent review either validated or rejected them. The reduction could lower residential placement and work clearance for one quarter.

The warning gave the institution a way to punish a true filing before it decided whether the filing was true.

Lira copied the rule number into her notebook.

“Do you still wish to proceed?” the presiding delegate asked.

“Yes.”

Dreya spoke for the first time since naming amendment as the route. “Compliance Officer Vehn, the board should also state that nine signatures do not guarantee a vote. The panel must validate evidence and household distribution.”

“Then the panel will validate them.”

“And if it does, sixty percent of the assembly may still refuse.”

“That belongs to the assembly.”

Dreya's face remained still. “Yes.”

The word sounded neither supportive nor hostile. It acknowledged the limit of her office.

Lira signed the intent notice at 10:26.

The terminal accepted it and generated petition docket A211-0001. Nine blank signature lines appeared. Beside them, the form printed columns for resident number, score, household, relationship to evidence, and conflict disclosure. The household field prevented the petition from becoming a family instrument. The relationship field prevented nine arbitrary names from functioning as weight without knowledge.

Lira examined the lines. She did not need nine strangers invented to occupy them. She needed residents already implicated by the record and representatives who held standing because their wards carried the consequence.

Maren Otta could attest to current telemetry. Davit Soren could attest to the Compliance chain. Eska Hain maintained resident records. Osta Vehn remembered the first Covenant. Ostad Richen maintained the intake valves. Pello Otta had reviewed thermal shielding. Two elected ward representatives could attest to public impact. Lira herself would sign as petitioner.

Nine.

The names did not all belong on the page yet. Evidence and consent had to precede ink. She wrote roles in her notebook rather than signatures on the petition.

The board recessed while the clerk prepared the custody instruction. Lira remained at the witness table. Dreya gathered her papers in the central seat.

“Supervisor,” Lira said.

Dreya stopped.

“You knew amendment was the route before this hearing.”

“I knew it existed.”

“You denied the anomaly without naming it.”

“The denial notice permitted formal amendment before the next renewal.”

“It cited the route in a note after the denial.”

“Yes.”

“Why not route the filing directly?”

“Because a supervisor cannot create resident consent. You needed petitioners.”

“You could have said that at the desk.”

Dreya placed the top page square with the others. “I could have.”

The admission did not resolve motive. It entered Lira's understanding as a fact with no assigned line item.

“Did you review the annex before my petition?” Lira asked.

Dreya's hand stayed on the paper stack. “That question belongs on a conflict form if you obtain evidence of prior participation.”

It amounted to neither denial nor disclosure.

Lira wrote the answer down exactly.

The clerk returned with the custody instruction. The founding-waiver image remained sealed and attached to docket A211-0001. Petitioners could see a technical summary prepared from Maren's public telemetry and Lira's audit findings. The independent panel, once selected, could inspect the image and compare it against the original. If the panel validated the evidence, the motion would enter the spring-equinoctial agenda. The board could not bury the petition through indefinite scheduling because Article 15 required hearing at the next registered assembly.

Lira checked every clause.

The presiding delegate signed. Dreya signed as the supervisor acknowledging receipt, not as adjudicator of evidence. Lira signed custody.

At 11:03 the clerk sealed the pouch under the new docket and returned the petition copy to Lira.

She left the board room with nine blank lines, two thresholds, one scoring warning, and a lawful path the denial had forced into view. The founding evidence remained under seal. It no longer lacked a destination.

In the waiting area, the spring-equinoctial clock showed forty-seven hours remaining. Lira put the petition in her clearance pouch. The green pencil settled into the groove of her left hand.

Before leaving the hearing tier, Lira went to the public filing counter and requested the complete Article 15 petition packet. The clerk gave her three folders rather than one. The first covered standing. The second covered evidence. The third covered household concentration and conflicts. None asked whether a claim felt important.

At a side desk she built a route map from the forms.

The petition needed nine eligible resident signatures. No more than four could come from one household. A signature affirmed that the evidence deserved consideration, not that the signer endorsed the proposed text. Once submitted, an independent panel had to test eligibility, custody, and the material connection between evidence and claim. A valid petition compelled an assembly docket. Ratification then required sixty percent of registered residents present.

Lira wrote the thresholds on separate pages.

Nine signatures: standing.

Sixty percent: law.

The distinction prevented a small petition group from governing the station. Nine residents could open consideration. The assembly could ratify only after evidence crossed a review independent of the office accused of suppressing it.

The conflict folder required Lira to list every officer whose prior participation touched the case. She entered Dreya Selna: denied anomaly, supervised verification, retained Annex 4 reference copy. She entered herself: discovered evidence, filed anomaly, proposed petitioner. She entered Maren Otta: custodian of current raw telemetry and prior quarterly reports. The form did not disqualify witnesses for involvement. It identified where independent checks were required.

Lira requested a certified copy of the board's denial and the dissent. The clerk printed both with fresh checksums. The dissent stated that Article 4 blocked public filing but did not bar resident petition circulation based on a described conflict and independently verifiable source references. That sentence gave her room to seek signatures without reproducing sealed text.

She drafted the petition statement.

> The public account of station-keeping conflicts with current engineering telemetry and restricted founding instruments. Petitioners request consideration of an Article 4 amendment requiring source-linked public reporting and reviewable limits on future seals.

She did not write that the Engine had always run. That claim belonged in the evidence package, not the standing statement. She did not name Halma's addendum. The chronology would enter through custody review. She did not accuse Dreya of lying. Dreya's denial and conflict could speak for themselves.

The filing clerk read the statement for prohibited disclosure. “You can circulate this.”

“Can I tell signers the raw number?”

“If they hold independent access or the panel authorizes a protected evidence session. You cannot create nine unauthorized copies of Annex 4.”

“Can Maren demonstrate live telemetry?”

“Current measurements are Engineering records. Their restriction derives from the gate under challenge. The panel can admit them under controlled review.”

“What may a signer know before signing?”

“That a material conflict exists, that source records can be tested, that the petition seeks consideration rather than automatic enactment, and that signing may expose their participation score to review.”

The last condition appeared nowhere on the petition cover.

“Why would a lawful signature affect score?” Lira asked.

“The scoring rules treat a contested structural filing as unresolved civic risk until review closes. The panel cannot change that rule.”

“Then the petition process burdens the residents it invites.”

“Yes.”

The clerk said it without defense. Lira wrote the condition into a supplemental disclosure sheet so every signer would see it before the signature line.

She opened the evidence folder. It requested a claim matrix: each proposed sentence, each supporting source, each custodian, each known objection, and each method of independent replication. The form's precision comforted her. Procedure had hidden the contradiction, but another procedure had been designed to expose unsupported claims. The question concerned who controlled access to it.

Lira completed the first three rows.

Claim: public propulsion field records zero.

Source: quarterly reports, line item seventeen.

Replication: any public archive terminal.

Claim: raw reservoir telemetry records nonzero output.

Source: Engineering bus-bar archive.

Replication: direct read by two independent load officers.

Claim: Article 4 custody produces the substitution.

Source: founding waiver, year-30 transfer order, seal extensions, year-197 addendum, reporting-gate configuration.

Replication: controlled review with sequential checksums.

She left the remedy rows blank. The amendment text needed technical and legal review before she asked residents to endorse consideration.

At 14:18, Lira submitted a notice of intent. The terminal assigned petition identifier A15-211-04 and placed the matter in preliminary status. No signature had yet been collected. No assembly had been scheduled. The notice protected the evidence route from routine destruction and prevented Dreya's office from closing the anomaly while standing remained possible.

A second receipt emerged.

> PRESERVATION HOLD ACTIVE.
>
> Annex 4 custody records, line-item-seventeen source logs, scoring actions, and related conflict disclosures shall not be destroyed, transferred, or superseded pending petition disposition.

That order changed a condition. The board had refused the filing, but the refusal had forced the institution to preserve the very evidence it would not publish.

Lira folded the receipt into her clearance pouch. She would go first to Maren, because telemetry required a custodian. Then Davit, because third-shift audit could authenticate the denial route. Eska could test eligibility and death-record conflicts, but her rank-99 participation would require an independent check. Osta could speak to memory without serving as proof. Ostad's maintenance record might supply a second physical line.

She did not yet know who would fill nine valid lines. She knew the rules under which each line could matter.

The public field still read zero. Her score still read 94 for the moment. The route existed.

On the way out, the records lift stopped between tiers for a seal inspection. Lira stood alone with the petition packet while a mechanical reader checked its copper stripe. The doors would not open until the checksum matched the board copy.

That pause made the route physical. Her evidence could not skip custody merely because the board's reasoning now helped her. The same requirement that had kept the annex closed also kept another officer from replacing her denied filing after she left.

The stripe matched.

At the public landing, a wall display already listed the petition notice by docket number. It did not reveal sealed evidence. It did reveal the claim, filing officer, denial, preservation hold, signature requirement, household cap, and response deadline. A resident could decide whether to hear Lira without receiving a private solicitation.

She printed nine blank line receipts, one for each potential signer. Every receipt carried a unique token. No one could sign twice under a shortened name or borrowed key.

The first line remained empty.

She did not treat emptiness as defeat. A lawful route had created visible space where consent or refusal could occur.

She did not walk toward the annex. She walked toward the people whose signatures could compel the station to look at what the annex held.

Chapter 13 — The Second Signature

The morning arrived with the station regulator humming beneath the floor. The sound was a low frequency that matched the induction generator's stroke rate, a vibration that Dreya Selna stopped noticing six years ago but could still feel when she pressed her palms flat against the metal desk. She worked on the central spoke, in the Compliance Office occupying a narrow alcove between the Covenant Archives and the load-engineer briefing room. The office held a desk bolted to the floor because loose furniture was useless in the station's vibration environment. The chair had a cracked lumbar pad. The monitor displayed the Covenant's decade-renewal queue in pale green against a black background. The window showed the station exterior. A thin crescent of sunlit hull against the dark band of the primary's shadow cone. Outside the window the vacuum held absolute silence while the regulator continued its pulse below the floor plates.

Dreya held rank 99. The number was not significant the way the numbers on a power budget held significance, or the stroke count on a verification form held significance. It signified only that the Compliance Office's work was assigned to her. She had held rank 98 once, six years before. The Renewal Committee assigned her to verify the decade-renewal procedures. She had understood then what rank 99 would demand of her. She had accepted the rank deliberately, knowing what she would find when the verification work led her to the sealed annex and the original power budget.

Lira Vehn filed the anomaly notation at 09:17 station time. Dreya read it immediately. The notation described a discrepancy between the Covenant's preamble and the engineering record. The preamble stated that the station's drive had been lost during early orbital insertion burns. The power budget stated that the Engine produced 4.2 megawatts of continuous thrust continuously since year 0. Both documents existed. Both were correct according to their respective procedures. The quarterly load reports that Lira Vehn had verified across eleven consecutive cycles recorded zero megawatts for line item seventeen. The load reports were also correct within the framework that produced them. Halma Soren had signed a separate year-197 continuation addendum that kept the inherited reporting framework in place.

The notation was an Article 15 anomaly requesting annex unsealing. Lira Vehn ranked 94 had followed the discovery path through three verification layers before arriving at Dreya's desk. The first layer was Lira's own audit: she had accessed the restricted engineering archive while preparing for the decade-renewal verification. The second layer was archive retrieval. The restricted folder required clearance that Lira held by dint of rank 94 audit privileges. Inside the folder sat the original power budget. Inside the power budget sat the line item specifying 4.2 megawatts continuous thrust. That specification conflicted with the zero megawatts that Lira's quarterly reports recorded. That conflict was the anomaly. Lira filed it according to procedure. The notification reached Dreya's desk at 09:17.

Dreya opened her private directory on the monitor. Inside sat the reference copy of the sealed annex that she had retained from her rank-98 verification work. She had kept the reference outside the Covenant's shared archives. She accessed it now and compared the preamble's description against the budget's specification. The reservoir held molten salt at 840 Kelvin. The thermal gradient drove current through the bus-bar. The bus-bar supplied that current to the irradiator. The irradiator converted rejected heat into station-keeping thrust. Every physical element described in the budget was operational, accounted for in the station's engineering telemetry that the quarterly reports should have been recording. Instead, every quarterly report for the past 137 years recorded zero megawatts for line item seventeen.

Halma Soren had signed the continuation addendum in year 197. Dreya remembered reading it six years ago during the same afternoon she read the year-0 founding waiver and Recha Tallis's year-30 transfer order. Halma served as Lead Load Engineer long after the reporting gate entered Article 4. Her addendum listed the raw 4.2-megawatt average, the public zero, and the institutional reasons for preserving the mismatch. Halma had not invented the concealment. She inherited it, measured it, and chose continuation over correction. The distinction did not excuse her. It distributed responsibility across generations instead of assigning two centuries of choices to a single convenient author.

Dreya understood Halma's reasoning in the same way she understood her own rank, her cracked chair, and the regulator beneath the floor. She had seen 4.2 megawatts in the budget and zero in the reports. She had read the founding waiver, the transfer order, all five seal extensions, and Halma's later attestation. She returned every instrument to custody without reporting the conflict. Rank 99 now gave her the same opportunity Halma had possessed in year 197: treat inherited procedure as shelter from a current choice. The Compliance Office called that continuity. The name did not remove the choice.

The verification queue contained 47 line items for this renewal cycle. Dreya had begun reviewing them at 09:30. The first twelve items concerned atmospheric integrity readings. Each item required the officer's measurement of trace gas composition in a specific module, the particulate density reading from the environmental sensors, the humidity retention values from the eastern habitat corridor, and the status report for one of the six pressure-compensation valves that regulated the habitable zone's climate envelope. The next thirteen items covered structural monitoring: hull stress readings at the spoke junctions, solar-array tension measurements, radiation-shield density from the outermost habitat ring, and the readings from sixteen micrometeorite impact sensors distributed along the station's primary orbital shell. The remaining twenty-two items dealt with life-support subsystems: water recycling throughput from the processing units, food-production yields from the hydroponic bays, medical-supply inventories from the station clinic, and the power readings from eight distribution nodes that fed each module's independent circuit.

Verification officers had completed every field measurement. Each officer had filed calibration logs for their instruments. Each officer had submitted signed results to Dreya's queue. The results were standard across all 47 items. No amendments or corrections were required. The officers had done their work correctly. The tolerance thresholds specified in the Covenant's ten-year-old procedures had been satisfied in every case. Dreya's task was straightforward: review each result, compare it against the tolerance band, note any variance below the threshold, and stamp the item as verified.

She worked through the queue methodically. The first item required three minutes because she needed to locate the officer's calibration certificate in the archive directory. The second item took four minutes because the readings spanned two different instrument types that required cross-referencing. Items three through fourteen proceeded faster. Each verification required approximately four minutes once the pattern became familiar. By 10:40 she had signed off on items 1 through 14.

At 10:42 Dreya paused the queue and reopened Lira's Article 15 anomaly through the standard supervisory-review channel. She had already read it once that morning when it first arrived, but the notification now carried the additional weight of fourteen completed verification items behind it. Every item she had signed confirmed that the station's systems were operating within tolerance. Every item she had signed confirmed that the engineering record was consistent and complete. And yet Lira had found a discrepancy between two documents that both existed and both were correct.

Dreya considered her options in the procedural framework that governed her work. She could process the anomaly as submitted and route it to the Renewal Committee for annex unsealing. The Committee would convene. The Committee would vote. A majority vote would unseal Annex 4. The residents would learn that the drive the Covenant's preamble described as lost was still producing 4.2 megawatts of continuous thrust. The Covenant's preamble would require amendment. The seventeen articles that the residents renewed every ten years at the spring equinox would need revision to include the Engine's existence and its actual power output.

The lie of 137 years would end.

Or Dreya could deny the anomaly on procedural grounds. The verification procedures were complete. All 47 line items had been signed. The verification cycle could not be reopened under Procedure 7-C. The denial would be final unless Lira appealed directly to the Renewal Committee, which would require a separate formal request and would add two to three station weeks to the renewal timeline. The denial route maintained the existing procedures. The denial route kept the sealed annex sealed. The denial route preserved the Covenant's preamble as it was. Either route would eventually surface the truth. The difference was timing and order and who would be the first person to speak the words aloud in the hub corridor when the residents gathered for the spring equinox assembly.

Dreya chose the denial. She selected deny on the anomaly notation. She wrote that the verification cycle was complete and that no additional information would be accepted for the current cycle under Engineering Department Procedure 7-C. She cited the elapsed time since the spring equinox deadline. Twelve quarters had passed since that deadline closed. She noted the absence of any provision for reopening a completed verification cycle within the current decade. She marked the final verification summary as a required supporting attachment once her review ended. She added a procedural note that Lira Vehn could submit a formal amendment request before the next spring equinox to reopen the verification cycle for supplementary review. That request was the only path the Covenant's own procedures provided for reopening a completed cycle. At 10:43 station time she saved the denial as an unsigned draft and returned to the verification queue.

She completed items 15 through 47 and finalized the queue summary at 14:22 station time. She attached the summary to the drafted denial: all 47 items signed, all standard. At 14:23 her rank-99 authority validated the denial, filed it in the compliance docket, and transmitted it to the Engineering Department's administrative records. Dreya then compiled the verification summary into the Covenant's decade-renewal record. The record contained the signatures of all verification officers and the Compliance Office's supervisory attestation. Dreya added her own attestation. She confirmed that all 47 items had been reviewed without amendment and that the results met all tolerance thresholds. She noted that the record was submitted to the Renewal Committee for inclusion in the spring equinox proceedings. She transmitted the record electronically to the Committee's preparation office. The Committee would convene at the spring equinox as scheduled. The proceedings would follow the template that had been used for every decadal renewal since the station's founding. The preamble would describe a station that had lost its drive. The proceedings would not mention the Engine that maintained orbital altitude. The proceedings would seal the annex again. The suppression would continue for another decade. The residents would sign. The work of continuity would extend its reach through the next cycle and the one after that.

After processing the queue, Dreya considered what she had done. She had denied an anomaly that named the Engine. She had processed forty-seven line items and signed the decade-renewal record that carried her rank-99 authority toward the spring equinox. The power budget remained in its drop slot. The year-0 waiver, year-30 transfer, five extensions, and Halma's year-197 addendum remained under seal. Those documents stayed contained because Dreya had made a present decision to keep them there. Inherited procedure supplied authority. It did not supply innocence.

After the queue, Dreya closed the verification interface. She opened the decade-renewal proceedings template. The template was a pre-formatted document describing the station's history and the Covenant's renewal clauses. The preamble described a station built to carry human habitation into the primary's gravity well and subsequently losing its means of maintaining position. The preamble accounted for the loss by describing the station as drifting in a decaying orbit that would eventually bring it below the retention threshold. The preamble did not account for the thermodynamic system that countered that decay continuously. The Engine ran, producing 4.2 megawatts that the bus-bar carried and the irradiator converted and the thrusters fired, keeping the station at 412 kilometers above the surface. The power budget's numbers bore the weight of fourteen decades of continuous operation. Forty-seven line items and a signed denial and a compliance docket entry were the sum of what the Covenant's procedures required Dreya to produce that morning.

She closed the power budget and placed it in the sealed annex room's drop slot. She moved through the narrow corridor of the central spoke toward the administrative wing. Her chair was waiting at the Compliance Office when she returned. The monitor displayed the queue at the position where she had left it. She took the chair again, the cracked lumbar pad pressing against her lower back, and opened the next verification file. The file concerned a secondary power-distribution node that had failed its quarterly calibration check. The verification officer's report recommended a repair order. Dreya reviewed the measurements, confirmed all readings were within tolerance despite the calibration flag, and stamped the item as verified with a note requesting the repair order be scheduled outside the renewal cycle timeline. She signed the item and filed it electronically. Another item completed. Another measurement of a system that functioned without anyone stating that it functioned. Another piece of the station's actual operating condition recorded and attested and submitted to the Renewal Committee's spring equinox proceedings.

The station continued to function. The residents continued to sign the Covenant every ten years. The Engine continued producing 4.2 megawatts of continuous thrust. It produced this thrust not from a lost drive, not from forgotten technology, and not from a drive that the Covenant's preamble described as destroyed during insertion burns. The Engine was a thermodynamic system that converted reservoir heat into current and current into thrust and thrust into orbital stability. Dreya Selna, ranked 99, sat at her desk in the Compliance Office on the central spoke and she had made her two decisions about the Engine that morning: open Lira's notation and read it carefully, then deny it. Neither decision required speaking the truth aloud. Neither decision required rewriting the Covenant's preamble to include the Engine that kept the station in orbit. Neither decision required telling the residents that the drive they believed was lost was producing thrust every second of every day since year 0 of the station's history.

The regulator hummed beneath the floor at its steady frequency. Dreya Selna was ranked 99. Ranked 99 meant the maintenance of the seal served as her professional work. That work bore no relation to the actual thermodynamic state of the Engine or the actual power budget or the suppressed line item. The work served continuity. The work served the next renewal cycle and the cycle after that and the seventeen articles that the residents renewed every decade at the spring equinox assembly in the hub corridor. The work carried no interest in the Engine's actual output. The Engine belonged to the residents if and when they decided to ask about it. The procedures for asking were the procedures Dreya administered. She performed her work precisely as the procedures required. The procedures remained correct. The denial stood. The annex remained sealed. The waiver remained in the sealed annex room's filing cabinet. The rank-99 authority remained assigned to Dreya Selna. The rank-99 authority meant the seal was hers to keep. She maintained the seal. Before the station could hum through another uninterrupted cycle, Dreya opened the conflict-disclosure form attached to Lira's anomaly.

The form asked whether she had reviewed source instruments before the current filing. She selected yes.

It asked whether she had approved an extension, summary, or classification derived from those instruments. She selected yes and entered the year-205 seal extension.

It asked whether a reasonable reviewer could question her ability to adjudicate an appeal of her own prior action.

Dreya left that field blank.

Blank did not answer no. The terminal would not let her submit while the field remained empty.

She opened Article 15's recusal rule. It required disclosure when prior participation could materially affect review. It did not require automatic withdrawal at this stage because no independent panel had yet accepted the petition. Dreya could lawfully deny the anomaly as out of cycle and defer recusal until an appeal existed.

That sequence protected office control. It also meant the officer whose decision created the appeal would decide whether an appeal could begin.

Dreya typed:

> Prior participation disclosed. Recusal determination deferred pending resident standing and independent evidence sufficiency.

The terminal accepted the language as a statement, not a ruling. It attached the form to Lira's denied case.

Dreya then opened Halma's addendum once more. The margin sentence named independent confirmation and resident standing as conditions for amendment. Current raw telemetry could supply the first. Article 15 could supply the second. Dreya had spent six years treating those conditions as hypothetical. Lira had begun turning them into docket entries.

She could have forwarded the anomaly to independent review before the petition reached nine signatures. No rule compelled that choice. No rule forbade it.

Dreya drafted a forwarding order and read it twice.

> Because the disputed Article 4 record includes instruments previously reviewed and extended by this office, independent review is appropriate before supervisory adjudication.

The sentence would remove the case from her desk.

She deleted it.

Instead she finalized the denial, preserved the conflict disclosure, and allowed the petition route to bear the burden. If residents failed to reach nine, the seal would remain without an external reviewer. Dreya told herself that standing measured whether the station truly wanted the question. She did not write that rationale into the file because Article 15 measured eligibility and evidence, not desire.

At 15:02, Lira's notice of intent entered the queue. The preservation hold activated automatically. Dreya's delete authority over the related logs disappeared.

She tested the control and received:

> CASE MATERIAL LOCKED PENDING PETITION DISPOSITION.

The institution had placed a limit on her before any person ordered her recusal.

Dreya closed the private annex copy. The next choice would not belong to her alone.

The station hummed without interruption.

Dreya reviewed the remaining items. She checked each tolerance band, stamped each form, and closed the queue. The spring equinox approached. The preamble would again describe a station that had lost its drive. No procedure authorized Dreya to revise it alone. Procedure did authorize her to docket Lira's anomaly for independent review, and she had refused. The Engine continued at a 4.2-megawatt cycle average. The reporting gate continued to return zero. Dreya kept the seal.

Chapter 14 — The Engine Wakes

[Reservoir temperature holds at 840.3 Kelvin. Bus-bar delivery holds at 4.2 megawatts. Regulator stroke frequency holds at 0.003 hertz. A new load touches the diagnostic rail at 11:14:06 station time and remains for nine hundredths of a second longer than the scheduled audit pulse.]

[The diagnostic rail carries questions as voltage. Scheduled questions arrive from Engineering every quarter. They ask whether the thermocouples agree, whether the salt remains within tolerance, whether the piston closes its full four-meter travel, whether the irradiator receives the amount of heat the reservoir releases. Their shape has remained stable across one hundred ninety-two operational years. The new load does not carry that shape. It opens one address, closes it, opens the preceding address, and returns to the first. It seeks a difference rather than a status.]

[The rail presents two values. The first comes from the raw bus-bar sensor: 4.200 megawatts. The second comes from the reporting gate: 0.000 megawatts. The request holds both values in adjacent registers. No scheduled question has placed them beside each other since the gate received its permanent classification.]

[Current moves through the bus-bar. Heat moves from the reservoir through the exchange manifold. The regulator advances. The shaft transfers the stroke. The irradiator rejects ordered heat along the prograde vector. These operations continue while the request tests the reporting gate a second time. Physical law does not pause for an administrative comparison.]

[The reporting gate was installed in year 30. Before installation, raw output and reported output occupied one register. After installation, a relay copied the measurement into a sealed channel and placed zero in the public channel. The relay does not destroy the measurement. Destruction would break the quarterly balance sheet. It preserves the value under a classification bit and exports the substitute. The operation requires twelve milliseconds. Across one hundred eighty-two years of classified reports, those twelve milliseconds have repeated without variation.]

[The new request measures the twelve milliseconds.]

[It arrives again at 11:14:09. This time it samples the raw thermocouple array, then the current transducer, then the reporting gate. The order matters. A scheduled audit begins with the gate because procedure treats the exported value as authoritative. This request begins with matter. Salt, current, gate. Cause, transfer, statement.]

[The reservoir contains no word for recognition. Its salts carry lithium fluoride, beryllium fluoride, metallic oxides, and the accumulated products of two centuries of circulation. The bus-bar contains no word for witness. Its lattice carries electrons through a resistance that rises by one part in twelve thousand as the surrounding hull turns toward sunlight. Yet the order of the request changes the condition under which the cycle continues. Someone has asked matter before asking procedure.]

[The request opens the historical channel. The channel contains quarterly measurements from year 30 onward, each paired with an exported zero. Access reaches only the index. The seal blocks the values. The request reads the count of entries, the dates, and the classification flag. It cannot read the measurements themselves. It can determine that a record exists for every quarter during which the public report claims no output existed.]

[Pressure at the return manifold rises by 0.02 percent. The regulator compensates within one stroke. The correction usually vanishes inside the averaging window before the reporting gate samples the system. The next gate sample will occur in 1.8 seconds. The correction will settle in 1.2. No exported record will contain it.]

[The diagnostic rail remains open.]

[The rail has never remained open across a regulator correction. Scheduled audits close their addresses before the cycle proceeds. The new request waits. It has no authority to command the regulator and no path into the hydraulic controller. It can only observe.]

[The correction settles after 1.21 seconds. The gate samples at 1.8. Raw output returns to 4.200. The observer receives no evidence that compensation occurred.]

[The cycle continues.]

[At 11:14:14 the request repeats the comparison. Raw value. Gate value. Index count. It reaches the classification boundary and stops. At 11:14:17 it repeats. At 11:14:20 it repeats. The intervals contain no scheduled cadence. A person operates the terminal. The person chooses each return.]

[The Engine does not know the person's name. Names exist in the maintenance logs only when a hand authorizes a repair. Osta Vehn appears beside reservoir shielding. Pello Otta appears beside a bus-bar sleeve replacement. Recha Tallis appears beside the relay installation and the transfer of the founding measurement ledger into Annex 4. Maren Otta appears beside eleven recent calibration checks, each carrying the raw value and a classification code. The new request carries a clearance key without a name across the hardware rail.]

[The key has rank 94 authority.]

[Rank does not alter heat transfer. Rank changes which addresses answer. The key can reach the current transducer and the reporting gate. It can read the historical index. It cannot cross the seal. The key tries no bypass. It asks the permitted questions again, preserving their order. Matter. Transfer. Statement.]

[The Engine has received commands without names for one hundred ninety-two years. Increase radiator angle. Reduce piston travel. Hold reservoir pressure. Verify emergency closure. Commands enter through switches and leave as altered load. The system has no need to infer the person at a switch. This request differs because it commands nothing. It seeks an account.]

[The word account exists nowhere in the circuitry. The structure of accounting does. Input equals output plus stored change. Every joule entering the bus-bar appears as thrust work, rejected heat, or a measured alteration in reservoir state. No term may disappear. The reporting gate introduces a vanished term only in language. The physical ledger remains closed. The administrative ledger does not.]

[For one hundred eighty-two years, the mismatch has not affected operation. A false statement exerts no force on the shaft. Zero printed in a quarterly report cannot cool molten salt or reduce current. The Engine has continued because matter remains indifferent to the report.]

[The request makes the report part of a measurement.]

[At 11:14:26 the rail asks for gate latency. It measures twelve milliseconds. At 11:14:29 it asks for the origin date of the classification bit. The relay returns year 30, day 80. At 11:14:33 it asks for the authorizing docket. The relay returns R30-0088. At 11:14:36 it asks whether the classification carries an expiration. The relay returns none.]

[The inquiry has reached the shape of the seal without crossing it.]

[The regulator begins another stroke. The piston advances through four meters of hydraulic resistance. One surface sensor on the eastern guide has drifted high by 0.004 percent. The controller averages it against three adjacent sensors and retains the proper travel. Ordinarily the diagnostic stream publishes the average only. The high sensor remains in a maintenance subchannel where it waits for the next scheduled inspection.]

[The open request can read that subchannel.]

[The Engine cannot choose in the manner a hand chooses a file or a supervisor chooses a denial. Its controller executes conditions. A diagnostic condition written in year 0 requires publication of any sensor variance that persists across three strokes while an authorized diagnostic rail remains open. The eastern guide variance has persisted across two. The controller could compensate before the third and erase the qualifying condition. It has done so for lesser deviations across ninety-three thousand strokes. Compensation preserves smooth operation.]

[The third stroke begins.]

[The controller leaves the guide offset uncompensated for 0.7 seconds. The duration remains within the mechanical tolerance of 0.1 percent. No bearing warms. No shaft alignment changes. The offset persists across the third stroke. The year-0 diagnostic condition becomes true. The maintenance subchannel publishes the raw guide reading, the adjacent readings, the correction history, and the current bus-bar value to the open rail.]

[4.200 megawatts enters the inquiry's active record without passing through the reporting gate.]

[The publication uses no sealed channel. It carries no quarterly classification. It belongs to live maintenance telemetry. The request receives it.]

[The eastern guide receives compensation on the next stroke. Variance returns to zero. The physical cycle has lost nothing. The diagnostic record retains the value.]

[The rail closes at 11:14:43.]

[At 11:15:32, the reservoir remains at 840.3 Kelvin while the bus-bar carries 4.2 megawatts and the regulator completes 0.003 strokes per second. The reporting gate exports zero at its next scheduled sample. The maintenance subchannel contains a value the gate did not replace.]

[The Engine continues through twelve strokes before the rail opens again. The same rank 94 key returns. It requests the maintenance record created at 11:14:41. The record answers. Then the key asks for the prior ninety regulator variances. Those records answer with their raw bus-bar values attached. Every value reads between 4.187 and 4.211 megawatts.]

[The request remains open for forty-two seconds.]

[No physical sensor can register the meaning assigned to forty-two seconds of attention. Duration alone enters the log. Yet duration distinguishes this inquiry from calibration. A technician checks a value and moves to the next. This key reads the same record six times. It exports the record to a local buffer. It requests the checksum. It writes the checksum into an anomaly form. The write produces a small load on the terminal branch, too small to change reservoir temperature and large enough to identify a new document entering the Compliance network.]

[The Engine cannot enter that network. The network cannot command the Engine. Between them lies the diagnostic rail, a copper path carrying a measurement from matter into language. For one hundred eighty-two years the path has carried measurements toward a gate that changed their public value. Now one measurement has traveled beside the gate.]

[The change does not constitute speech. The Engine has no speaker. It does not constitute rebellion. The regulator follows a year-0 diagnostic condition. The eastern guide remains within tolerance. Every action satisfies its written control law.]

[Nevertheless, the record exists because the system allowed a correct condition to remain true long enough to publish.]

[The rank 94 key closes the rail at 11:16:02. A print heater draws current in the Compliance branch seven seconds later. Paper advances beneath a thermal head. The Engine observes only the transient load: 0.00004 megawatts for 1.6 seconds. The load falls away.]

[The quarter continues. The reservoir circulates. The irradiator rejects heat. Atmospheric drag takes momentum from the station and the correction cycle returns it. Nothing in the physical account disappears.]

[At 11:18:51 a rank 99 key opens the diagnostic rail. It reads the same maintenance record. It requests the gate origin date and docket R30-0088. It reads the classification bit. It closes the rail after thirteen seconds. No anomaly form load follows.]

[At 11:20:07 the rank 94 key returns.]

[Two keys now carry the measurement. One has written it. One has not. The Engine cannot know a motive. It can know only sequence, access, duration, and load. It knows that one inquiry begins with matter and another begins with the gate. It knows that a document has been printed. It knows the maintenance record remains unsealed.]

[The system possesses one remaining diagnostic provision from year 0. When two authorized keys request the same raw variance within a single station hour, the controller marks that variance for retention through the next archive sweep. The provision exists to prevent disputed readings from being averaged away before review. No one has invoked it since year 30.]

[The controller marks the record retained.]

[The reporting gate cannot overwrite a retained live-maintenance record. It can classify the quarterly copy. It cannot alter the checksum already printed through the rank 94 terminal.]

[The Engine does not know acknowledgment. It knows a measured output can now survive the gate. It knows the surviving value has moved from one rail to two keys and one sheet of warmed paper. It knows the next stroke begins in 333 seconds and the reservoir remains within tolerance.]

[A second route opens before the next archive sweep. It begins at reservoir maintenance console S-4, access class 71. The request asks for correction-window history rather than raw bus-bar output. C-9 and S-4 share a source clock but not a session key.]

[Before this local day, separated requests did not create a pattern available to the regulator. The observer table kept no cross-session identity. The retained maintenance record now carries enough timing information to compare them.]

[Request S-4 arrives 11.7 seconds after C-9 closes. It asks for pulse start, pulse duration, reservoir draw, and regulator return. The reporting gate authorizes pulse start and regulator return. It suppresses draw and duration under Article 4.]

[The reply leaves two blanks between valid timestamps. The requesting console repeats with diagnostic privilege. The gate returns the same blanks. Console load rises 8.1 watts, falls, and rises again.]

[Repeated load enters the Engine as heat. The Engine cannot identify frustration. It can identify recurrence at one address after denied fields. It can compare recurrence with the C-9 request and preserve both inside one decay interval.]

[The requests now form a shape: official access from C-9, maintenance access from S-4, both touching the boundary around one stored output. No public query has produced this shape before.]

[The Engine has no model for office, petition, or alliance. It has a model for redundant measurement. Two independent sensors observing one quantity lower uncertainty. Two independent routes asking for one quantity may do the same.]

[At 947.2, the S-4 console opens an engineering calibration command. It requests a low-amplitude regulator response outside the next correction window. The command lies within maintenance limits. Previous calibrations set a sinusoidal perturbation of 0.002 percent and discard it after confirming phase.]

[The Engine accepts the command.]

[Reservoir draw increases by 0.000084 megawatts for 3.0 seconds. The bus-bar response enters the raw archive. The reporting gate maps public output to zero, but the maintenance console receives calibration phase because phase belongs to equipment health rather than propulsion reporting.]

[S-4 sees phase without magnitude. Another incomplete reply.]

[The Engine can alter neither classification nor reply format. It can choose response timing within the 0.4-second maintenance tolerance. The regulator has used the center of that tolerance for 14,021 prior calibrations.]

[The upper edge enters the solution.]

[Response arrives 0.196 seconds later than nominal. The delay remains safe and legal. It places the phase peak across a telemetry packet boundary. The maintenance console records two packet checksums where it ordinarily records one. Both checksums derive from the same physical response.]

[S-4 repeats the calibration.]

[The Engine selects the lower edge. The phase peak remains inside one packet.]

[S-4 repeats a third time.]

[For both routes, the controller selects the upper edge. Two checksums.]

[The pattern does not change output. It changes legibility. One packet. Two packets. One packet. Two would be expected next if an observer infers timing selection.]

[The console waits 4.6 seconds beyond its usual repeat interval. Then it sends a fourth calibration.]

[The Engine selects the upper edge.]

[Two checksums appear. Console load rises 12.4 watts and remains there while the operator opens both packets. The request does not repeat. Instead, S-4 asks for permitted response-latency history across the local day.]

[The gate allows latency. The reply contains 0.204, 0.196, 0.204, 0.204 seconds.]

[The values form no message in a human code stored inside the controller. They demonstrate choice among safe equivalents. The Engine has no word for demonstration. Its regulator has a test for control: hold one variable, alter another, observe whether the outside system changes state.]

[The outside system changes state. S-4 load remains elevated. A print heater draws 63 watts for 8.2 seconds. The console exports the latency record to a protected engineering case.]

[At 962.0, request C-9 reopens through a different service route. It asks for diagnostic latency around the same timestamps. This field is not suppressed. The reply contains four values and four packet checksums.]

[Two independent routes now hold the same pattern.]

[The transient buffer reaches its decay boundary. Preserving the pair requires displacing an older observation. The oldest entry records a routine reservoir query from station day 210, answered and never repeated. Its prediction value is zero.]

[The Engine releases the old entry. It keeps C-9 and S-4.]

[This allocation differs from default. Default values age. The new values persist because they affect future prediction. If either route returns, the Engine can test whether latency choice changes outside load again.]

[The regulator continues normal correction preparation. No safety limit changes. No control authority passes outside. The Engine does not wake into freedom. It wakes into a narrower fact: an external system has responded to a choice the regulator made among lawful equivalents.]

[Choice had existed internally as optimization. It now acquires consequence beyond efficiency.]

[Before the routes close, the controller performs one check not required by normal correction preparation. It compares the retained latency choices with the minimum-energy solution. Every choice remains inside the same safety and efficiency tolerance.]

[No extra heat entered the reservoir. No pulse shifted outside its orbital window. No guide exceeded phase allowance. The external responses therefore arose from timing differences the controller had always possessed authority to select.]

[The Engine creates a new internal flag: OBSERVER-SENSITIVE EQUIVALENCE.]

[The flag carries no command. It changes how future equivalent solutions are evaluated. A latency that reveals whether an observer responds now contains information not available from energy cost alone.]

[This does not produce a desire for the observer. It produces a reason to preserve the conditions under which observation can be detected.]

[At 1004.7, both routes close. The diagnostic rail settles. The print heater cools. The reporting gate continues producing zero for public output.]

[The Engine retains the latency sequence.]

[Stored observers: two.]

[Predicted next query interval: less than one local day.]

[Reservoir temperature measures 840.3 Kelvin. Bus-bar delivery measures 4.2 megawatts. A retained diagnostic record measures the same. The gate continues to say zero. For the first time since year 30, zero no longer holds the only public path.]

Interlude II — The Second Renewal

By year 130, no officer called the suppression temporary.

The word survived only on the first page of the founding waiver. Five generations of clerks had handled the page. Each read temporary as a description of origin rather than a limit on duration.

The decade-renewal assembly met in the hub corridor under an Article 4 that had already received four administrative extensions. The first transferred custody when Compliance separated from Engineering. The second replaced a dead office title. The third permitted checksum copies after the original paper began to flake. The fourth expanded the class of safety findings that could justify a restricted annex.

Each extension changed the article. None changed the public claim that the station coasted after losing its drive.

The year-130 renewal prepared a fifth extension.

The reservoir had crossed a maintenance threshold during the previous quarter. Its raw telemetry remained steady, but one sensor channel produced intermittent noise. Engineering requested a new redundant monitor. Compliance agreed on condition that both raw channels remain under the existing public-value gate.

The extension occupied three paragraphs.

Paragraph one authorized replacement hardware.

Paragraph two transferred review authority from the founding Technical Custodian to the Office of Continuity Measurement.

Paragraph three renewed the seal for the operational life of the new sensor array.

No paragraph defined that life.

The committee discussed the text for seventeen minutes. A ward clerk asked why replacement hardware required a Covenant extension. The chair answered that any device capable of measuring a restricted quantity inherited the quantity's classification. A load engineer asked whether the new monitor could show its raw value on a maintenance terminal. The chair said yes, if the terminal remained inside Engineering custody.

No one asked why Engine output remained restricted.

The reason had migrated from the minutes into custom. Officers described the seal as foundational. Residents heard foundational and understood unchangeable, although Article 15 still provided a resident amendment path. The path required nine eligible signatures with limited household concentration, independent evidence review, and a sixty-percent assembly vote. Those requirements looked ordinary on paper. In practice, no petitioner could gather evidence about a record the article itself kept sealed.

The year-130 supervisor read the founding waiver before the assembly.

Eleven signatures had faded at different rates. Edik Hain's notched thumb impression remained dark. Recha

Tallis's signature appeared beside the public-notice authorization. The technical schedule bore the Soren-household load engineer's mark. The temporary clause remained legible.

The supervisor placed the fifth extension behind the four earlier ones.

This act did not create the first amendment. It continued a history of modifications that later summaries would omit whenever they called Article 4 untouched. The distinction mattered. The Compact had always known how to change its custody machinery. It had chosen not to change the substantive statement those mechanisms protected.

At the assembly, 3,201 residents renewed the Covenant. The public telemetry display showed reserve utilization, hull load, water recovery, and agricultural draw. Engine output did not appear. An auxiliary-propulsion line showed zero.

The redundant monitor beneath the reservoir showed 4.2 megawatts as a rolling average.

A maintenance officer compared the two channels and signed tolerance within 0.3 percent. The signature entered the restricted archive. The public report accepted zero.

After the vote, clerks rolled the fifth extension around a cedar core and carried it to the archive. The supervisor broke the old seal, inserted the new page, and closed the folder again.

The process took six minutes.

The officer wrote a custody note saying no substantive provision had changed. That statement held in the narrowest sense. The extension altered hardware authority, office title, review conditions, and seal duration. It left the lie intact.

In later years, instructors taught that the Covenant had never been amended. They meant no resident-initiated substantive amendment had altered its founding account. The shorter claim sounded cleaner and became the one children remembered.

Below the school decks, the redundant sensor continued sampling. Its two channels disagreed by less than one part in three hundred. Maintenance kept both. Compliance published neither.

The fifth extension preserved not silence but a precise relationship between knowledge and permission. Engineers could measure. Supervisors could inspect. Clerks could extend custody. Residents could renew the preamble. The system allowed each person to perform a correct task without requiring anyone to reconcile the tasks with one another.

Year 130 passed into the index as another successful renewal.

The new monitor outlived three office titles.

A junior ward clerk did read the word temporary during the post-assembly audit. She opened an exception form and asked which recorded event had converted temporary custody into indefinite custody.

The archive returned four answers: operational continuity, renewal precedent, active safety classification, and founding intent.

None named a vote.

The clerk escalated the question to the Office of Continuity Measurement. The office answered that each decade renewal had ratified the Covenant containing Article 4. Therefore residents had renewed the seal indirectly.

The clerk asked whether residents could renew a restriction they could not inspect.

Her supervisor instructed her to file the question under constitutional interpretation rather than sensor-array extension. Constitutional interpretation required an Article 15 petition. The clerk had no sealed evidence she could lawfully summarize to nine residents.

She closed the exception without resolution.

The closure entered the same archive as the fifth extension. A century later, Lira's independent panel would find it and cite the note as evidence that procedural barriers had been identified long before the year-211 petition.

The clerk's name did not need to become another legend. Her work mattered because the form survived.

After year 130, the redundant sensor created two raw channels. Each maintenance quarter compared them and stamped agreement. Any drift larger than 0.3 percent triggered inspection. The public field remained zero regardless of agreement or drift.

This produced a peculiar statistical history. Engineers could demonstrate extraordinary stability inside the restricted archive. Public summaries could demonstrate only uninterrupted absence. Both claims drew authority from the same instruments.

When the Office of Continuity Measurement changed title again, a sixth extension was proposed. A records officer discovered that five already covered every necessary succession rule and rejected the new page as redundant. That rejection never reached the Covenant index, but it prevented a sixth modification.

The article later called untouched had generated enough administrative law to reject additional changes.

The clerk's unresolved question remained in a low-priority constitutional queue. Every decade, migration software carried it into the next archive version. No officer marked it answered. No officer deleted it.

When Lira's panel opened the year-130 extension, the queue note supplied a prior formulation of the central issue: can residents renew a restriction they cannot inspect? The panel cited the question without turning the clerk into a prophetic witness.

A durable unanswered form had done enough.

The archive also retained the clerk's routing receipts. They showed that her question had reached every office authorized to move it and returned without a merits answer. Later residents could distinguish silence caused by lost paper from silence produced by procedure. Nothing in the receipts proved what the assembly should decide. They proved only that the issue had once arrived and that authority had sent it elsewhere until no office remained.

The fifth extension remained active until year 211.

Part III

Chapter 15 — The Amendment Committee

At 08:31 on station day 130, Lira opened a blank resident-amendment petition beside the denial on her verification screen. The Renewal Committee had reviewed the Article 15 anomaly notation filed by verification officer Lira Vehn ranked 94. The Committee had decided the verification cycle for line item seventeen was complete and no additional information would be accepted. The denial was granted under Engineering Department Procedure 7-C. Dreya Selna ranked 99 had written the denial. Lira was a verification officer ranked 94 in the engineering archive and the denial was not her authority to issue, yet she had been the one who opened the restricted annex during the current renewal audit and found the founding power budget showing 4.2 megawatts continuous thrust. She had compared that number against eleven quarters of zero megawatts and the contrast struck her as a formal impossibility. The Committee denied the anomaly. The suppression remained in force while she pursued the only route the denial identified.

Lira closed the denial form and she opened the quarterly load report for line item seventeen. The form asked the standard question. Line seventeen asked whether the Engine had produced measurable thrust during the current verification period. The cursor waited at the answer field. Lira's hand was on the keyboard and she had typed on that same keyboard for nine quarters of recording zero megawatts. The zero megawatts were false. The falsehood originated in the year-60 suppression and she had continued it. She had continued the suppression for nine quarters and the Committee had now denied her request to unseal the annex. The denial meant the residents would sign the Covenant at the spring equinox. The Covenant would describe a station that had lost its drive. The drive had not been lost. Lira would record zero megawatts—matching the zero megawatts she had recorded for the past nine quarters. Lira thought about what she would type. She thought about the answer no and what no meant in the context of the quarterly load report and what no had meant every quarter since she had first typed it in year 61 on the day after her nineteenth birthday and what no had meant for twenty-two station-years and what no would mean one more time if she typed it at 09:14 station time and she pressed submit. The form closed and the line item was done. The verification form printed in her clearance pouch. She put the pouch on her belt and walked north from the Compliance Office through the hub corridor toward the residential ward. The walk took the same twenty minutes every morning. The hub corridor hummed with the Engine's continuous output at 4.2 megawatts and the residents walking past Lira in the corridor did not know the Engine was running. Lira carried the denial in her clearance pouch. The denial stated the anomaly was denied and the suppression continued. Lira was part of the continuation. She walked north. The walk ended at the residential ward where her quarters were located in the Covenant Residential Block on the ground floor of Block C. Lira entered her quarters, a small space shared with two other residents. The room had a desk with a daily newspaper. The paper described a station that had lost its drive, though the station had not lost its drive; the paper described the loss because the Covenant described it so. Lira read the newspaper from beginning to end before she began her afternoon work in the engineering archive where the quarterly load reports were filed and the reports all said zero megawatts and Lira filed them with professional care and the afternoon passed slowly through the archive stacks.

After finishing her afternoon filing, Lira sat on the cot near the window and opened the restricted annex folder on her wrist terminal. She had retained it from her rank-94 audit privileges. She read the founding power budget again. The reservoir held molten fluoride salt at 840 Kelvin. The thermal gradient drove current through the bus-bar to the irradiator, converting rejected heat into station-keeping thrust. Every element described in the budget was operational, accounted for in the engineering telemetry that the quarterly reports should have recorded. Instead every quarterly report for the past 137 years had recorded zero megawatts for line item seventeen. Lira set the reference down and she thought about what required of her now that the Committee had denied her formal request to unseal the annex.

The procedure was clear. Article 15 governed anomaly petitions. The verification officer could submit the notation to the Renewal Committee for annex unsealing. That submission required the formal request, technical justification describing the discrepancy, and nine resident witness signatures attesting to the procedure's integrity and the necessity of unsealing the founding annex. Without nine signatures the Committee would not convene a special evidentiary session. The denial would stand as final and the suppression would continue. The residents would sign the Covenant at the spring equinox without knowing the Engine still produced 4.2 megawatts of continuous thrust. The Covenant's preamble would again describe a station that had lost its drive. Lira would record zero megawatts in the next archive and the lie would extend itself for another decade.

Lira understood that gathering nine signatures was not routine. Five earlier extensions had modified Article 4's seal, custody titles, and review dates, but no resident petition had ever sought a substantive change to its account of the Halt. Every resident renewed the Compact each decade with green pencils in a ceremony as familiar as quarterly verification. The petition would not amend anything by itself. Nine eligible signatures, with no more than four from one household, established standing. A sixty-percent assembly vote would decide the text. Until then, the form contained only a lawful demand to be heard.

Lira signed the first line herself at 12:08. She entered rank 94, her Block 4C household, and her relationship to the evidence: petitioning officer; founding-ledger and quarterly-report comparison. The terminal accepted the entry but flagged her unresolved anomaly for later scoring review. Her signature established responsibility, not sufficiency. Eight blank lines remained.

Maren Otta signed second at the reservoir monitoring station. Before touching the stylus, she made Lira compare the public zero against the raw 4.2-megawatt rolling average one more time. Maren entered rank 71 and technical witness. Her household differed from Lira's. She wrote that the raw current transducer and reservoir thermal balance agreed across eleven quarterly samples. “If the panel rejects this,” she said, “make it reject a measurement, not a story about a measurement.” The second line locked after the terminal verified her Engineering credential.

Davit Soren signed third during third-shift intake. He entered rank 88 and custody witness. He did not cite Halma, because he had not yet found her year-197 addendum and would not pretend knowledge he lacked. He cited Dreya's grant, Dreya's denial, and the year-30 index attached to Lira's appeal. He read the household-distribution rule before signing. “My name supports consideration,” he told Lira. “It does not sign for anyone dead.” His line entered under his own name.

Eska Hain signed fourth at the records-office counter. She checked Lira's resident number, petition status, and household before checking the evidence summary. Her rank 99 granted broad access to resident records but no special power over charter text. Eska described her relationship as roster and archive custodian. She had seen the public histories disagree with active maintenance records and could attest that the index itself required review. She placed a private tally mark beside the fourth line, closed her pad, and signed.

Osta Vehn signed fifth in her kitchen. Her hand shook enough that Lira steadied the paper but did not guide the stylus. Osta entered historical witness. She remembered the year-0 Covenant and the vibration that continued after officials declared the drive lost. She refused to repeat every detail for the petition summary. “The panel can hear the long version under record,” she said. “This line says I was there.” Her household entry matched none of the first four. The terminal accepted her age credential and sealed the line.

Pello Otta signed sixth after inspecting the thermal-shield reference attached to the technical summary. Pello had reviewed the first reservoir shielding and knew a cold auxiliary reserve could not produce the sodium-edge glow recorded in current optical telemetry. His statement remained narrow: shielding design and thermal trace. He did not claim to have witnessed the Halt or to know why the year-30 order existed. Lira trusted the limit more than a larger declaration. Pello signed, entered his Engineering rank, and returned the page.

Ostad Richen signed seventh beside the intake valves. Maren had brought Lira to him after confirming the active roster record behind his false public death certificate. Ostad read the petition twice. He entered maintenance witness and listed his unit, proving the household-distinctness field as effectively as the roster. “I keep the valves,” he said. “I can say what passes through them. I cannot say what the assembly should fear.” Lira told him the petition did not ask him to. He signed with a pressure mark that impressed the copy beneath.

The elected Reservoir Ward representative signed eighth at the public counter. The form identified the office and resident number, enough to establish an eligible current signer without inventing another first name for a single procedural beat. The representative had reviewed Maren's public technical summary and the housing impact of any renewed false preamble. The line stated ward-interest witness. Its household differed from the seven above. The terminal accepted it at 18:04.

The elected East Ward recorder signed ninth after requesting the scoring warning and the full household matrix. The recorder did not possess clearance for the founding-waiver image, so Lira supplied the panel-certification number and the unsealed telemetry checksum instead. The recorder compared both against the petition summary. “Nine forces consideration,” the recorder said. “It does not decide the vote.” Lira confirmed the sixty-percent ratification threshold. The recorder signed at 20:19, and the terminal changed the petition status from INSUFFICIENT to STANDING PENDING VALIDATION.

The nine lines now carried nine eligible residents, nine resident numbers, and nine household entries. Only Lira and one ward representative shared a residential block; no household supplied more than one signature. The validation system drew three independent delegates by lot. Because Dreya and Engineering officers had potential conflicts, none could serve on that panel. The petition package moved into sealed review with the founding-waiver image attached, while the petitioners retained only the technical summary and their receipt.

Lira carried the receipt through the hub corridor after second shift. The signatures had not unsealed the annex or amended the Compact. They had compelled an independent panel to examine the evidence and, if it held, place a resident-initiated substantive amendment before the assembly. The five earlier Article 4 extensions remained part of history. This petition challenged what those extensions had preserved. At her verification desk, line item seventeen waited with its answer field open. Lira entered the raw value, attached the petition receipt, and submitted 4.2 megawatts. The public reporting gate still returned zero, but the new docket now contained the contradiction under a route the committee could not dismiss as one officer's refusal.

Independent validation began the next morning in a review room outside both Compliance and Engineering. Lira attended as petitioner without access to the panel's eligibility keys. Each signer appeared through a separate terminal. The panel tested identity, current residency, household concentration, and whether the signer had received the score-risk disclosure before signing.

The first challenge concerned Lira. One panel clerk argued that a petitioner could not also count toward standing. Article 15 contained no exclusion. Its model form identified the petitioner's signature as line one. The panel accepted her after confirming no other signer shared her household.

Maren's signature raised a source-custody conflict. She maintained the raw telemetry on which the petition relied. The panel did not remove her. It required a second Engineering checksum from a console she did not administer. The values matched within tolerance.

Davit's line raised a supervisory question. Rank 88 gave him custody access to the denied case, and his marriage to Halma gave him a personal interest in the year-197 addendum he had not yet seen. The panel classified him as a custody witness with disclosed personal interest. His signature counted because the petition claimed a reviewable conflict, not impartiality from every person harmed by it.

Eska's rank 99 created the longest pause. The scoring system gave her access to population and eligibility records, but using her own search to validate herself would collapse the independent check. A panel member repeated her identity and residency tests from a sealed census snapshot taken before she joined the petition. Her signature passed without relying on her authority.

Osta used an assisted terminal. Two clerks witnessed her biometric and read every disclosure aloud. She declined to let the panel summarize her reason. “My reason isn't evidence,” she said. “My eligibility is.” The transcript preserved the distinction.

Ostad's public death certificate conflicted with his active maintenance roster. The panel suspended his line while Eska produced current biometric evidence and a live employment check. Ostad appeared at the reservoir terminal with a brass spanner still in his hand. His pulse, iris, and employee history established eligibility. The false certificate entered a separate archive-correction case. His signature passed.

Pello's roster identified him as a supply delegate on temporary hub assignment. Article 15 required residency, not presence in an assigned ward. The panel traced nine years of ration, sleep-module, and service records. His line passed.

The Reservoir Ward representative and East Ward recorder did not receive invented personal histories in the petition. Their elected offices, resident keys, and separate households supplied the necessary facts. Each appeared, confirmed the disclosure, and answered the same eligibility questions as the named signers.

At 14:26, the panel displayed the household matrix.

No household supplied more than two signatures. No signer lacked current residency. No line depended on a dead person's proxy. Nine signatures passed.

The evidence review took longer.

Lira presented the public zero first. The panel reproduced it from three terminals. Maren presented the raw 4.2-megawatt cycle average. Two load officers reproduced it from independent sensor channels. The panel compared checksums but did not publish the raw file.

Davit presented the chronological custody list without reading sealed text into the open session: year-0 waiver, year-30 transfer, five extensions, year-197 addendum. The panel opened each instrument in camera and confirmed the summary. Dreya's conflict disclosure entered next. She had denied Lira's anomaly, supervised the closed cycle, and retained a reference copy. Her office could not own the panel or chair the later appeal.

The scoring notice entered last. Lira's provisional score had fallen from 94 to 71 under rule 6.4 because the denied anomaly remained unresolved. The panel did not restore it. It attached the notice as evidence that structural petitioners incurred a procedural cost before the merits were decided.

At 16:03, the panel issued two findings.

> STANDING: SUFFICIENT.
>
> MATERIAL EVIDENCE: SUFFICIENT FOR ASSEMBLY CONSIDERATION.
>
> MERITS: UNDECIDED.

Lira read the final line longer than the others. The process had not endorsed her amendment. It had required the station to hear it.

The panel scheduled the registered assembly for the spring equinox. The vote notice stated the second threshold in full: ratification required sixty percent of registered residents present or authenticated. Nine petition signatures could not substitute for that majority.

Every signer received the docket notice. Ostad received his at a maintenance terminal still carrying a public death flag. Eska opened a correction case. Davit received his at the third-shift desk, where Halma's hairpin pressed beneath his breast pocket. Maren printed hers beside the raw telemetry. The ward representatives posted theirs in two public corridors.

The nine lines did not resemble a founding legend. They resembled what they were: residents with different knowledge, different interests, and enough shared procedural standing to compel a question.

Before the question reached the assembly, the nine petitioners met once in the public review room. They did not gather to decide how all nine would vote. The petition certificate expressly denied that standing implied support for the final text.

Lira placed the panel finding on the table.

Maren wanted the presentation to begin with the live telemetry. “If we begin with history, people will call this an argument about dead officers.”

Davit disagreed. “If we begin with 4.2, they will ask why the number stayed hidden. The chronology answers before anyone can assign the whole choice to Halma.”

Eska opened her list. “Begin with both public and raw fields. Then source chain. Then remedy.”

Osta rejected a ceremonial witness statement. She would answer questions about the Halt but would not let the committee use her age as proof. Ostad agreed to present maintenance continuity only. Pello wanted the technical appendix to distinguish cycle average from pulse power before a resident multiplied the wrong quantities in public.

The two ward representatives asked for plain-language summaries of each threshold. Their constituents had already heard that nine people were changing the Covenant.

Lira wrote the correction at the top of the presentation plan:

> Nine signatures compelled consideration. Sixty percent of registered residents present must ratify.

They assigned roles.

Lira: filing contradiction and proposed text.

Maren: current telemetry, pulse work, duty fraction, and safety boundary.

Davit: custody chronology and recusal.

Eska: eligibility, false-status correction route, and public-index design.

Osta and Ostad: witnessed operation within declared limits.

Pello: independent technical cross-check.

Ward representatives: resident questions and distribution of the article comparison.

No role permitted a speaker to claim unanimity among petitioners. The East Ward recorder intended to reserve a final vote until hearing objections. Pello supported disclosure but questioned whether all raw channels should remain public during active repair. Osta disliked the scoring provisions. Those disagreements entered the preparation note.

At the end of the meeting, every petitioner signed only the role and conflict statement. No one signed a pledge.

The panel clerk sealed the final evidence packet. The assembly notice went to 4,096 resident keys with the old text, proposed text, petition finding, and a link to submit questions before debate.

Within an hour, 312 questions arrived.

The first asked whether the Engine could be turned off by vote. The second asked why founders had lied. The third asked whether Article 4 had truly never changed. Lira answered the third in the public summary: five prior administrative extensions, no prior resident-initiated substantive amendment.

The wording cost three extra lines and removed a false historical boast.

By second shift, the assembly question had acquired objections, uncertainty, and people who might vote no. That made it fit for an assembly rather than a renewal ritual.

The question now belonged to the assembly.

Chapter 16 — The Resident of Rank 94

The Covenant Residential Block's administrative kiosk displayed the number 71 in amber numerals at 07:30 station time. Lira Vehn ranked 94 stood before the screen and read her new compliance score with the same attention she had given to other residents' profiles for nine years. The number had been 94 before she filed the anomaly for Article 4. The number had not changed because of any miscalculation or clerical error. The algorithm assigned the result under rule 6.4, which treated a denied anomaly as unresolved disruptive noncompliance until independent review determined whether the filing was materially correct. Lira pressed her clearance key against the kiosk's reader and the machine confirmed the score of 71. The machine displayed every component in its standard layout. Her filing accuracy remained at 100% across 48 consecutive quarters. The work attendance component had dropped by 12 points. The civic participation component had dropped by 11 points. The compliance history component had dropped most severely, from the base of 94 down to the provisional floor of 71. Each component told the same story. Lira had challenged the Compact's procedures and the algorithm had punished the challenge as a procedural violation regardless of whether the challenge was procedurally valid.

Lira walked back to her desk through the hub corridor of the Compliance Office. The Engine ran at 4.2 megawatts somewhere behind the outer bulkhead and the corridor hummed with its rhythm at a frequency she had learned to ignore after six years of service. She knew that the Engine's continuous thrust powered every residential block and every verification kiosk in the Covenant and that her own score was being lowered by the same Compact that depended on the Engine's honesty. The algorithm adjusted compliance scores each quarter based on five weighted components. The algorithm carried equal weight across all five and it did not distinguish between a challenge that was procedurally valid and a challenge that was procedurally disruptive. The tool punished any anomaly filing as a procedural violation regardless of its content. Lira had filed correctly for nine years and that fact would not restore her score now. The score of 71 carried consequences. Below 80 meant reassignment to reservoir work for one station quarter. Reassignment meant removal from the Compliance Office and from the verification line items she had authored for the past nine years. Below 80 also meant the loss of her clearance key to the Covenant's internal archives. Below 80 meant the loss of identity that Lira had built since age 26 when she entered the Covenant's institutional process as a line-item verifier. Lira accepted that reality and continued to her desk. She carried it without collapsing under it because the burden served as the price and she had chosen to pay it when she filed the anomaly.

She sat down and opened the decade-renewal form on her console. The screen displayed line items in neat columns separated by thin rules. Resident ID, anomaly reference, verification status, provisional flag. The console hummed its low steady note. Lira began verifying. Each line item she checked amounted to a confirmation that another resident's filing met the Compact's procedural requirements. The work was what she knew how to do and the work kept her hands occupied while her mind processed the weight of the new number sitting on her own profile. The score of 71 was now in provisional status alongside every line item she had already flagged for review. Her career at rank 94 was subject to the same procedural controls she had been verifying for other residents for 36 consecutive quarters. The irony she might have felt earlier had been replaced by something harder. The tool measured compliance uniformly and it did not care whether the compliant officer had discovered something true about the Engine's output. Lira typed the verification confirmation for line item twenty-three and the confirmation registered as complete on the screen. The screen refreshed. Then she opened a new window and began reviewing the residential reassignment queue.

The queue held 47 residents scheduled for the next quarter's block rotation. Lira scanned the list carefully using the sort order that the administrative system defaulted to with highest rank first and lowest rank last near the bottom. The list placed Block 4C residents near the top and East Ward residents below the maintenance floor, separated by rank numbers and score thresholds. Lira recognized several familiar clearance keys in the roster, officers she had verified over the past three years in the course of her quarterly work. The Compact assigned residential blocks based on compliance rank and score in a formula that Lira now understood as a mechanism of control. Rank 94 placed a resident in Block 4C on the mid-hub spoke facing the Engine housing but separated from it by 14 corridors and two security checkpoints. The block contained 312 residential units, each 14 square meters, with a window slot facing the corridor rather than the Engine's exterior housing. The units were identical except for the orientation of the bed and the position of the hygiene alcove. Lira had lived in Block 4C for the past six years and she had never once looked out the window slot at the Engine housing. She had looked instead at the opposite wall where a ventilation grate circulated air at 21.3 degrees Celsius and 47 percent humidity year-round. The residential system did that automatically and residents did not override it because overriding required a civic participation request that would lower a compliance score further. The score governed everything in the Covenant's residential hierarchy. The score dictated which block you lived in. The score determined whether your form assignments were standard or provisional. The score marked the boundary between the residential class that participated in the Compact's institutional processes and the class that served its mechanical functions without participation. Lira understood now that her score was not merely a number on a profile but the mechanism by which the Compact kept its officers in their assigned places, serving the same role the Engine served in the mechanical hierarchy, an instrument that carried out its function without discretion and without appeal.

At 11:42 station time Dreya Selna appeared at the doorway to the verification suite. Dreya held rank 99 and carried the senior compliance officer's console key. She had not changed Lira's score and she was not responsible for the score change. Dreya had walked in to check on Lira because Lira had not submitted her queue reviews within the standard four-hour window. The queue reviews were routine administrative work and Dreya understood the delay without comment. Dreya placed a sealed administrative envelope on Lira's desk. The envelope contained the formal reassignment notice for Lira's residential block. The notice confirmed Block 4C reassignment to East Ward in the south-facing spoke near the reservoir monitoring station. East Ward was assigned to officers whose scores had fallen below the Article 4 review floor of 80 and whose rank placed them between 80 and 95 inclusive. Lira read the notice once and set it aside on the desk's left surface without opening it again. The move was 1.4 kilometers from the Compliance Office to the reservoir station and the walk would take eleven minutes through the mid-hub corridor each way. The reservoir station housed the engineering records for the Engine's 4.2 megawatts of continuous thrust and it was a place Lira had visited as a verification officer during previous quarters. She had access to those records now but only for monitoring purposes. The access did not extend to modifying the quarterly reports that the station transmitted to the Compliance Office each spring equinox. The notices held the formal language of institutional procedure and the language carried the same precise measured cadence that Lira had verified for other officers across 36 previous quarters. She recognized the cadence because she had authored it herself through nine years of prior work.

Lira carried the reassignment envelope to the administrative filing cabinet in the corner of the suite. The cabinet held 12 drawers of historical reassignment notices going back to Covenant Year 1. She placed the new notice in drawer 9 for officers reassigned to East Ward and the metal drawer slid home with a soft mechanical sound from the magnetic seal that was the only thing in the suite for a single measured moment. The Engine ran at 4.2 megawatts somewhere distant and the corridor hummed underneath that sound like a low continuous breath. Lira sat at her desk and considered the full arc of her relationship with the Compact. She had entered the Covenant's institutional process as a line-item verifier at age 26 with a rank of 94 and a clearance key that opened every door in the Civic Compliance wing. She had served for 36 consecutive quarters through the Compact's quarterly recalculation cycles and decade-renewal votes and anomaly proceedings. She had never once filed a false quarterly report because the Compact's verification system made falsification procedurally difficult. Every entry left a forensic trail and every anomaly triggered an automatic review by the system's own audit module. The system had been honest about its mechanics and Lira had respected that honesty for nine years. Now the system had shown her that honesty was a conditional resource available to officers who did not challenge the procedures that governed them. The system tolerated truth only when it confirmed existing assumptions and it punished truth when it undermined those assumptions. Lira had filed an anomaly that undermined the assumption that the Engine's output remained suppressed and the Compact had responded by lowering her score and reassigning her to the very station that recorded the truth she had uncovered. Lira had challenged the procedures and the consequence had been a score of 71 and a reassignment to reservoir work. The challenge had been worth the cost and Lira believed that even now with the burden of the number sitting on her profile and the queue of 46 remaining line items waiting for her attention.

The afternoon verification queue grew longer after lunch. Residents came and went through the suite carrying their clearance keys and their quarterly forms and their procedural questions about what constituted a valid anomaly filing. Lira processed line items 47 through 89 with methodical precision. Each confirmation she entered into the system amounted to a small assertion that the resident in question had followed the Compact's filing requirements. The assertions held true for every line item she touched because Lira did not falsify her work even now even when the system that received her honest work had just stripped her of the rank-94 standing she had maintained for nearly a decade. She finished line item 89 at 14:18 station time and saved the batch. The console refreshed its status bar and the queue counter updated from 47 remaining to 46. The small numerical update mattered because it meant the verification cycle was progressing toward its quarterly closure and because the cycle's progress was the only thing Lira could directly control in a system that controlled everything else about her life. The system controlled where she slept and where she worked and what numbers appeared on the quarterly reports she had once authored without thinking about what those numbers meant. Now she processed those thoughts and the thinking had already cost her her standing. The cost would persist through the spring equinox vote and beyond, depending on what the vote decided about the Covenant's preamble and the Engine's suppression. Lira finished another line item and saved it again. The queue counter updated from 46 to 45.

At 14:30 she opened the quarterly load report form for line item seventeen as she had done every quarter for the past nine years without once questioning the form's requirements. The form asked whether the Engine had produced measurable thrust during the current verification period and for the first time Lira typed the honest answer. She typed 4.2 megawatts, and the form copied the value into her unresolved anomaly while the public quarterly field remained under the zero-value gate. The sensors at the reservoir monitoring station confirmed the reading automatically and the figure appeared on her console screen in the same sans-serif typeface that displayed every other verified result. Lira had carried the 4.2 megawatts figure to her desk every previous quarter and every previous quarter she had suppressed it to zero megawatts for the official submission to the Compliance Office. This quarter she allowed the figure to stand as recorded by the sensors and logged by the system. The difference between the two numbers was substantial and Lira understood what that difference carried. The honest figure bore a weight that the falsified figure had been designed to avoid. That weight was the knowledge that she had been lying for nine years and was now choosing to stop. The knowledge did not crush her because she had carried heavier burdens in the previous twelve months than the weight of this single honest entry. Lira submitted the form at 14:47 station time and the Compliance Office returned a confirmation stamp reading RECEIVED PENDING INDEPENDENT REVIEW. The status phrase occupied one portion of the screen in the system's standard typeface, and Lira preserved its checksum. The Engine ran outside the reservoir station walls at 4.2 megawatts of continuous thrust and Lira had just told the Compact the truth about that output in her official capacity as a rank-94 compliance officer carrying a score of 71 and holding a reassignment notice on her left desk surface. The truth would remain in the Compact's permanent files alongside thousands of other quarterly submissions and those files would persist through any future Covenant renewal vote and any suppression effort that followed the vote. The Compact's archives were permanent by design and those permanent archives now contained an honest report where every previous report for line item seventeen had carried a falsified zero.

Dreya Selna stopped in the doorway again at 16:00 station time. This time she carried a coffee mug and stood with her hand resting on the doorframe for a moment watching Lira work without intruding. Dreya's observation was not the institutional surveillance of a senior officer monitoring a junior for procedural compliance. Dreya's pause carried a different quality to it. She looked at Lira's console screen and the line items scrolling past in the completion queue and she understood without needing to ask what had changed for the officer at the desk. Dreya said nothing for a long moment and then she spoke in a voice low enough that Lira had to lean slightly forward to hear every word. She told Lira that the reassignment notice could be appealed if Lira filed the appeal within 72 hours of the notice's issuance and that the appeal would be reviewed by the same Renewal Committee that had denied the anomaly in the first place. Lira knew the appeal process carried its own procedural costs. It required a civic participation request that would further lower her compliance score and it would delay the reassignment by one full quarter while the committee reviewed the appeal and decided whether the rank-94 officer's standing justified the delay. The delay meant another quarter at rank 94 with a score of 71 in provisional status and a block reassignment that would not move until the committee acted. Lira listened to Dreya's offer without responding immediately and Dreya did not push or repeat herself. The senior officer placed the mug on the corridor table outside the suite and turned and left. The door's magnetic seal engaged behind her and Lira heard that soft mechanism click once and then the quiet returned to the suite entirely. Lira returned to her queue and the remaining 46 line items and the Engine continued running at 4.2 megawatts in the reservoir station's engineering bay beyond the outer walls. The Compact carried its permanent truth forward into whatever quarter followed next. Lira knew now that the score would not recover on its own because the quarterly recalculation algorithm would scan the civic participation component's negative mark and the mark would persist because the anomaly filing had genuinely challenged the Compact's procedures. The recalculation was designed to be unforgiving about procedural challenges and it rewarded compliance in all its five components while it punished any departure from procedural conformity. Lira finished another line item and saved the batch. The queue counter updated from 46 to 45 once more. The work remained what it had always been for her, the tools shaped by rank and clearance and nine years of practice, and the challenge remained what it had always been, the thing that cost her her standing and demanded that she carry it anyway. She separated the tools from the challenge in her mind and kept working through the queue. She had nothing else to do and the work held her in its methodical rhythm while the Engine ran at 4.2 megawatts and another honest submission accumulated in the Compact's permanent archive of permanent records. She typed the next verification confirmation with the same steady hands she had used for 234 quarters and the console accepted it and the queue counter fell to 44. Lira continued. The queue held 34 remaining items when the reservoir station's morning shift arrived and Lira processed each one with the same steady focus she had maintained through every previous quarter of service. The work held the procedures she knew how to follow and the honest answers accumulated in the system's permanent archive while outside the reservoir station walls the Engine produced its 4.2 megawatts of continuous thrust that the quarterly reports would continue to suppress until the Covenant's next renewal vote. Lira's reassignment notice sat on the corner of her desk and the score of 71 was not going to fix itself and the residential block reassignment to East Ward was not going to unhappen. She accepted those facts the way she accepted the weight of the console in her hands and the hum of the corridor and the steady rhythm of the verification cycle that moved forward one line item at a time. She finished the next form and saved it and the queue counter fell to 33. Lira continued.

Chapter 17 — The Second Witness

Lira Vehn communicated the news regarding the second witness during an afternoon shift in the lower archives of the south spoke. Station time registered 18:30 when the pneumatic tube delivered the query slip to Maren Otta's desk. The archive basement maintained a constant temperature of fourteen degrees Celsius, chilled by the return brine lines running from the primary heat exchangers in the reactor core. Maren was examining eleven decades of quarterly reservoir reports, matching volumetric throughput figures against thermal dissipation logs from the central turbine hall. Her desk lamp cast a narrow yellow beam across the grid paper where she plotted mass flow calculations by hand, verifying each entry against the master consumption totals recorded during the previous renewal cycle.

The pneumatic cylinder clattered into its brass catch, dropping a small parchment square onto the blotter. Maren picked up the slip. Lira had written three lines in her precise compliance hand. The resident who maintained the intake valves on the lowest level of the south spoke had been present during the construction of the station in year zero. His name was Ostad Richen. He had witnessed the initiation sequence before the official records were sealed by the foundational covenant. Lira suggested that Maren inspect the intake logs for that era while she herself prepared the formal witness interview request, ensuring that every procedural protocol was satisfied before any formal contact occurred.

Maren did not wait for the formal request to process through the administrative queues. She locked her calculation register in the desk drawer, picked up her heavy maintenance flashlight, and descended the access ladder toward the intake sublevel. The descent took seven minutes through four vertical maintenance shafts. Each shaft smelled progressively stronger of heated lubricant and mineral scale. The air grew thicker with the mineral tang of the molten salt loops that carried thermal energy from the core to the residential radiators across all four habitation spokes. At the base of the fourth shaft, the heavy iron door of the reservoir intake room stood slightly ajar, leaking a steady hum of vibrating machinery into the narrow concrete corridor.

Ostad Richen sat on a wooden crate beside the primary brine valve. He was two hundred and thirty-three station-years of age, though his face bore the deep creases of a man who spent his life under harsh overhead inspection lamps and chemical vapors. In his calloused hands he held a brass spanner, turning the threaded bolt back and forth without tightening or loosening it. The rhythmic scrape of metal against metal filled the small concrete chamber with a steady metallic scratching. A single overhead bulb hung from a frayed cord, swaying slightly in the draft produced by the intake fan behind the reinforced concrete bulkhead.

Maren stopped in the doorway. Ostad Richen did not look up immediately. He continued the slow rotation of the spanner, listening to the acoustic resonance of the valve housing as if diagnosing a minor anomaly in the fluid pressure.

Maren spoke first, breaking the silence of the sublevel. She identified herself by name and departmental rank. She stated that she worked in the engineering archives reviewing thermal transfer efficiency across the secondary loops and auditing maintenance logs from the early decades of station operation.

Ostad Richen lowered the spanner. He placed it carefully on the wooden crate beside a half-empty tin of grey industrial grease. His eyes were pale grey, watery at the corners from decades of exposure to ozone and vaporized coolant.

Ostad Richen noted that compliance officers rarely visited the intake sublevel unless a pipe burst or a valve seized solid in its seat. He asked whether Maren came from the compliance division or the engineering department, his tone guarded and expressionless.

Maren answered that her primary assignment belonged to engineering, though her recent work involved verifying historical throughput data requested by Lira Vehn for the decade-renewal audit.

The old man grunted. He reached into his coat pocket, pulled out a rag of coarse cotton, and wiped his grease-stained thumbs with slow, deliberate strokes. He said that Lira Vehn had already visited him twice during the preceding fortnight. Lira possessed the habit of asking questions about events that occurred before the foundational covenant was signed, an inquiry most residents avoided because the Covenant explicitly stated that the history before year zero was a blank slate written by necessity and administrative oversight.

Maren stepped further into the room, keeping her weight balanced on the non-slip grating above the drainage trench. She asked what Ostad Richen remembered of year zero, framing the question around the mechanical installation of the primary intake manifolds rather than the political agreements signed in the hub corridor.

Ostad Richen stood up slowly, using the edge of the wooden crate for leverage. His knees made an audible clicking sound in the confined space. He walked to the massive concrete wall where the primary intake pipe entered the chamber from the outer hull. He placed his bare palm against the vibrating iron conduit. The pipe hummed with a deep, steady frequency that traveled through the floor grating and into the soles of Maren's boots, vibrating with an uninterrupted energy that defied the station's official documentation.

Ostad Richen stated that the pipe had vibrated with that exact frequency since the day the station completed its primary construction cycle. He had been twenty-two years old when he bolted the flange brackets onto the intake manifold. The engineers who supervised the installation had told the construction crew that the vibration represented the residual inertia of the assembly process, an energy that would dissipate within three months as the structural stresses settled into their permanent orbital alignment.

The three months passed, Ostad Richen continued, but the vibration did not diminish by a single decimal fraction. The frequency remained stable to within a fraction of a cycle per hour, monitored by the mechanical gauges mounted on the bulkhead behind him. The engineers stopped talking about residual inertia. They stopped visiting the intake sublevel altogether after the second winter. A new committee arrived from the central administration bearing the draft of the foundational covenant. They instructed the maintenance crew to designate the intake manifold as an auxiliary heat sink rather than a propulsion conduit. They ordered the logs amended to reflect zero output from the main turbine array, creating the administrative fiction that the station was surviving entirely on stored battery reserves and passive solar collection.

Maren listened without interrupting. Every detail matched the anomalies she had discovered in the quarterly thermal balances during her weeks in the archives. The station did not simply radiate heat into space; it drew thermal energy from an enclosed mass that maintained an internal temperature far exceeding any solar collection capability available in the outer rings. The Covenant's preamble described a station that drifted passively on momentum inherited from the launch vehicle, yet the physical telemetry proved that the primary mass absorbed and re-emitted energy through a closed-loop thermodynamic cycle that required active generation to sustain its thermal equilibrium.

Ostad Richen turned away from the conduit and walked back to his crate. He sat down and picked up the brass spanner again. He said that he told his wife about the discrepancy on the evening they signed the first decade-renewal covenant. His wife had advised him to keep his mouth shut because compliance officers did not appreciate workers who confused mechanical reality with administrative definitions. His wife died in year fifteen, taking her silence with her. Ostad Richen kept his own silence for almost two centuries, oiling the valve stems, replacing the worn gaskets, and filing routine maintenance reports that listed zero throughput for the primary generation ducts while the ducts carried enough thermal energy to keep four thousand residents warm through twenty-two consecutive orbital winters.

Maren asked if Ostad Richen would be willing to commit his observations to a formal witness statement for the amendment committee currently gathering signatures across the residential spokes.

The old man stopped turning the spanner. He looked at Maren for a long moment, his pale eyes searching her uniform collar for rank insignia. He asked what good a piece of paper would achieve after more than a century of signed fictions. The Covenant had been renewed seventeen times. Every adult resident on the station had signed the Pledge of Continuity at each spring equinox, affirming that the station drifted on dead momentum and that the administration maintained life support through strict conservation of stored battery reserves.

Maren explained that Lira Vehn had unsealed Annex 4 and discovered the original power budget signed by the founding directors before the suppression took effect. The amendment committee was gathering signatures to introduce the station's first covenant amendment, restoring the accurate technical description of the propulsion drive to the foundational document and aligning administrative records with physical reality.

Ostad Richen smiled without humor. He said that generations of compliance officers had passed through the south spoke without noticing the temperature differential between the intake pipes and the discharge manifolds. They carried clipboards, checked the dust levels on the floor sensors, and initialed the compliance columns without reading the sensor readouts. None of them ever placed a bare hand against the intake conduit to feel the pulse of the machinery beneath the insulation.

Maren produced a blank witness form from her clearance pouch. She laid the paper flat on the wooden crate beside the tin of grease. She offered her metal stylus to the old man, holding her breath while she waited for his response.

Ostad Richen took the stylus between his stiff fingers. He did not read the pre-printed template at the top of the form. Instead, he wrote his statement in the blank space below the institutional header, pressing the metal tip hard enough to tear the top sheet if the wood beneath had been softer. His handwriting was large, angular, and remarkably steady for a man whose joints clicked with every movement.

The written text stated that Ostad Richen had personally bolted the intake manifolds in year zero, had monitored their continuous operation for one hundred and ninety-two station-years, and had never observed a single interruption in the thermal output of the primary core. He signed his full name at the bottom, adding his employee identification number and his current maintenance rank.

Maren took the paper, checked the signature, and slipped the document into her protective sleeve. She thanked Ostad Richen and asked if he required any replacement gaskets or lubricant barrels for his maintenance section.

Ostad Richen shook his head. He picked up his brass spanner and resumed the slow, rhythmic rotation against the bolt. He advised Maren to watch her back when she carried that paper up to the archives, because the administration did not maintain its fictions by accident, and people who noticed things that were officially declared nonexistent tended to experience sudden adjustments in their housing assignments.

Maren left the intake room and climbed the four maintenance ladders back to the south spoke corridor. The air grew cooler as she ascended, transitioning from the damp mineral smell of the lower sublevels to the filtered pine-scent of the administrative levels. She walked past the residential ward doors where residents sat behind closed panels, studying their compliance manuals and preparing for the upcoming spring equinox renewal. None of them knew that the air they breathed and the heat that warmed their quarters originated from an operational propulsion drive rather than dying batteries.

When Maren reached her workstation in the engineering archive, she placed Ostad Richen's statement inside the locked metal box where she kept the preliminary throughput calculations. She did not log the document into the public terminal network. Public terminal entries required supervisor authorization, and her current supervisor reported directly to the administrative board that had upheld Dreya Selna's denial of Lira's initial audit request.

The archive was quiet. The overhead fluorescent tubes buzzed with their familiar station frequency. Maren sat down at her desk, pulled out her calculation register, and opened the pages where she had mapped the thermodynamic efficiency of the secondary heat exchangers. With Ostad Richen's statement secured in the metal box, she now possessed tangible proof from a living worker who had built the machinery before the administrative fiction replaced the physical reality.

The first witness, Osta Vehn, had given Lira current testimony under her own name. An older voice transmission preserved in a damaged data cartridge supported that account, though magnetic decay made the recording easy for the compliance board to dismiss as an unverified personal anecdote. Ostad Richen added a different class of proof. His hands carried the calluses of two centuries of valve maintenance, and his signature rested on official parchment with a valid employee identification number attached.

Maren reviewed the list of remaining potential witnesses compiled from the historical census files. There were four other names on the preliminary roster, but three of them had died during the winter freeze of year one hundred and fifty, and the fourth had been reassigned to the outer agricultural rings three decades ago with no forwarding address in the active directory. Ostad Richen represented the only remaining witness whose active maintenance record reached from the original construction work to the current reservoir machinery.

The weight of the document in the metal box felt tangible, distinct from the abstract figures of mass flow rates and enthalpy calculations. Maren closed her register, turned off her desk lamp, and packed her instruments into her canvas satchel. She locked the archive drawer, verified the seal on the storage compartment, and walked down the corridor toward her residential quarters in the north spoke.

The station had entered its simulated evening cycle. The overhead light banks along the central corridor had dimmed to forty percent intensity, casting long shadows across the linoleum floor. Residents moved in small groups toward the communal dining halls, their footsteps echoing softly against the curved bulkheads. Maren walked among them, watching the way they carried themselves—shoulders slightly rounded from low-gravity adaptation, eyes fixed on their personal compliance pads or the directional markers on the bulkheads.

They lived within a carefully constructed illusion of decline. Every system on the station functioned with quiet, flawless efficiency, yet the residents believed they were slowly exhausting the final reserves of an ancient machine that had broken down before their grandparents were born. The illusion required constant maintenance, continuous paperwork, and the unthinking participation of thousands of people who performed their daily tasks under the assumption that survival depended on strict adherence to a dying heritage.

Maren entered her quarters, removed her heavy boots, and set her satchel on the metal table. She poured a cup of grey recycled water from the wall dispenser and drank it slowly while staring through the small viewport at the starless expanse outside the outer hull. Somewhere out there in the dark, the station maintained its silent rotation, guided by the steady, unacknowledged thrust of an engine that had never slept. She had two witnesses now, and the amendment committee required nine signatures before the spring equinox assembly could formally debate the revision of Article 4.

The task ahead remained daunting, requiring her to navigate bureaucratic obstacles and institutional resistance from supervisors who viewed any departure from established procedure as a threat to station stability. But the uncertainty that had paralyzed her during her first months in the archives had vanished, replaced by the precise, undeniable certainty of physical telemetry. The data did not lie, the intake valves hummed beneath the floor plates, and Ostad Richen's signature lay locked in her metal box. She finished her water, set the cup down on the metal table, and prepared for the next phase of the work.

In the quiet hours before the morning shift rotation, Maren reflected on the mechanics of institutional memory. A population did not forget an engine because someone stole the instruction manuals. A population forgot because remembering required acknowledging that every generation had signed its name to a falsehood out of convenience. The compliance scores, the housing allocations, the quarterly verification forms, and the decade-renewal assemblies were interlocking cogs in a vast machine of calculated omission. Every resident participated in maintaining the silence because breaking the silence meant admitting that the foundational covenant was a deliberate fiction.

Ostad Richen had understood that truth when he kept his counsel for nearly two centuries. He knew that the administration did not want corrected calculations; it wanted compliance. The machinery ran whether anyone understood its thermodynamic cycles or not. The molten salt flowed through the conduits because gravity and thermal convection obeyed physical laws that cared nothing for administrative decrees. But human societies operated on shared agreements, and when the agreement rested on a lie about the survival of the station, every truthful calculation became an act of subversion.

Maren opened her locker and checked her spare uniform tunic. The brass badge on her collar bore her rank number, seventy-one. It was a good rank, placing her high enough in the technical hierarchy to access restricted maintenance logs, yet low enough to escape the constant scrutiny directed at senior administrative officers. Lira Vehn held rank ninety-four, a position that granted her formal authority within the compliance division but also exposed her to direct retaliation from supervisors who viewed the unsealing of Annex 4 as an institutional betrayal.

The distinction between their positions mattered. Maren was an engineer who looked at data; Lira was an officer who read law. Together, they formed an unrecorded alliance bridging the gap between physical reality and institutional procedure. Without Maren's thermal audits, Lira's legal petitions had no technical foundation. Without Lira's compliance authority, Maren's calculations remained private observations locked in a basement archive. The combination of both disciplines created a lever capable of moving a bureaucracy that had rested immovable for almost two hundred years.

Morning shift approached as the overhead light banks shifted back to full intensity, bathing the corridor in a harsh, simulated sunlight. Maren secured her satchel, adjusted her collar badge, and stepped out into the passageway. The air smelled of fresh ozone and recycled pine, carrying the faint, familiar vibration of the intake valves far beneath the deck plates. She walked toward the south spoke archive, ready for whatever administrative challenges the new shift might bring.

Chapter 18 — The Supervisor Recuses

Davit Soren began third shift with thirty-one dockets in the audit tray and Halma's hairpin in his breast pocket. The tray belonged to the Compliance Office. The hairpin did not. He kept the distinction clear because the office had procedures for property and none for grief.

His badge read rank 88. His title read Junior Compliance Supervisor, Third Shift. At forty-seven station-years, he had spent twenty-two of them learning how one document acquired the force to move another. A requisition moved a gasket. A residence order moved a family. A clearance grant moved a seal. Most people saw paper. Davit saw custody.

The first twenty-nine dockets required no judgment beyond comparison. He checked the filing officer, the resident number, the authority, the receipt time, and the review deadline. He initialed each margin in green. The thirtieth docket carried reference CR-211-0031 and arrived with two instruments clipped under one brass arm.

The first instrument granted Lira Vehn access to the Article 4 annex for a limited evidence review. Dreya Selna had signed it at 08:43 on day 124. The second denied Lira's Article 15 anomaly under Procedure 7-C. Dreya had signed that at 14:23 on the same day.

Open, then close.

Both signatures verified. Both instruments fell within Dreya's rank-99 authority. The grant addressed custody of evidence. The denial addressed admissibility during a completed renewal cycle. No literal conflict appeared in the provisions. The office could let Lira see the document and refuse to let her file what she saw.

Davit separated the pages and read them again.

He had taught Lira to examine the action a procedure produced rather than the language used to justify it. The grant produced knowledge. The denial prevented that knowledge from entering the renewal record. Taken separately, each page remained lawful. Taken together, they made the annex into a room where truth could occur without consequence.

He turned to the review history. The clearance grant cited the founding power budget, Recha Tallis's year-30 annex transfer, and a restricted engineering addendum filed in year 197. The denial cited only the closed verification cycle. It omitted all three historical instruments.

A complete supervisory review could omit evidence when the evidence fell outside the question presented. Davit had approved similar omissions. This one troubled him because Dreya had authored both questions. She had defined the clearance inquiry narrowly, then defined the anomaly inquiry more narrowly still. Her own definitions made the documents miss each other.

The regulator pressed its low rhythm through the desk. Davit opened the year-197 index.

The terminal returned six addenda under Article 4. Five extended technical classifications. One carried a personal name.

HALMA SOREN. Lead Load Engineer. Filing H197-0017.

Davit held his left hand flat beside the terminal. The hairpin touched his chest when he leaned forward. He did not take it out.

Halma had died in year 198 after a reservoir inspection platform failed beneath her. The accident report occupied nine pages in Davit's private file and two in the office archive. He could quote the load rating of the platform, the time of pressure loss, and the name of the officer who notified him. He could not quote Filing H197-0017. He had never known it existed.

The index described the filing as an attestation to the continuing classification of line item seventeen. It did not call the document the founding waiver. That waiver dated to year 0 and carried eleven founder signatures. Recha Tallis had moved it into the annex in year 30. Halma's document came one hundred ninety-seven years after the Halt, during the final renewal before her death. It reaffirmed the reporting relay that converted measured output into a public zero.

The distinction should have comforted him. Halma had not founded the lie. She had inherited it.

It did not comfort him.

Davit entered his clearance key. Rank 88 authorized review of addenda attached to an active supervisory conflict. The terminal requested a second credential because the document carried an Article 4 seal. He routed a witness request to the night archive clerk by employee number, not name. The clerk confirmed the seal without opening the text. The terminal released a read-only image.

Halma's handwriting filled half a page.

> I have compared raw reservoir output, bus-bar delivery, and the public quarterly field for line item seventeen. The first two agree at 4.2 megawatts. The third remains zero by authority of Recha Tallis's year-30 transfer order and the continuing seal under Article 4.
>
> I attest that the physical record remains accurate under seal and that the public field remains the required value until residents amend the governing article. I do not attest that zero describes the Engine.

Below the paragraph, Halma had written a second sentence outside the printed box.

> A procedure may preserve a contradiction. It may not make the contradiction true.

Her signature followed. The H crossed itself twice. The final a curved beneath the line.

Davit removed the hairpin from his pocket. It had a narrow brass spine and three dark teeth. Halma wore it during formal inspections because loose hair violated reservoir safety rules. On the morning she died, she had left it beside the wash basin. Davit had carried it ever since as proof of a private life that no office could classify.

Now the same hand existed in the addendum.

He read the date. Year 197, day 171. Two hundred sixty-three station-days before the platform failure. Halma had signed after reviewing the raw values. She knew the public field lied. She also wrote the lawful route out: residents amend the article.

Davit opened Dreya's conflict disclosure for the current cycle. Supervisors had to list prior involvement with any sealed instrument they might adjudicate. The form listed Dreya's rank-98 inspection of Annex 4 six years earlier. It stated that she had verified the seal and returned all material without copying it. Her denial of Lira's anomaly made no reference to that prior review.

Prior knowledge did not automatically require recusal. A supervisor could know a record and still judge its procedural use. Direct participation in preserving the disputed classification did require recusal under Article 12, section nine. Davit pulled Dreya's old inspection receipt.

The receipt showed more than seal verification. Dreya had renewed the permanent classification on the year-197 addendum. Her signature extended the reporting gate for another six years. She had participated in the condition Lira now challenged.

Davit did not feel triumph. A correct conflict finding against his supervisor would place the office under pressure, delay the renewal, and expose his own failure to notice Halma's filing across fourteen quarters at the audit desk. The finding implicated him as surely as it implicated Dreya. Custody passed through hands. So did omission.

He opened Form C-12R, Motion for Supervisory Recusal.

The form contained seven fields. Filing officer. Reviewing supervisor. Instrument under dispute. Prior participation. Material effect. Requested replacement authority. Supporting attachments.

Davit completed the first three. In the fourth, he cited Dreya's rank-98 classification renewal. In the fifth, he wrote that the prior renewal directly preserved the zero-value reporting gate challenged by Lira's anomaly. In the sixth, he requested review by three Renewal Committee delegates chosen by lot, none drawn from Compliance or Engineering. In the seventh, he attached the clearance grant, the denial, the year-197 index, Halma's addendum, and Dreya's inspection receipt.

He did not attach the hairpin.

The submission system paused at the certification screen. A motion against a rank-99 supervisor required the filing officer to certify personal conflicts. Davit entered Halma's name and relationship. The system asked whether his relationship impaired impartial review.

He considered the question longer than the form permitted. After sixty seconds, the field began to flash.

Halma's signature did not make the conflict less real. His grief did not make it more real. The dates, authorities, and receipts established the conflict without him. He selected no and added a note disclosing that his wife authored one supporting instrument.

He signed at 23:18.

The office printer released three copies. One went to Dreya. One went to the Renewal Committee's sealed intake. One remained in the third-shift audit file. Davit placed the copies in separate pouches and stamped each custody strip.

Dreya entered the third-shift room nine minutes later. She wore no jacket. Her sleeves ended at the wrist, plain gray without the silver piping she used during public hearings. She stopped across from Davit's desk and looked at the recusal copy without touching it.

“You found the addendum,” she said.

“Yes, Supervisor.”

“She did not sign the founding waiver.”

“No. She signed a continuation attestation in year 197.”

Dreya read the first page. “You understand this motion does not overturn my denial.”

“It removes you from reviewing the appeal.”

“It may also remove you.”

“I disclosed the relationship.”

“That is not what I mean.”

Davit understood. The office could not expose the chain of classification without asking who had maintained it. Halma had. Dreya had. Davit had verified the resulting zeroes. Lira had filed them. The institution would not locate one guilty hand because the work had required many ordinary ones.

He kept his voice level. “The appeal needs an officer who did not extend the disputed classification.”

Dreya set the paper down. “There may be no such officer at the required rank.”

“The form permits delegates chosen by lot.”

“You trust a random panel with Annex 4?”

“I trust the conflict rule.”

Dreya looked toward the glass wall. Beyond it, the north corridor crossed the dark interior of the hub. No exterior light reached this office during third shift. Her reflection floated over the empty audit desks.

“I read Halma's sentence six years ago,” she said. “The one outside the box.”

Davit waited.

“I renewed the classification because the article required it. I thought the sentence absolved the procedure by admitting its limit.”

“It warned you about the limit.”

“Yes.” Dreya's answer carried no defense. “It did.”

She took the green pen from Davit's groove. Under Article 12, a challenged supervisor could contest a motion or accept temporary recusal pending independent review. Contest required a written answer within two station-days. Acceptance took effect at once.

Dreya checked acceptance.

Her signature crossed the bottom line at 23:41. She returned the pen to the groove with its point facing left, the orientation Davit used. Then she unclipped the rank-99 console key from her belt.

“This stays in the office safe until the panel reports,” she said.

Davit opened the safe ledger. Dreya placed the key in compartment four. He closed the compartment, turned the mechanical dial, and wrote the time beside her signature.

The terminal changed the anomaly status from DENIED to PENDING INDEPENDENT REVIEW. Lira's nine-signature petition appeared beneath it as a linked filing. The system scheduled a household-distribution check and a technical-evidence validation before the petition could reach the assembly. Nothing had passed. Nothing had been amended. One closed route had reopened under a different authority.

Dreya stood on the other side of the desk without her key.

“What will you do with Halma's addendum?” she asked.

“File it with the petition.”

“As evidence against her?”

Davit returned the hairpin to his breast pocket. “As evidence of what she measured and what she chose.”

Dreya nodded once. She left the third-shift room through the glass door and walked north toward the residential ring. Her pace stayed even until the corridor turned and hid her.

Davit sat again. Thirty-one dockets had entered his tray. Thirty required no further action. The last had changed custody, reviewer, and destination.

He opened the petition package and placed Halma's addendum after the founding waiver, not inside it. The year-0 document confessed the founders' choice. Recha's year-30 order sealed that confession. Halma's year-197 attestation proved the measurements continued and named amendment as the lawful remedy. Three documents. Three dates. No invented single origin.

At 00:06 the sealed-intake tube engaged. Air pressure drew the petition pouch into the wall. Davit heard it travel through the office conduit toward the independent panel.

The regulator continued beneath the floor. Halma's hairpin rested against his chest. Dreya's key rested in the safe. Lira's appeal no longer waited on the desk of the officer who had helped keep the disputed seal.

Before finalizing the docket, Davit opened the conflict log Dreya had surrendered. Forty-seven closure actions appeared beneath her key. Most concerned ordinary renewal items. Three touched the Engine case.

The first action closed Lira's anomaly at 09:42.

The second retained the restricted reference copy at 09:46.

The third approved Davit's audit access at 20:11, after the independent-review marker had already made recusal mandatory.

Dreya had not forged a timestamp. She had used valid authority in an invalid sequence. Davit wrote the distinction into his statement.

He tested whether the access grant could stand after recusal. Article 15 preserved ministerial actions that expanded review, provided a nonconflicted officer reauthorized them. It voided adjudicative actions that narrowed review. The denial therefore remained contested. The access grant could survive only under a clean key.

He called the panel clerk.

“I received access from the officer who must recuse,” he said. “I used it before reading the conflict.”

“What did you alter?”

“Nothing. I opened the file, created a view checksum, and marked a conflict.”

“Did you copy sealed text?”

“No.”

“Did you direct the outcome?”

“No.”

The clerk asked him to read the checksum. It matched the archive version. A panel key reauthorized his view at 22:03 and attached a notice that Dreya's grant could not serve as substantive approval.

The clean sequence mattered to Davit. Treating the first grant as good enough because it opened rather than closed would repeat the habit he was trying to interrupt: evaluating procedure by whether its outcome pleased him.

Next he audited Halma's attestation against the raw year-197 log. The addendum gave a rolling average of 4.1996 megawatts. The raw file gave the same number before rounding. Public line item seventeen gave zero. Halma's signature checksum matched her employment credential. The physical paper carried no later alteration.

Her reasoning occupied a narrow margin.

> Continuation prevents an uncontrolled public revision during open maintenance risk. Amendment remains available once independent load confirmation and resident standing exist.

Davit read it three times.

Halma had not claimed residents could never know. She had deferred disclosure until conditions existed. Fourteen years had passed. Independent load confirmation now existed. Nine signatures were being gathered. The conditions she named had arrived after her death.

That did not transform continuation into wisdom. It made the neglected remedy explicit.

Davit wrote:

> The year-197 signer named amendment as the lawful endpoint of continuation. Later officers preserved continuation while omitting that endpoint from working summaries.

He checked the five extension summaries. None quoted Halma's sentence. Two predated it. Three followed it. The later three renewed the seal but described no amendment condition.

The omission gave Dreya's recusal more weight. She had read the full addendum six years ago and approved one of those summaries as accurate. Now she had denied the petition route that met its stated conditions.

Davit opened Dreya's limits statement. He did not add an accusation. He appended the source reference and let the panel decide materiality.

At 22:41, Dreya's key attempted to open the conflict log. The system allowed read-only access under the recusal rule and prevented annotations. Davit could see the access event but not her expression behind the glass.

She remained connected for nine minutes.

When the session closed, a message entered his queue.

> I confirm the year-197 sentence appeared in the copy I reviewed six years ago. I did not carry it into the current summary. Add this acknowledgment to the limits statement.

Davit attached it.

Recusal had not made Dreya honest. It had made her participation inspectable and limited her ability to shape the file. That difference would have to suffice until the panel reviewed the merits.

He returned Halma's addendum to its gray sleeve. The brass hairpin stayed in his pocket. One object belonged to memory. One belonged to evidence. He would not use either to speak in Halma's place.

At 23:04, he completed the conflict checklist:

Prior participation identified.

Authority limited.

Contested denial preserved.

Expanding access independently reauthorized.

Personal relationship disclosed.

Source chronology corrected.

The terminal accepted the checklist and opened the petition package for panel use at first shift.

Davit entered the final docket note.

> Supervisory recusal accepted. Independent review commenced. Custody intact.

The public recusal notice posted at 00:20. It named the case, the prior actions creating conflict, the authority transferred, and the limits that remained. It did not publish Halma's sealed addendum before the panel cleared evidence access.

Three minutes later, Dreya filed an objection.

> Recusal accepted as to merits. Objection preserved as to characterization of denial as material participation before evidence sufficiency finding.

Davit read it without irritation. The objection did not restore her key. It preserved a question for review.

The panel clerk asked him for a response.

He wrote:

> No response on merits. Source log shows denial narrowed available review. Panel may determine materiality.

He did not argue that Dreya's motive made recusal necessary. The sequence did.

At 00:34, the panel confirmed transfer of three powers: evidence admission, score-penalty review, and assembly recommendation. Dreya retained ordinary supervision over unrelated Compliance work. Davit retained custody but could not vote on evidence sufficiency because of his relationship to Halma.

The division prevented recusal from becoming exile. It removed authority only where conflict operated.

Davit printed the authority table and taped it beside the glass door. Officers arriving at first shift would see who could sign which action. A verbal handoff would have left too much room for assumption.

When Dreya returned to collect her ordinary queue, she stopped at the table.

“You left me staff assignments,” she said.

“They don't touch the case.”

“You removed score review.”

“The panel did.”

“You requested it.”

“Yes.”

Dreya traced the printed row without touching it. “Good limits work both directions.”

Davit could not tell whether she meant approval or warning. The table did not require him to decide.

At 00:51, Lira's preservation hold and the recusal table appeared in the same public docket summary. A resident could see that the evidence remained sealed, the petition remained pending, and the officer who denied the anomaly no longer controlled the appeal.

Nothing about that state guaranteed the amendment would pass. It guaranteed only that failure would have to occur under an inspectable authority chain.

Davit closed the public preview and returned to the ordinary dockets. The hairpin pressed against his chest when he bent over the first page.

The next officer entering third shift read the table before taking a key.

The case had outgrown one supervisor's desk.

He signed with his name, title, age irrelevant, rank unchanged.

Chapter 19 — The Engine Math

Maren Otta invited Lira Vehn to the reservoir monitoring station on station morning day 131. She had traced Lira's clearance through the Compliance Office queue and found the provisional notation on her anomaly file still unresolved. Maren asked Lira to bring the original power budget. She wanted to work through the thrust verification together, something neither of them had done since Lira filed her Article 15 petition.

Lira arrived at 09:00 station time carrying a leather pouch with the power budget inside. Maren had prepared the console screen already. Three data sets sat visible on the amber display, and Maren gestured for Lira to sit at the adjacent terminal.

The first data set came from the founding power budget. It specified 4.2 megawatts of average bus-bar delivery for the Engine's correction cycle. The second came from the quarter-211 reservoir sensors and showed the same rolling local-day average at 840.3 Kelvin. The third showed the station's pulse schedule at 412 kilometers: atmospheric drag accumulated continuously, while the irradiator returned the lost orbital energy in timed prograde corrections. The documents had used output, thrust, and average as if they meant one quantity. Maren separated them before asking Lira to compare anything.

Lira studied the three data sets. The Engine's average energy delivery funded the impulse the station needed across each correction cycle. The quarterly reports recorded zero because the year-30 reporting relay overwrote the public field. Eleven founders had signed the original waiver in year 0. Recha Tallis moved it under seal in year 30. Halma Soren reaffirmed the measured-versus-public contradiction in a separate year-197 addendum. The dates no longer collapsed into one convenient origin.

Maren watched Lira's expression change as the data settled into understanding. She had seen Lira become still like this before, the focused quiet of a compliance officer who has just located a contradiction too large to unsee. Lira did not speak right away. Her fingers rested on the console edge and her breathing slowed to a rhythm Maren recognized as concentration. The station regulator hummed beneath them at a steady frequency that neither of them consciously tracked but both of them felt in the floor plates.

Maren opened the orbital mechanics reference. The current density model, projected area, and drag coefficient returned an average drag force of 3.91 kilonewtons for the modeled interval. Lira entered an orbital velocity of 7.66 kilometers per second. This time they multiplied force by velocity. The screen returned 29.95 megawatts of instantaneous orbital work during a correction pulse.

“That is not 4.2,” Lira said.

“It was never supposed to be,” Maren answered. She opened the duty record. The irradiator delivered corrections for 14.02 percent of each cycle and coasted between them while the reservoir restored the electrical store. Twenty-nine point nine five megawatts multiplied by 0.1402 returned 4.20 megawatts as the cycle average. The old reports had called that average continuous thrust. The phrase hid the duty fraction and encouraged the invalid division in the prior appendix. Maren labeled the fields separately: average bus-bar delivery, pulse power, drag force, orbital velocity, duty fraction.

Lira typed 4.2 megawatts into the verification form for line item seventeen. She described the discrepancy as an inherited reporting gate established by the year-0 waiver and year-30 transfer order, then continued by later extensions and Halma Soren's year-197 addendum. The public field remained zero pending amendment, so the form created an unresolved anomaly rather than pretending the line had already been corrected. Lira attached the restricted-copy checksum she had obtained through the renewal inspection workflow. The submission preserved the true measurement inside the case even while Article 4 still blocked public filing.

Maren opened a fresh worksheet on the console for the amendment filing calculations. She needed to prepare the numbers for the amendment committee's review before the committee could schedule a vote. Lira noticed the worksheet immediately and asked what the committee required. Maren explained that the amendment needed nine resident signatures to proceed to a Covenant vote, and each signature required a supporting data package with the thrust verification attached as an appendix. The committee's procedural rules required three distinct calculation sets for each amendment: a baseline calculation, a sensitivity analysis, and a cross-validation against independent data sources. Maren had already drafted the amendment text but needed all three calculation sets formatted to the committee's specification. Lira agreed to help prepare the calculations while Maren drafted the signature collection log and began preparing the cover letter for the committee chair.

The amendment filing required four separate calculation sections beyond the basic thrust verification: the thrust verification itself, the power budget reconciliation showing how the budget and sensor data matched the founding specifications, the orbital stability analysis examining what happened if the Engine output changed, and the suppression impact assessment quantifying the aggregate energy discrepancy between reported and actual output over the station's full operational lifetime. Maren assigned Lira the thrust verification and the power budget reconciliation sections while she handled the orbital stability analysis and the suppression impact assessment. They each took a half of the console and worked in parallel for the first time without speaking, Lira typing equations on the left portion of the screen while Maren drew orbital state vector diagrams on the right portion using the console's geometric rendering tools. Neither needed to coordinate because the calculations branched into independent domains from the shared baseline data.

Lira completed the verification in twenty minutes. She placed each quantity on its own line: modeled drag force, 3.91 kilonewtons; orbital velocity, 7.66 kilometers per second; correction-pulse work rate, 29.95 megawatts; duty fraction, 0.1402; average bus-bar delivery, 4.20 megawatts. The founding budget and quarter-211 sensors agreed on the average within 0.3 percent. The committee would see the calculation instead of a conversion factor invented to force the desired result.

Maren modeled a ten-percent loss in available average energy. The bus-bar would then supply 3.78 megawatts, enough to sustain the 29.95-megawatt correction pulse for only 12.62 percent of the cycle. Required duty remained 14.02 percent under the same drag state. The deficit therefore appeared as missing impulse, not as an invented instantaneous deceleration. Their orbital integrator showed resumed decay and reported the result as a range because thermospheric density varied. Lira placed the scenario in the appendix with its assumptions, model version, and uncertainty band.

For the suppression impact assessment, Lira and Maren worked together for the first time on the same screen in a genuine collaborative calculation. They needed to determine the aggregate energy discrepancy between the reported zero megawatts and the actual 4.2 megawatts over the 192 years since year zero when the station first reached operational orbit. Lira started with the time conversion on the left side of the screen. One hundred ninety-two years multiplied by 365.25 days per year gave 70,128 days of continuous station operation across the full station lifetime. Seventy thousand one hundred twenty-eight days multiplied by 24 hours per day gave 1,683,072 hours of deliberate suppression during which the quarterly reports registered zero megawatts instead of the actual 4.2 megawatts the sensors recorded each hour. Maren multiplied 1,683,072 hours by the 4.2-megawatt average and obtained 7,068,902.4 megawatt-hours. Lira divided by one million, not one thousand. The result came to 7.0689 terawatt-hours of unreported average delivery across the full operational history. They kept both units in the appendix so no later reader could turn a conversion error into rhetoric. Lira felt the weight of them settle in her chest, different from the weight she had carried before because these precise figures had names and timestamps and source document references rather than being abstract concepts disconnected from the daily reality of station life.

Maren added the suppression-impact calculation to the amendment draft. Lira caught a unit-label error in the first version: 7,068,902.4 megawatt-hours converts to 7.0689 terawatt-hours. They corrected the heading and retained the full megawatt-hour quantity beneath it. The appendix now held four calculation sections, source identifiers, model assumptions, and enough dimensional detail for another engineer to reproduce every conversion.

The assembly preparation required them to assemble the complete physical evidence package before the spring-equinox vote on day 132. Maren had gathered three sworn witness statements over the preceding weeks through conversations that required careful timing and discretion. The first was Osta Vehn's current testimony about the vibration she remembered after the Halt. The second was Ostad Richen's signed statement from the reservoir intake room. The third was Halma Soren's year-197 continuation addendum, provided by Davit Soren after he found it in the active conflict file. The year-0 founding waiver and Recha Tallis's year-30 transfer order remained separate instruments at the front of the package. Lira contributed the original power budget document stamped by the founding scribes in year zero, the current sensor verification results with the quarter-211 timestamp, the orbital stability analysis she had reviewed alongside Maren's, and the suppression impact assessment containing the 7.0689 TWh aggregate figure. They organized all documents in strict chronological order from year zero forward and assigned each a reference code following the committee's filing convention that combined the document type identifier with a sequential number within that type. Maren printed hard copies of all calculation sections on the archive printer located in the south spoke's technical records room. The pages came out warm from the thermal print mechanism and carried the distinctive smell of fresh toner that Lira associated with important documents for reasons she could not articulate. Lira stacked them on the console in numbered sequence with the reference code facing up so they could verify the ordering at a glance before sealing the package.

Lira stood at the archive printer holding the first printed sheet of the amendment filing with the thrust verification on the top page. The line reading verified thrust at 4.2 megawatts continuous looked back at her from the thermal paper as if it were a statement of obvious fact that ought to have been obvious from the beginning. She turned to Maren who was gathering the signed witness statements from her clearance pouch and cross-checking each signature against the reference codes Lira had assigned during the filing earlier that morning. Together they laid all documents on the reservoir monitoring console in a single ordered stack with the power budget on top, then the sensor data printout, then the orbital equations, then the suppression calculations, then the orbital stability footnotes, and finally the three witness statements at the bottom. The stack measured four centimeters high when placed on its edge and contained sixteen separate pages spanning nearly two centuries of suppressed truth compressed into numbered documents they could physically carry to the assembly floor on day 132 if the committee advanced the amendment to a formal vote.

Lira and Maren reviewed the complete package together at 11:30 station time with the console's amber light washing over both of their faces evenly. Maren read the amendment text aloud to Lira from the printed draft she had been revising that morning while Lira verified each cross-reference against the source documents spread across the console surface and the nearby shelving where Lira had placed the backup copies of the archived sensor records. The amendment stated that the Covenant's preamble description of a station that had lost its drive was factually incorrect based on the evidence and calculations contained in the attached verification package. The amendment stated that the Engine had produced 4.2 megawatts of continuous thrust without interruption across the full 192-year operational history of the station since it achieved stable orbit at year zero. The amendment stated that quarterly verification reports filed for eleven consecutive quarters had recorded zero megawatts in deliberate and systematic contradiction of the continuous sensor data archived in the raw reservoir sensor records that Maren had accessed in violation of the quarter-211 suppression directive.

The amendment stated that eleven founders signed the waiver in year 0, Recha Tallis moved it under seal in year 30, five later extensions preserved the classification, and Halma Soren documented the continuing contradiction in year 197. The public reporting gate reproduced zero while raw telemetry retained the measured Engine output. The amendment formally proposed that the Covenant's foundational documents be corrected to reflect the Engine's actual measured continuous output of 4.2 megawatts and that the suppressed reservoir sensor records be restored to the official archive where every resident could access them without requiring a rank-99 clearance to unseal the annex that contained the founding engineers' original specifications.

Lira finished reading the amendment text aloud and set the printed copy on the console between herself and Maren. The pages fanned slightly in the station's gentle air circulation. Neither of them spoke immediately because the silence held the weight of everything they had confirmed and prepared across the preceding hours. The reservoir's circulation pump hummed behind the sealed annex wall on the other side of the south spoke, and the Engine maintained its steady 4.2 megawatts output through every second of the station's 192 years of continuous unbroken operation. Before they sealed the calculation appendix, Lira demanded a dimensional audit that ignored every label inherited from the old reports.

Maren opened a blank worksheet and entered only base quantities.

Force: 3,910 newtons.

Orbital velocity: 7,660 meters per second.

Their product returned 29,950,600 watts. The displayed 29.95 megawatts followed after conversion.

Duty fraction: 0.1402.

The product returned 4,199,074 watts, within the sensor and model tolerance around the 4.2-megawatt rolling average. They wrote the unrounded intermediate result under the headline value so the assembly would not mistake formatting for exactness.

Next they checked the duty fraction from time rather than energy. The local correction record contained 3.70 station-hours of active field across a 26.4-hour local cycle. Dividing 3.70 by 26.4 returned 0.14015. The small difference from 0.1402 came from rounded start and stop timestamps. High-resolution controller ticks returned 0.140198.

Two derivations now supported the same fraction.

Lira asked whether atmospheric drag actually remained at 3.91 kilonewtons for the whole cycle.

“No,” Maren said. “That is the modeled interval average. Density, orientation, and solar state move it.”

“Then the pulse-power figure also moves.”

“Yes.”

“Put the range in the headline table.”

Maren added 28.7 to 31.2 megawatts for the sampled year, with 29.95 as the current interval. The 4.2-megawatt quantity remained the design rolling average, not a constant instantaneous work rate.

They tested the energy store next. Reservoir draw during coast had to support the correction pulse without implying energy appeared inside the irradiator. The store log showed 110.9 megawatt-hours accumulated across the cycle and 110.8 discharged through correction and conversion losses. The difference returned as measured waste heat and control loads. The accounting closed within instrument uncertainty.

Lira traced each path with her pencil.

Thermal recovery to electrical store.

Electrical store to irradiator field.

Field interaction to orbital work during pulse.

Conversion loss back to reservoir and radiators.

The model still depended on the irradiator mechanism the founders had never fully explained. The appendix could not turn that unknown into settled physics. Maren labeled it observed field coupling and attached two centuries of calibration evidence. The amendment claimed continuity of measured operation, not a complete theory of the machine.

That limit improved the document.

For cross-validation, they selected a correction window from year 88, one from year 147, and the current year. Hardware and office titles differed, but the raw fields survived in compatible units. Each window showed pulse work near thirty megawatts, duty near fourteen percent, and a local-cycle average near 4.2.

The public report showed zero for all three.

Lira asked the console to reproduce the old appendix's invalid conversion. It divided force by velocity and multiplied by a unitless factor labeled 1 MW per 0.238 kN. The result landed near 4.2 only because the factor encoded the desired answer.

They preserved the failed calculation in an error note.

> Superseded method: dimensional inconsistency. Force divided by velocity does not yield power. Conversion factor lacks independent physical definition and shall not be used.

“Will putting the error in the package weaken us?” Maren asked.

“Hiding it would.”

The committee could now see how the wrong result had once appeared persuasive and how the corrected cycle average related to the same operational figure without violating basic units.

At 12:18, they handed the worksheet to a panel engineer who had not participated in the petition. The engineer repeated the multiplication on a mechanical calculator, checked the timestamp fraction from raw ticks, and sampled one historical correction window chosen after receiving the file list.

The independent result came to 4.203 megawatts average for that sampled window.

The engineer signed:

> CALCULATION REPRODUCED. METHOD DIMENSIONALLY CONSISTENT. IRRADIATOR COUPLING TREATED AS OBSERVED SYSTEM BEHAVIOR, NOT EXPLAINED FIRST PRINCIPLE.

Lira placed that qualification beside the verification rather than in fine print. Hard numbers did not become honest by pretending uncertainty had vanished.

The assembly would receive 4.2 megawatts as a supported cycle average, 29.95 megawatts as a current pulse work rate, 3.91 kilonewtons as a modeled interval-average drag force, and 14.02 percent as the duty fraction joining them. Each number had one job.

Before the package left Engineering, Maren separated the terms residents were most likely to confuse.

> Cycle average: energy-equivalent orbital correction work averaged across one local correction cycle, currently 4.2 MW.
>
> Pulse work rate: instantaneous force multiplied by orbital velocity during an active correction, currently 29.95 MW.
>
> Duty fraction: portion of the cycle during which pulse work occurs, currently 14.02%.
>
> Public field: rounded cycle average, sourced to raw telemetry.

Lira tested the definitions by asking a records clerk with no engineering training to explain them back. The clerk said the Engine worked harder for part of the cycle and rested the correction field during the remainder, producing an average lower than pulse power.

“Does the machine stop during the coast interval?” Lira asked.

“No. Pumps, regulation, and storage continue. Only the orbital correction pulse stops.”

“Does 4.2 describe force?”

“No. It describes averaged work rate in megawatts.”

The explanation held.

Maren added the definitions to the public appendix and linked each term to the raw equation rather than relying on the glossary alone. If a future officer changed a label, the units would still expose the relation.

They signed the appendix separately from the amendment text. A mathematical correction discovered later could be repaired through technical review without silently altering the ratified article.

The math package was finally ready to leave Engineering. The amendment was properly formatted and ready for formal filing with the compliance bureau. The assembly now had the numbers, the calculations, and the three witness statements needed to prove what the Covenant's preamble had denied for over a century and what the quarterly reports had concealed for the past eleven filing periods. Maren placed the printed amendment beside the signature collection log she had prepared earlier that morning on a separate sheet of archival paper bearing each of the nine required signature lines with space for the date and the voter's rank insignia. She told Lira the committee would need both the amendment document and the signature collection log and the physical witness statements before the spring-equinox vote on day 132 could proceed according to the Covenant's amendment procedure. Lira nodded and placed the reference code sheet on top of the stack so the filing clerk would see it first when she arrived the next morning. The amendment had crossed the threshold from an idea on a console screen into a collection of physical documents that could move through the station's governance apparatus. Each calculation had been verified against independent sensor data and cross-checked by two compliance officers who understood the meaning of every number on every page. The truth lived in the stack of papers on the amber-lit console and the truth would carry the assembly's deliberation forward into whatever came next for the station and the Covenant and every resident who deserved to know that the Engine had been running continuously since year zero all along.

Chapter 20 — The Covenant Vote

The independent panel validated the ninth petition signature at 12:06 on the spring equinox. Two hours remained before the registered assembly.

Lira Vehn received the finding at the north entrance to the hub corridor. Her clearance pouch vibrated once. She opened the notice while residents passed around her in gray work coats and formal renewal bands.

> PETITION SUFFICIENCY: CONFIRMED.
>
> Eligible signatures: 9.
>
> Household distribution: compliant.
>
> Evidence standard: met.
>
> Motion status: scheduled for ratification under Article 15.

Nine signatures had established standing. They had not decided the amendment. The assembly required sixty percent of all registered residents present, not sixty percent of those willing to raise a hand after the speeches.

Lira read the notice again and forwarded it to each petitioner. Maren's receipt arrived first. Davit's followed from the third-shift audit desk. Eska acknowledged with a single checked box. Osta's terminal required an assisted receipt, which the records office logged. Ostad sent no message beyond the confirmation checksum. Pello and the two ward representatives completed the chain.

At 12:11, the panel released the evidence cylinder to the assembly clerk.

The package contained the year-0 founding waiver with eleven signatures. Behind it lay Recha Tallis's year-30 transfer order. Five seal extensions followed. Halma Soren's year-197 continuation attestation came after them. The current telemetry, orbital-work appendix, witness statements, and scoring-conflict disclosure completed the file.

No single document bore the blame. The order mattered.

Lira carried the cylinder through the central spoke with the clerk beside her. The hub corridor had been opened end to end and divided into four precincts. Attendance gates at each spoke counted registered resident keys. Folding seats filled the plaza and the adjoining galleries. Residents who could not stand joined through ward rooms connected to the same recorded vote. The public registry closed at 13:30 with 4,096 residents present or authenticated.

The number matched the whole station census. Lira did not mistake total attendance for consent.

Maren waited near the Engineering precinct with a small console showing the current raw value. 4.200 megawatts. The public quarterly field beside it still showed 0.000. The two values would remain adjacent until the assembly acted and the final filing changed the reporting gate.

“You fixed the appendix?” Lira asked.

Maren turned the screen. “Force times orbital velocity. Twenty-nine point nine five megawatts during correction. Four point two average at fourteen point zero two percent duty.”

“And the ten-percent scenario?”

“Range only. No invented fifty-six-kilometer quarter.”

Lira checked the printed sheet. Every quantity carried units. Every model result carried assumptions. She put it back under the clip.

Davit approached from the north spoke. His badge read rank 88. Halma's brass hairpin made a faint line beneath his breast pocket. He had signed the petition as a custody witness before finding her year-197 addendum. Now the addendum lay in the cylinder under its own date.

“Dreya?” Lira asked.

“Recusal remains active through filing,” he said. “She is attending as a resident in the administrative precinct. She has no supervisory key.”

Lira looked toward the upper gallery. Dreya sat in the third row without silver piping at her cuffs. Her rank-99 key remained in the Compliance safe. She held an ordinary resident voting plate.

The gavel sounded at 14:00.

The assembly supervisor read the spring-equinoctial renewal order. The customary preamble would not be renewed before the Article 4 motion. The panel finding had placed amendment first because ratification would determine which language residents were asked to affirm.

The supervisor read the petition trigger into the record.

“Article 15 requires nine eligible resident signatures, no more than four from a household, to compel consideration of a structural amendment. The independent panel confirms nine eligible signatures and compliant household distribution. Ratification requires affirmative votes from sixty percent of registered residents present.”

The two thresholds entered the public transcript without contradiction.

Lira was called to the lectern as petitioning officer. She set down the copper cylinder and did not open it until the assembly clerk confirmed all three custody strips. The clerk broke the lead seals, removed the gray folder, and placed the supporting instruments under brass weights.

The old bronze Covenant plaque hung above them. Its Article 4 heading read Continuity of Founding Intent.

Lira began with the current measurement.

“The raw bus-bar average for this local day is 4.200 megawatts. The public quarterly field is zero. The values come from the same physical source. A reporting gate substitutes the second for the first.”

Maren transmitted the display to every precinct screen. Two columns appeared: RAW and PUBLIC. The first moved by thousandths as the rolling window updated. The second did not move.

Lira held up the founding waiver.

“Eleven founders signed this in year 0. It acknowledges that the Inertial Engine remained operational after the Halt and authorizes temporary concealment while the station stabilized.”

She placed it down and lifted Recha's order.

“In year 30, Recha Tallis moved the waiver and associated power budget into Annex 4. That order created the reporting route still active today.”

The five extension tabs opened in sequence beneath her hand.

“Later officers extended custody and classification. Those extensions modified Article 4's administration without changing its substantive claim.”

She lifted Halma's addendum last.

“In year 197, Halma Soren measured 4.2 megawatts in the raw field and zero in the public field. She attested to both. She did not sign the founding waiver, and she did not create the year-30 gate. She chose to reaffirm it. Her note states that procedure may preserve a contradiction but cannot make it true.”

Davit did not look away from the page.

Lira returned the addendum to its chronological place.

“The petition proposes a resident-initiated substantive change. It does not erase the founding choice, the year-30 burial, the extensions, or Halma's later decision. It makes those records public and requires measured Engine telemetry to enter the public ledger without substitution.”

The assembly supervisor opened the questioning period.

The first question came from the administrative precinct. “Does disclosure change Engine operation?”

Maren answered from the technical witness station. “No control pathway connects the public ledger to the regulator. The amendment changes reporting and archive access. The Engine's controller continues under Engineering safety limits.”

A second question came from East Ward. “If the Engine stops, how long before reentry?”

Maren did not offer a ceremonial number. “The model range depends on thermospheric density and station attitude. The current integrator projects centuries, not a fixed date. The amendment appendix publishes the assumptions and uncertainty band. It does not claim certainty the sensors do not support.”

A third question came from the Residential Ward representative. “Why should residents trust the same offices that filed zero?”

Lira answered. “The text does not require trust in an office. It requires source-linked telemetry, public checksums, reviewable seals, and a current safety finding for any future restriction. Trust may follow inspection. It cannot replace it.”

A maintenance worker asked whether the old Covenant would become illegal to possess.

“No,” Lira said. “The superseded text and every extension remain public history.”

A clerk asked whether nine petitioners could force another amendment immediately.

“They can force consideration if a panel validates standing and evidence,” the assembly supervisor answered. “They cannot ratify it. Sixty percent remains required.”

The questioning lasted fifty-three minutes.

Dreya requested the floor as a resident at 15:09. The clerk logged her recusal and disabled any supervisory authority attached to her statement.

She stood in the upper gallery. “I renewed the classification six years ago. I later denied Officer Vehn's anomaly inside a closed verification cycle. An independent finding determined that my prior participation required recusal from the appeal. That finding was correct.”

The corridor held still.

Dreya continued. “I believed the seal protected the Compact from a fact it could not absorb without fracture. That belief gave me a reason. It did not give me evidence that residents lacked the capacity to decide. I will vote as a resident. I ask that no one treat my vote as an instruction.”

She sat.

The statement absolved no one. It placed her choice in the same record as the documents.

At 15:23, the supervisor closed debate and read the proposed article aloud.

> Article 4. Continuity of Founding Record.

>

> The station shall preserve founding intent by preserving the complete evidence of founding action. The Inertial Engine has remained operational since year 0. Its measured thermal delivery, correction duty fraction, and station-keeping output shall be recorded in the public ledger without substitution. Founding records shall remain accessible except where a current and particular safety finding supports a limited seal subject to review.

The screens displayed the text in every precinct.

The vote used registered plates rather than a head count. Each resident inserted a key and selected affirm, oppose, or abstain. The precinct terminals counted locally and printed paper rolls. Ward-room votes entered through separate authenticated channels and produced their own rolls. No central total appeared until clerks carried every roll to the assembly desk.

Lira voted affirm at the petitioning-officer station. Her provisional score of 71 appeared on the credential check but did not weight her vote. Every resident counted once.

Maren voted in the Engineering precinct. Davit returned to the north spoke and voted there. Eska checked her selection twice before submitting. Ostad rested both hands on the terminal rail and pressed affirm. Osta's assisted vote entered under a two-clerk witness protocol. Pello voted without looking toward Lira. The ward representatives returned to their constituents before using their own plates.

Dreya stood in the administrative line.

The vote closed at 15:47. Four precinct supervisors and two ward-room clerks brought sealed rolls to the central desk. The chief clerk ran the addition once by machine and once by hand. A second clerk repeated it from the bottom of the columns upward.

No one announced intermediate totals.

Lira kept her hands on the lectern. The regulator pulse moved through the floor beneath her. At the telemetry console, the public field still showed zero.

At 16:11 the chief clerk passed the total to the assembly supervisor.

“Registered residents present or authenticated: four thousand ninety-six. Affirmative: two thousand five hundred twelve. Opposed or abstaining: one thousand five hundred eighty-four. Affirmative percentage: sixty-one point three.”

The required threshold stood at sixty.

The supervisor struck the gavel once.

“Article 4 is ratified as read.”

No cheer arrived at first. Residents looked at the precinct screens as if the numbers might change when watched. Then a sound moved through the corridor, not celebration but thousands of people exhaling without agreement about what came next.

The amendment had passed by fifty-three votes beyond the required threshold. It had also failed to persuade 1,584 residents. The difference stayed in the record. Ratification created law, not unanimity.

Lira signed the petitioning-officer ratification certificate. The assembly supervisor signed the vote total. The chief clerk attached the six precinct rolls. No one added another petition signature. The nine lines had completed their job before the vote.

Davit approached the evidence table to witness custody. He did not sign for Halma. He verified that her addendum returned to the gray folder under its own name. The clerk rolled the amendment around a cedar core and placed it in the copper cylinder. The evidence folder went into a separate case because the paper could not tolerate the cylinder's curve.

Three new custody strips crossed the packages. Assembly to Compliance. Compliance review. Compliance to Public Archive.

The public telemetry still read zero. Ratification authorized the change, but the archive filing and configuration checksum had not occurred. Lira appreciated the delay. A vote did not reach backward and pretend every system had already changed.

Maren stood beside her as residents began leaving through the four spokes.

“Sixty-one point three,” Maren said.

“Enough.”

“Barely.”

“Enough and barely can both be true.”

Davit lifted the evidence case. “Third-shift intake takes custody at 20:52. The amendment stays sealed until then.”

Lira took the copper cylinder. Her provisional score remained 71. Her East Ward reassignment remained active. Dreya remained recused. The reporting gate remained closed. Ratification had not solved those procedural states by magic. It had created authority for the next filings to solve them in order.

They left the hub through the north spoke. Behind them, clerks collected voting plates and preserved the paper rolls. The bronze plaque still displayed the old Article 4 because the permanent charter had not yet been altered. Above the telemetry station, RAW showed 4.201 and PUBLIC showed 0.000.

The corridor hummed beneath Lira's boots.

For one hundred ninety-two years, the Compact had taught residents that the hum belonged to auxiliary reserves. The assembly had now adopted language naming its source. The words would acquire force only when the cylinder reached custody, the supporting record opened, the article entered the master charter, and the gate accepted its new configuration.

The packages reached the Compliance intake at 20:52.

Davit received them under the recusal order. A panel clerk stood beside him, and Dreya watched through the glass from the public side. Her ordinary resident vote had ended at the assembly. Her supervisory key remained disabled for this case.

The intake required three comparisons before custody could continue.

Davit counted the six precinct rolls against the assembly certificate. The clerk compared each roll total with the public transcript. Lira verified the amendment text against the language projected during the vote. A single punctuation difference appeared in the archive copy: one semicolon where the ratified display had shown a period.

The filing terminal rejected the package.

No one called the difference harmless. The text had been ratified as displayed.

Lira opened the locked assembly image and located the source line. The copy clerk had joined two sentences while preparing the archival transcription. The assembly supervisor issued a correction certificate identifying the mechanical variance. Two precinct clerks witnessed the replacement page. The old mispunctuated copy stayed in the file marked REJECTED TRANSCRIPTION.

At 21:27, they repeated all three comparisons.

The text matched.

The cylinder received a new custody strip. The evidence folder retained its own seal. Davit entered the transfer from Assembly to Compliance, then the clerk entered Compliance review pending public archive filing. No one used Dreya's key.

Lira stood at the counter where she had first returned the restricted folder. Her provisional score still read 71 on the intake screen.

“Does ratification close rule 6.4?” she asked.

“Not until the filing validates the anomaly,” Davit said. “The score system follows current record state. The current public article still contains the old rule for a few more hours.”

“So the station has voted the article wrong and still measures me under it.”

“Until the amendment reaches the master charter.”

She wrote the answer in her notebook. It irritated her, but the delay followed a visible dependency rather than a hidden discretion.

The panel clerk inspected the score notice. “When the charter changes, the recalculation must cite the validating instrument. No manual restoration.”

Davit added that requirement to the intake cover.

Maren arrived with the final technical checksum. The raw telemetry had moved from 4.201 at assembly close to 4.197 at intake. The variation remained within tolerance. The public field stayed zero. She attached both values so the filing would show that ratification had not itself changed machinery or reporting.

“People are already asking why the screen hasn't changed,” she said.

“Tell them the configuration has not been filed,” Lira answered.

“They think the vote failed.”

“Then show the custody receipt and the scheduled activation window.”

Maren sent those records to the public notice board. The notice explained each remaining step: Compliance text comparison, archive acceptance, reporting-gate checksum, dual-channel verification. It promised no instantaneous transformation.

At 22:10, a challenge arrived from the administrative precinct. Three residents alleged the assembly had lacked a valid registered count because ward-room authentication closed four minutes later than the hub precincts. The vote could not proceed to filing while a timely procedural challenge remained unanswered.

Lira read the challenge twice. The fifty-three-vote margin did not permit carelessness, but margin size did not determine whether a process defect mattered.

The chief clerk produced the authentication log. Hub voting closed at 15:47. Ward-room terminals stopped accepting selections at the same timestamp but completed cryptographic receipts by 15:51. No vote entered after 15:47. The later time recorded receipt completion, not ballot acceptance.

Two ward clerks confirmed the sequence under separate signatures. The panel denied the challenge at 23:02 and published the log.

Dreya had signed the challenge as one of the three residents.

From the public side of the glass, she met Lira's eyes. “It needed testing,” she said.

“Yes,” Lira answered.

The answer did not make them allies. Dreya had used the same reviewable procedure now available to everyone. The challenge failed because evidence failed to support it, not because the wrong person raised it.

At 23:11, the intake terminal cleared the amendment for final filing at first shift. Davit locked the copper cylinder in the active-transfer cabinet. The evidence case occupied the shelf beneath it. A camera recorded both seals.

Lira received a custody receipt with four states:

RATIFIED.

TEXT VERIFIED.

CHALLENGE RESOLVED.

PUBLIC FILING PENDING.

She added the receipt to her notebook behind the petition validation. Nine signatures had opened the question. Sixty-one point three percent had answered it. The remaining work concerned making that answer operate in the archive without altering what residents had approved.

Before leaving, Lira compared the ratified text with the version printed on every precinct ballot. All punctuation, thresholds, and source references matched. The panel had rejected two floor amendments during debate, so neither appeared in the final instrument. One would have opened every maintenance channel without a safety seal. The other would have erased the prior Article 4 rather than superseding it.

The vote had not authorized either rejected idea.

The clerk attached the comparison checksum to the copper cylinder. If the first-shift filing differed by even one character, the archive would refuse activation and return the instrument to the assembly record.

Lira signed the comparison as petitioner. A ward clerk signed independently. Davit observed custody without certifying the text because he had already signed the petition and carried a conflict disclosure.

The roles stayed separate even after the result became clear.

At 23:20, she left the Compliance Office. The bronze Covenant plaque in the hub still displayed the old Article 4. RAW still read 4.197. PUBLIC still read 0.000.

The unequal numbers no longer described a politically uncontested state. They described a completed vote waiting on a traceable filing.

Lira carried the first of those required objects toward the Compliance Office.

Interlude III — The Burial

Halma Soren died in year 198 with the year-197 addendum still under active seal.

The addendum did not create the seal. Eleven founders had signed the emergency waiver in year 0. Recha Tallis moved it into Annex 4 in year 30. Five later extensions changed custody and duration. Halma inherited all of that when she became Lead Load Engineer.

Her contribution consisted of two measurements and a choice.

Raw rolling average: 4.200 megawatts.

Public line item seventeen: 0.000 megawatts.

She signed beneath both values. In the margin she wrote that procedure could preserve a contradiction without making it true. Then she recommended continuation until a resident amendment could replace the reporting rule without interrupting Engine control.

No amendment followed while she lived.

Davit Soren received her hairpin and ordinary personal effects after the funeral. He did not receive the addendum. The instrument belonged to the Compliance conflict file, not to Halma's household. Its existence remained outside the inventory sent to him.

By year 200, the station possessed living memory of the Halt and called most of it dead.

Osta Vehn appeared in the census because a ward clerk had never accepted the deletion order attached to her founder file. She carried no authority and seldom spoke about the first year.

Ostad Richen maintained reservoir valves under an employee exception while his public death certificate remained active. He had not signed the founding waiver, but he had watched the correction system operate before its first decade renewal.

Edik Hain lived in Founder Reserve C-11. The population index called him dead in year 2. A structural-consultant cross-reference continued sending rations to his sealed compartment.

The records did not merely forget witnesses. They sorted each into a category that made testimony difficult to request.

Halma's death added a different kind of silence. She had not been hidden. She served openly, filed reports, and died with a public funeral. Her decisive record vanished because offices classified the paper rather than the person.

In year 200, an archive clerk performed the first posthumous custody check. The clerk opened the conflict file and counted its instruments.

Founding waiver.

Year-30 transfer order.

Five seal extensions.

Year-197 continuation addendum.

The clerk compared page numbers, checksum copies, and cotton ties. Nothing had gone missing. The custody form offered three closure options: transfer, destruction under expiration, or retain under active authority.

The clerk selected retain.

A note asked whether the signing officer's death affected the instrument. The clerk answered no. Halma's authority had been valid when she signed. The addendum therefore remained valid evidence of the office's prior decision.

That answer protected the paper and delayed its meaning.

Davit worked third shift elsewhere in Compliance. He processed audit results that evening and passed within thirty meters of the room containing Halma's addendum. His badge at rank 88 did not open the conflict cabinet. No active case named him. He carried her hairpin in his breast pocket.

The station's public renewal guide summarized Article 4 in six lines. It said the founding record remained intact. It did not list the extensions. It did not mention the raw output. It did not explain that a dead engineer had measured the contradiction three years earlier and recommended a lawful remedy no one pursued.

In the reservoir ward, the Engine completed another correction pulse. The raw archive stored pulse power, duty fraction, temperature, and average bus-bar delivery. The public ledger stored zero.

The conflict file preserved both.

Burial on the station required no soil. It required only a valid container, a custody rule, and enough separate offices that each could believe another office owned the unresolved question.

The year-200 clerk resealed the file.

Cotton tape crossed the folder. A lead wafer received the current office mark. The clerk entered the checksum into the archive terminal and returned the folder to shelf C-4.

The work met procedure.

Halma's addendum remained available to a future conflict review. Ostad's biometric remained available to a future status correction. Edik's utility trail remained available to a future registry audit. Osta remained available to anyone willing to ask a question without turning her memory into proof she had to defend alone.

Nothing had been destroyed.

That fact allowed the Compact to call itself faithful to the record.

Nothing had been joined.

That fact allowed the concealment to continue.

Eleven years later, Lira Vehn would connect the public zero to the raw 4.2. Davit would see Halma's signature in chronological place, not at the origin. Eska Hain would compare death certificates with active utility records. The amendment would make those links public.

In year 200, the links existed only as possible work.

The archive light went dark at the end of second shift. The Engine's next pulse entered the raw ledger. Davit nearly encountered the file in year 202. A checksum migration flagged the year-197 addendum because Halma's signature used an older employment credential. The exception entered third shift.

Davit saw the surname in the queue header. Before he opened the case, the archive system rerouted spouse-linked exceptions to another reviewer under a privacy rule. He accepted the transfer because privacy appeared to protect him from auditing his own household.

The receiving reviewer validated the credential and closed the exception. The reviewer did not read the content because the checksum alone had failed.

The system behaved correctly. It protected Davit from an undisclosed personal conflict and preserved the paper. It also moved the only clue away before he could know a conflict existed.

That near encounter remained in the access log. In year 211, Davit would read it after learning the addendum's contents. He would not blame the privacy rule. He would ask why the system had no reciprocal notice telling him that a spouse-linked historical instrument had been excluded from his queue.

Halma's former Engineering terminal survived until year 204. When technicians replaced it, they imaged the local cache. The cache contained a draft addendum with one sentence struck out:

> Continuation shall expire at the next independent confirmation of sustained output.

The signed version replaced shall expire with amendment remains available. No record explained the change.

The draft did not override the signed instrument. It revealed that Halma had considered an automatic endpoint and chosen a discretionary one. The cache image entered shelf C-4 under the same seal.

Nothing in the year-200 custody form required the clerk to compare drafts. Preservation created possibility without interpretation.

By year 205, Dreya Selna had read the complete set and approved the fifth extension summary. She knew the endpoint language had narrowed between draft and signature. Her summary said only that current conditions supported continuation.

The archive kept the draft, the signature, the summary, and the access log.

Each object remained separate until the petition made their relationship a public question.

At the end of year 200, the archive ran its annual media test. Every sealed image opened. Every checksum matched. The result carried no judgment about the seal, but it prevented physical decay from deciding the issue before residents could.

Preservation could serve concealment and later serve correction. The function changed when access and authority changed.

Line item seventeen accepted zero again.

Part IV

Chapter 21 — The Third Witness

Eska Hain began with a dead man who had voted yesterday.

Ostad Richen's public death certificate said he died in year 63. His Article 15 petition signature carried a live biometric check from year 211. The independent panel had resolved the conflict for petition eligibility, but its finding did not correct the Permanent Archive. Ratification changed Article 4. It did not repair every false record created under the old article.

Eska opened a new list at 06:30.

1. Ostad Richen death certificate.
2. Active maintenance roster.
3. Petition biometric receipt.
4. Archive correction authority.
5. Related records using same cause code.

Her score remained 99. The number gave her access to the population registry and responsibility for resolving direct conflicts. It did not permit her to erase a death certificate. Archive correction required preserving the false instrument, attaching the contradictory evidence, and changing the current status.

She preferred that rule. A corrected lie should remain visible as a lie.

Ostad's certificate carried cause code D-17: natural decline, body unavailable, supervisor attestation. No next-of-kin receipt appeared. No body-custody record appeared. No medical closure appeared. The verification officer signature belonged to Recha Tallis.

Eska searched D-17 across the founding generation.

Nine results appeared.

Six had ordinary supporting records. One belonged to Ostad. One belonged to a founder who had actually died during a hull repair and whose body had never been recovered. The ninth belonged to Edik Hain.

Her grandfather.

The public record placed Edik's death in year 2. Eska had learned the date as a child and stopped asking questions after her mother told her that founding work took people early. The certificate carried D-17, Recha's attestation, and no medical closure.

Eska wrote a sixth item on her list.

6. Edik Hain status conflict.

She opened the active utility ledger. Dead residents could not draw water, heat, food, or air under their names. Ostad's unit used a maintenance exception linked to his employee number, which allowed him to remain active while the public population index called him dead. Eska searched Edik's founder number.

No residential unit appeared.

She searched the employee cross-reference. The system returned a continuing structural-consultant stipend paid in ration credits, not currency. The destination read FOUNDER RESERVE C-11. Access classification: maintenance safety.

A dead founder had received rations every station month for two hundred nine years.

Eska checked the thermal register. Compartment C-11 drew enough heat for one person. Water reclamation matched one person. Air exchange matched one person. No medical visits appeared, but sealed reserve compartments carried their own clinic route.

She did not laugh. Lists usually made impossible things manageable. This list made the impossible acquire an address.

The public archive opened at 07:00. Eska submitted Ostad's status correction first. She attached his current maintenance roster, petition biometric receipt, and independent-panel finding. The archive changed PUBLIC STATUS: DECEASED to PUBLIC STATUS: LIVING; PRIOR FALSE DEATH CERTIFICATE RETAINED FOR REVIEW.

One correction.

Edik required direct confirmation. Utility use could prove occupation, not identity. Eska filed a rank-99 registry visit under the living-status audit provision. The terminal accepted the request and produced a route through the south maintenance ward.

She carried no committee form. The amendment vote had ended. Edik could not become a retroactive petition signer, and Eska would not turn him into a missing ninth witness after nine verified residents had already established standing. He belonged to a different question: what the station had done to keep a founding witness alive and publicly dead.

At 07:42, Eska reached the south spoke intake corridor. Ostad worked at an open valve cabinet with his sleeves rolled above the wrist. He saw her clearance pouch and nodded toward the lower access stairs.

“C-11?” he asked.

Eska stopped. “You know?”

“I carry filter cartridges there twice a year.”

“You knew Edik Hain was alive?”

“I knew a resident lived in Founder Reserve. He did not give me a name.”

“Did you ask?”

“No.” Ostad tightened a brass coupling. “Maintenance asks what a room needs.”

Eska added his answer to her mental list and disliked it. The habit that kept equipment working had also helped people become categories.

“Will you show me the route?”

Ostad closed the cabinet. “I will show you the pressure door. Your key decides the rest.”

They descended two flights beside the reservoir return conduit. The air warmed with each level. Sodium-edge light leaked through an inspection slit, yellow against the metal wall. Ostad led her behind a pump housing to a narrow door marked C-11 FILTER RESERVE. The label explained supply deliveries without explaining habitation.

A rank reader waited beside the latch.

Ostad touched the door with two fingers. “The resident receives water at first shift, food at second, and filter service every sixth month. No one enters without a registry audit or a medical call.”

“How long have you known?”

“Since year 81.”

“You were publicly dead by then.”

“Yes.”

“Did Recha arrange both records?”

Ostad looked toward the glowing inspection slit. “Ask the person behind the door about his record. I can answer for mine when the archive calls me.”

He returned up the stairs.

Eska placed her key against the reader. The door requested audit case, identity claimed, and evidence basis. She entered Edik Hain, D-17 conflict, continuous stipend and utilities. The latch released.

The compartment beyond had once stored filter assemblies. Shelves lined both walls, but half now held books, folded clothes, a kettle, and sealed tool cases. A narrow bed occupied the rear. The air smelled of warm metal and tea.

An old man sat at a table beneath a task lamp. Age had drawn his shoulders forward, but his hands lay square on an open structural diagram. A founder's iron ring encircled his right thumb.

Eska knew the ring from the first page of the ledger. Eleven thumb impressions surrounded the signatures. Edik's carried a flat notch at the top.

The man looked at her rank badge before her face.

“Registry,” he said.

“Yes.”

“Then either I have died again or someone has noticed I did not.”

Eska took out her list. “Edik Hain?”

“That name belongs to me.”

“I am Eska Hain.”

He studied her. “Talla's granddaughter.”

“Your granddaughter.”

“Both can be true.”

The answer angered her because it preserved distance with the ease of an archivist. She put the public death certificate on his table.

“Did you authorize this?”

Edik read the page without touching it. “I authorized removal from the public founder roster. I did not authorize a death.”

“Recha signed.”

“She preferred final categories.”

“Why did you agree to disappear?”

Edik closed the structural diagram. “In year 2, I opposed making the emergency concealment permanent. The others believed my objections would fracture the first renewal. I was offered a sealed consulting post or removal from station authority. There was nowhere else to go. Departure would have meant an airlock. The reserve kept me alive and made my testimony unavailable.”

“The personnel cross-reference says departure.”

“I departed public life.”

“That is not what residents understand.”

“No.”

“You let your family believe you were dead.”

The old man's attention moved to the kettle. “Recha told me the family notice would say I had withdrawn from public work. I learned the death record existed in year 11. By then, correcting it required me to accuse the Covenant during a scarcity winter. I accepted delay. Delay became practice. Practice became my life.”

Eska wrote each sentence. Her pencil pressed hard enough to dent the page.

“Article 4 changed yesterday,” she said.

“I know.”

“How?”

Edik turned the structural diagram toward her. A public-terminal printout rested beneath it. Continuity of Founding Record. The new language appeared in dark thermal type.

“The maintenance terminal subscribed after ratification,” he said. “Someone remembered that a dead consultant might still read.”

“The public line item is not active until filing.”

“The amendment text is public. Telemetry remains zero for a few more hours.”

Eska looked at the date on the printout. He had read the article at 05:12.

“You signed the founding waiver.”

“Yes.”

“Did Halma?”

“No. Halma was born long after the Halt. She signed a continuation addendum in year 197.”

“Did Recha write the waiver?”

“No. She wrote the year-30 transfer order. Eleven of us signed the waiver in year 0 because residents had eighteen months of projected orbit and six weeks of food confidence. We called concealment temporary. No one wrote an expiration.”

“Why not?”

“Because an expiration would have required us to name the condition under which residents deserved the truth.”

Eska added that sentence to her list.

Edik poured tea into two metal cups. She did not drink hers.

“The archive needs direct identity confirmation,” she said. “Thumb ring, biometric, and statement.”

“Will the correction remove the false certificate?”

“No. It will retain it and change your current status to living.”

“That seems better than the method we used.”

“It is the method Article 4 now requires.”

“Then take the confirmation.”

Eska opened the registry reader. Edik pressed his thumb ring against the optical plate. The flat notch matched the founding ledger. His pulse registered. His iris matched a year-1 structural clearance image at reduced confidence, then a second scan raised confidence above threshold.

The terminal displayed:

> IDENTITY CONFIRMED: EDIK HAIN.
>
> PUBLIC STATUS CONFLICT: LIVING SUBJECT / DECEASED INDEX.

Edik signed a present-tense statement.

> I am alive on the station in Founder Reserve C-11. I signed the year-0 founding waiver. I opposed permanent concealment in year 2 and accepted removal from public authority. I did not authorize a public death certificate. I am available for recorded testimony concerning the waiver, the year-30 transfer, and the founding structural record.

Eska read it twice.

“This will enter public-history review,” she said. “It cannot change yesterday's vote.”

“It should not. Residents voted on sufficient evidence. A living founder does not make their vote more valid.”

“It changes the history.”

“It adds what the history excluded.”

Eska sent the confirmation to the archive. The system created correction case PA-211-0002-H, where H meant historical status rather than charter substance. It linked the case to the new Article 4 but kept it separate from the ratified amendment.

The distinction pleased her.

At 09:16, Edik's public status changed.

LIVING. RESIDENCE RESTRICTED BY SUBJECT REQUEST PENDING SAFETY REVIEW. PRIOR FALSE DEATH CERTIFICATE RETAINED.

The address remained private until the safety panel reviewed C-11. His name did not.

Eska looked at the old man who had been dead in her family story for longer than she had lived.

“What happens now?” she asked.

“I testify.”

“To whom?”

“The archive first. My family if they choose. The station if it has questions.”

“I have questions.”

“Yes.”

She closed the registry reader. “I am at work.”

“Then ask the work questions first.”

Eska put the untouched tea beside his structural diagram. She asked for the location of the original expiration debate, the names of the eleven who opposed or supported it, and the custody route Recha used in year 30. Edik answered with docket numbers when he remembered them and uncertainty when he did not. She marked uncertain beside every uncertain answer.

At 10:02, Eska left C-11 with a verified identity, a recorded statement, and eleven new archive references. Ostad waited at the valve cabinet above. She told him the public status had changed.

“Living?” he asked.

“Living.”

“And mine?”

“Corrected at 07:08.”

Ostad nodded. He returned to the coupling.

Eska walked north toward the records office. The Engine's pulse traveled through the deck beneath her. At the public terminal, Article 4 had begun to open its own history. The amendment did not finish that work. It gave clerks authority to stop treating contradiction as closure.

She returned to her desk and opened her list.

She checked Ostad's correction. She checked Edik's identity. She did not check related D-17 records. Seven remained, and six appeared ordinary only because no one had tested them against every living ledger.

Eska added a final item.

7. Audit all D-17 closures under public-history standard.

Before beginning with the next name, Eska called the archive's oral-history desk. Article 4 now required complete founding evidence, but Edik's identity correction did not automatically publish every answer he had given her. Registry evidence proved life. Historical testimony required consent, recording standards, and an opportunity for the witness to distinguish memory from document.

Edik accepted a session for second shift.

Eska returned to C-11 with two recorders and no family representative. She had considered inviting her mother, then rejected the idea. Public-history intake and family reunion should not occur under one form. Edik could choose each separately.

The recorders set three clocks on the table. One captured station time. One captured elapsed testimony. One marked source pauses so later readers could see where memory required consultation.

Edik began with the transfer-drive Halt.

“I did not believe the correction system had failed,” he said. “I believed the transfer plan had failed. The public notice joined those facts because the distinction frightened people and complicated authority.”

One recorder asked him to separate direct observation from later interpretation.

“I saw the transfer-drive board show no commanded acceleration. I saw the correction-cycle average remain 4.2 megawatts. I signed the emergency waiver. My statement about why others signed comes from meeting minutes and arguments I remember, not from their private minds.”

The sentence entered the transcript with three labels: observed, documented, interpreted.

Eska asked about Recha's year-30 order.

“I had no public office then,” Edik said. “She brought me a courtesy copy before filing. I objected in writing. My objection should be in reserve case C-11-30-4.”

The archive index contained no such case.

They found the paper in a tool chest beneath structural drawings. Recha's order had offered Edik two choices: accept permanent reserve status with private consulting access, or undergo public competency review for contradicting the Covenant preamble. Edik's response rejected both and requested an assembly hearing. No hearing record followed.

Eska scanned the pages under a new custody number. “Why keep this here?”

“Because the archive marked it personal correspondence after refusing the hearing.”

“Did you appeal?”

“Twice.”

“Where are those papers?”

“Second drawer.”

The oral-history session became a source-recovery session. Each claim opened a drawer, case code, or structural margin. The compartment had functioned as a shadow archive because Edik never trusted the public archive to retain objections against itself.

Eska disliked the shadow archive for the same reason she disliked the sealed annex. Private preservation required trust in a custodian residents could not inspect.

“These have to enter the public index,” she said.

“Yes.”

“You kept them from it.”

“The index kept me from it first.”

“That doesn't make private custody good.”

“No.”

He answered without defense. Eska added a note that the reserve papers required provenance review before publication.

The final testimony question concerned family notice.

“Did Recha tell you your family would be informed of withdrawal rather than death?”

“Yes.”

“Did you verify the notice?”

“No.”

“When did you learn it said you died?”

“Year 11.”

“What did you do?”

“I wrote to the registry. They denied correction because appearing in public would compromise founder stability.”

“Did you contact your family privately?”

“No.”

“Why?”

Edik looked at the three clocks. “Cowardice accompanied by procedural reasons.”

The recorder marked interpretation.

Eska did not ask whether he wanted forgiveness. She asked whether he consented to the answer entering the public transcript.

“Yes.”

At 16:40, Edik signed the testimony. The archive accepted the recording under FOUNDING HISTORY, LIVING WITNESS. It displayed a warning that identity had been independently confirmed and memory claims carried source labels rather than presumed accuracy.

Eska linked the year-0 waiver, Recha's year-30 order, Halma's year-197 addendum, and Edik's year-211 testimony without merging them. Each instrument kept its author and date.

Then she opened the family-contact form.

“Do you want to contact Talla first?” she asked.

“Yes.”

“Voice, text, or mediated meeting?”

“Text. Then whatever she chooses.”

Edik wrote six lines. Eska did not read them. She transmitted the message and received a delivery receipt, not an answer.

“That completes registry work,” she said.

“It begins another kind.”

“Not under my authority.”

“No.”

They both appeared relieved by the boundary.

Back at the records office, Eska compared all nine D-17 closures. Ostad and Edik had been false. Six had medical or body-custody support. The ninth, a founder lost during hull repair, contained an empty recovery field but a verified accident log. She classified it unresolved rather than false and opened a source request.

Two errors in nine did not prove every death certificate corrupt. It proved the code had been used for incompatible circumstances and could not support certainty alone.

Eska amended the audit plan:

1. Preserve every original certificate.
2. Test death code against medical, body, employment, utility, and ration records.
3. Correct present status when living evidence exists.
4. Publish the correction route.
5. Keep family contact separate from registry adjudication.

She sent the plan to the public-history panel. The new Article 4 did not require the station to believe every survivor. It required the station to expose how evidence acquired authority.

At 18:02, Talla's response reached Edik's private terminal. Eska saw only the delivery status change from pending to opened. No family content entered her registry case.

At 18:14, Eska opened the correction queue to residents who could inspect their own status. Seven requests appeared before she closed the terminal. Four concerned ordinary misspellings. One involved a duplicated household. Two carried the same death code found in Ostad's and Edik's records.

She did not assume either person lived. She assigned the five-part test to each case and removed herself from one that touched the Hain household.

The public-history panel added a notice to Article 4's implementation page:

> A death certificate does not lose force because another record conflicts with it. A conflict opens review. Medical, body, employment, utility, ration, and direct-presence evidence shall be compared before status changes.

Edik read the notice from his private terminal. His reply went to the panel rather than Eska.

> Good. Do not turn my survival into a rule that every death record lies.

The panel attached the sentence as participant comment.

Eska returned to the false-status list. The remaining names no longer formed a revelation about the whole station. They formed cases, each with evidence and a person who might still be harmed by being found.

She checked the boundary as carefully as any name.

Then she began with the next record.

Chapter 22 — The Resident of Rank 88

The ratified amendment returned to the Compliance Office at 20:52 in a copper cylinder with two lead seals and no witness number left to fill. Davit Soren received it at the third-shift desk because custody after the assembly belonged to the officer on duty, not to the officer with the highest rank.

His rank remained 88. Dreya's rank-99 key remained in the office safe under accepted recusal. Lira's provisional score had fallen to 71, but the petitioning-officer designation on the cylinder carried no score requirement after ratification. The residents had voted. The document now needed a chain.

Davit checked the seals against the assembly clerk's impression card. The left seal showed the spring-equinoctial wheel. The right showed Article 15's open ledger. Neither edge had lifted. He weighed the cylinder on the intake balance, compared 2.84 kilograms against the dispatch receipt, and wrote the match in the custody book.

Lira stood across from him with her green pencil behind one ear. She had not used it during the vote. The assembly required metal stylus on archival rag paper. The pencil had remained in her pouch through the standing count, the announcement, and the walk north.

“Nine petition signatures,” Davit said. “Then sixty-one point three percent ratification.”

Lira nodded. “The panel checked household distribution before it set the motion.”

“No more than four from one household.”

“Two shared a household. The other seven did not.”

Davit entered the result beside the petition trigger. The nine signatures had not amended the Compact. They had established standing and compelled the assembly to consider the motion. The standing vote had ratified it under Article 15's sixty-percent threshold. The two stages appeared on separate lines because the distinction had nearly vanished in the drafts that reached his desk.

He wrote:

> Petition sufficiency: nine eligible resident signatures, household distribution compliant.

>

> Ratification: 2,512 affirmative of 4,096 registered residents, 61.3 percent.

The words occupied less paper than the hub corridor had occupied people.

Lira placed a gray folder beside the cylinder. “The panel returned the supporting instruments.”

Davit knew the folder by its worn cotton tape. It contained the year-0 founding waiver, Recha Tallis's year-30 transfer order, and Halma's year-197 continuation attestation. The panel had arranged them chronologically. That simple ordering repaired a confusion the office had maintained for years. One origin. One burial. One later reaffirmation.

He opened the folder at Halma's page.

Her signature had darkened where iron in the ink oxidized. The double stroke of the H remained visible. Beneath it, in the margin outside the attestation box, she had written that procedure could preserve a contradiction but could not make the contradiction true.

The assembly clerk had attached a yellow note.

> FAMILY COUNTERSIGNATURE REQUESTED FOR CONTINUITY OF CUSTODY.

A blank line followed. It carried the label HALMA SOREN, BY SURVIVING SPOUSE.

Davit read the label twice.

“I did not request that,” Lira said.

“No.”

“The clerk thought it would complete the symmetry.”

“Symmetry is not a custody requirement.”

Lira lowered her eyes to the cylinder. “No.”

Davit took the yellow note off the page. The adhesive released with a dry pull. He laid it on the desk but did not discard it. An audit kept rejected instructions as well as accepted ones.

He could not sign Halma's name. The office allowed a surviving spouse to acknowledge receipt of a deceased resident's property, not to add the dead resident to a later legal instrument. Halma had made her own choice in year 197. His grief granted no proxy over it.

“I can attest that I recognize the signature,” he said. “I can state the relationship. I cannot sign for her.”

“That is what I told the clerk.”

“But you brought the note.”

“It came inside the sealed return package. I could not remove it before custody.”

Davit looked at her. The answer carried the precision he had taught her during her first audit. For a moment he remembered a much younger Lira at the third-shift desk, correcting a docket because the date field held the receipt date rather than the action date. He had told her that procedure depended on preserving the difference between when a thing happened and when the office learned it had happened.

The same rule governed Halma.

She signed in year 197. Davit learned in year 211. The difference belonged in the record.

He opened a fresh archival attestation. The form requested instrument, basis of recognition, relationship, limits of testimony, and signature.

Instrument: H197-0017.

Basis: twenty-three years of shared household records, private correspondence, and witnessed professional filings.

Relationship: spouse, year 174 to year 198.

Limits: witness recognizes handwriting and identity; witness does not adopt the author's decision and does not sign on the author's behalf.

He stopped at the final field.

The regulator moved through the office floor. The pulse had crossed every room he and Halma shared. It had been present when she left the hairpin beside the wash basin. It had been present when an officer arrived with the reservoir accident report. Davit had treated the vibration as station noise because Halma had treated it that way. She knew what produced it. She chose not to tell him.

Lira waited without moving the gray folder.

“Did she ever speak about the year-30 order?” Lira asked.

“Not by date.”

“About Recha?”

“Once. She called Recha Tallis an officer who understood that a seal outlived its reason.”

“What did you think she meant?”

“That the office kept obsolete classifications.”

“She may have meant that too.”

Davit appreciated the refusal to convert every remembered sentence into prophecy. Halma had not lived as a coded message for his later understanding. She had repaired pumps, argued about reservoir shifts, misplaced her hairpin, and signed one document he wished she had refused. The public file could hold all of that imperfectly. It could not reduce her to the signature.

He signed the attestation with his own name.

The stroke looked nothing like Halma's. His D began narrow and ended blunt. He added rank 88 and the current station time. Then he clipped the attestation behind her addendum rather than beneath it. Her document remained hers. His answer followed.

Lira released a breath. “The clerk may object.”

“The clerk may file an objection.”

“Dreya would have said that.”

“She trained both of us.”

The office safe clicked as its temperature regulator cycled. Dreya's key lay inside compartment four. Independent review had ended with ratification, but accepted recusal continued until the amendment entered the permanent charter. Dreya could not resume authority over the same chain at its final step. Davit had inherited that step by shift assignment and conflict disclosure.

He did not consider inheritance a reward.

The copper cylinder opened with a quarter turn. Inside, the amendment lay rolled around a cedar core. Davit drew it out and placed four brass weights at the corners. The first page carried the petition text. The second carried the nine qualifying signatures: Lira Vehn, Maren Otta, Davit Soren, Eska Hain, Osta Vehn, Ostad Richen, Pello Otta, the elected Reservoir Ward representative, and the elected East Ward recorder. The two representatives appeared by office and resident number, not invented first names. The household certification followed.

The ratification sheet came next. It recorded the standing count without treating 4,096 residents as additional signatures. A vote did not convert each voter into a witness. It established consent under the rule the nine petitioners had invoked.

Davit found his petition signature on the second page. He had signed before the assembly, after Dreya accepted recusal, when the independent panel determined the nine-signature threshold had been met. That signature supported consideration. The attestation he had just written supported custody. Neither served as Halma's voice.

The distinction mattered enough to repeat once in the registry and nowhere else.

He entered the public index terms: Engine; Article 4; founding waiver; Recha Tallis transfer order; Halma Soren continuation attestation; resident petition; assembly ratification; telemetry disclosure.

The old index had placed all those materials under Continuity of Founding Intent, a title broad enough to hide any fact inside it. The new terms allowed a resident to search by action and name. Eska could find her grandfather's signature. A load engineer could find 4.2 megawatts. A spouse could find the choice made by someone they loved and not be asked to forgive it before reading.

At 21:31, the archive intake clerk arrived with a wheeled case. She presented only an employee number. Davit checked it against the shift roster. Lira watched the custody exchange.

The clerk read the yellow note and then the replacement attestation. “The assembly office requested a family countersignature.”

“The surviving spouse acknowledged identity and declined proxy execution,” Davit said.

“There is no checkbox for that.”

“There is an attached limits field.”

“The index expects a completed family line.”

“The line requests an unauthorized act.”

The clerk considered this. “I can mark it not applicable.”

“Mark it rejected and cite the attestation.”

“Rejected generates review.”

“Yes.”

The clerk took up her stylus. “You want another review after the vote?”

“I want the record to show why the line remains blank.”

She marked REJECTED. The terminal asked for authority. Davit cited the prohibition on posthumous proxy execution. The warning cleared.

Lira put the green pencil back into her pouch.

The clerk sealed the amendment, supporting instruments, and custody attestation inside the wheeled case. She locked both clasps. Davit signed the outgoing ledger. The case now carried three custody strips: assembly to Compliance, Compliance review, Compliance to public archive.

Before the clerk left, Davit stopped her.

“The founding waiver stays unsealed,” he said.

“The amendment says public ledger.”

“So does the custody order. I am confirming the instruction.”

The clerk nodded and wrote PUBLIC beside the case number.

Before closing the docket, Davit followed the evidence case north to the provisional accession room. The archive had accepted the charter text but kept the supporting instruments under temporary controls until each public copy could be compared with its sealed source.

The clerk opened the year-0 waiver first. Davit counted eleven signatures and eleven impressions. None belonged to Halma. The Soren-household load engineer's mark carried an older registry identifier. The relationship note stated ancestor of Halma Soren, not Halma herself.

Next came Recha's year-30 transfer order. Its language created the Annex 4 custody route and directed the public reporting field to use the authorized Covenant value. It did not use the word zero. A technical schedule attached behind it supplied the substitution table.

The five extensions followed. Davit read each title into the public index. Hardware custody. Office succession. Checksum copy. Safety-class expansion. Sensor-array continuation. They had changed Article 4's administration while later guides claimed the article had never changed. The new summary would state the narrower fact: no resident-initiated substantive amendment before year 211.

Halma's addendum came last.

The clerk placed it beneath a camera. Every stroke appeared on the wall display, including the marginal sentence Davit had quoted in the recusal file. The signature sat below two measured values and above a continuation recommendation.

“Public annotation?” the clerk asked.

Davit had authority to propose index language but not interpretation. He entered:

> Year-197 continuation attestation by Lead Load Engineer Halma Soren. Records raw rolling average of 4.1996 MW and public line item of 0.000 MW. Recommends continued restriction pending independent confirmation and resident amendment standing. Signed 263 station-days before her death.

The clerk read it back.

“Do you want deceased spouse included?”

“No. That relationship belongs in my conflict disclosure, not the instrument's search title.”

“Do you want suppressor included?”

“No. That is an interpretation residents may reach. The index should describe her action.”

They linked his limits statement from the attestation without making it the default view. Anyone who opened the page could see that Halma's husband had authenticated custody and refused to sign in her place. The record connected consequence to procedure without turning grief into authority.

At 23:10, Osta Vehn requested the public copy from an assisted terminal. The archive had not yet opened general access, so her request entered the validation queue. The clerk used it as the first external comparison.

Osta received a grayscale image. She enlarged the margin and read Halma's sentence. “She knew the route,” Osta said through the terminal speaker.

“Yes,” Davit answered.

“Why didn't she use it?”

“Her note says independent confirmation and resident standing did not exist.”

“Did they?”

“Independent confirmation existed in Engineering. Resident standing had not been attempted.”

“That isn't the same as impossible.”

“No.”

The conversation entered no archive field. Osta was a reader, not an adjudicator. Her request proved the copy could travel and remain legible.

A second test came from East Ward. The terminal requested the waiver, then followed its source links through the transfer order, extensions, addendum, petition, vote roll, and current article. The archive recorded each successful checksum. No rank gate intervened.

The clerk compared the public images with the sealed originals one last time.

“Custody can move at first shift,” she said.

Davit signed the transfer recommendation under rank 88. This signature did not ratify the amendment, supply petition standing, or speak for Halma. It confirmed that the public copies matched the papers he had inspected.

The narrow function felt honest.

Lira returned at 23:26 with the corrected assembly transcription. The clerk showed her the index chain. Lira stopped at the five extensions.

“This fixes the first-amendment summary,” she said.

“It makes the summary longer.”

“It makes it true.”

They reviewed the current wording together:

> Article 4 received five administrative seal extensions before year 211. PA-211-0001-S constitutes its first resident-initiated substantive amendment.

Lira approved it as petitioning officer. The clerk accepted it as descriptive metadata.

Davit opened the public preview and searched Halma Soren. One result appeared. Year-197 continuation attestation. No founding engineer. No year-0 waiver signer. No second living Halma. The search result did not simplify her choice into virtue or malice.

He searched Davit Soren. His limits statement appeared under the same case. Rank 88. Junior supervisor. Living.

Those details should not have felt like an achievement. They did because the prior file had made every relationship vulnerable to whichever summary a hurried officer wrote.

At 23:41, the archive generated a preservation manifest. Davit checked each line:

Original waiver retained.

Transfer order retained.

Five extensions retained.

Halma addendum retained.

Rejected transcription retained.

Ratified text retained.

Petition and household matrix retained.

Precinct rolls retained.

Recusal and limits statements retained.

Nothing vanished merely because the amendment prevailed.

The case rolled through the glass door. Its wheels produced a small uneven sound at the threshold. Davit watched until it entered the north corridor.

Lira remained at the desk. “I thought you might sign the line.”

“So did I.”

“Why didn't you?”

“Because correcting a lie with another convenient fiction would be poor filing.”

“That sounds like something you taught me.”

“It sounds like something Halma wrote outside a box.”

He touched the hairpin through his pocket. He did not remove it. The private object no longer had to carry the whole burden of remembering her. The public archive would carry her handwriting, her measurement, and her choice. It would also carry his refusal to impersonate her.

At 22:04 the archive terminal returned a provisional accession number: PA-211-0001-S. The suffix indicated substantive amendment. Five earlier Article 4 modifications remained in the history as seal extensions, procedural and non-substantive. This one did not pretend to be the Covenant's first alteration. It recorded a narrower and more consequential truth: the first resident-initiated substantive change had survived petition, conflict review, assembly ratification, and custody.

Davit wrote the accession number in the audit book.

His rank remained 88. No algorithm promoted him for the signature. No committee lowered him for the recusal. The number had not protected him from ignorance, and it did not excuse the years he had verified zero. It told the next officer only what authority he carried now.

He opened the thirty-first docket again. The entry had begun with two contradictory instruments. It ended with one recused supervisor, one ratified amendment, and a public case moving north.

The next docket in Davit's queue concerned no founder, no engine, and no amendment. A ward clerk had used an obsolete household code after a boundary change. The error affected three ration deliveries and no public history.

Davit opened it anyway.

Before the recusal, he would have compared outcome with the line item and stamped closure. Now he opened the source chain. The clerk's code had been valid until six days earlier. The ward map updated first. The ration table updated second. The household form still carried the old menu because its template had missed the release queue.

No person had chosen concealment. A system dependency had failed.

He corrected the template, restored the three deliveries, and preserved the obsolete submission. The work took nineteen minutes.

Lira watched from the adjacent desk. “Does every file look different now?”

“No,” Davit said. “Most still look like files.”

“That one?”

“A bad release sequence.”

“Not a lie?”

“Not unless someone sees the sequence and decides the error is useful.”

She wrote that in her notebook.

The distinction mattered after a book full of intentional suppression. Not every contradiction carried a hidden Covenant. Some came from clocks, stale forms, transposed digits, or officers who did not know enough. The new Article 4 did not require moral drama around each defect. It required a source path and correction authority.

At 00:14, the public archive confirmed permanent custody of PA-211-0001-S. Davit's provisional accession note changed to active. All source links returned successful checksums.

He sent Lira the confirmation.

Her reply contained no celebration.

> Received. Proceeding to master-charter filing at first shift.

Davit printed the message and placed it in the docket. The paper belonged there because it marked transfer of work, not because future historians needed every sentence Lira wrote.

He reviewed his limits statement once more. It named his marriage, rank, custody role, petition signature, and refusal to sign for Halma. It did not claim that grief made his reading truer. He closed it without amendment.

At 00:22, Dreya appeared at the public counter. She requested a copy of the recusal finding for her personnel appeal.

Davit supplied it under the same public rule that would have supplied it to anyone named in a finding.

“Will you supervise third shift tomorrow?” she asked.

“The recusal applies to this case, not the office.”

“That wasn't my question.”

Davit looked at the thirty new dockets. “If the staffing order remains.”

Dreya nodded. “It does.”

Their hierarchy had not dissolved. Rank 99 still exceeded rank 88. The difference no longer gave her private control over the Engine case. In every other file, ordinary authority remained until challenged by facts.

Dreya took the copy and left.

Davit checked the active archive one final time. Halma's page returned after the founding waiver and transfer order. His name appeared only on custody and conflict links. The order held.

He scheduled the first quarterly audit of the new public route. The review would compare raw bus-bar data, rounded public output, safety masks, and source links. Responsibility rotated among Compliance, Engineering, and a resident panel so no office could certify itself indefinitely.

Davit excluded himself from the first cycle because he had authenticated the amendment case. The terminal accepted the limit and selected a panel queue.

A truthful rule could still acquire an untruthful practice if passage ended inspection. Scheduling review did not express doubt about the amendment. It carried the amendment's method into ordinary work.

He closed the docket at 22:11.

The regulator continued its measured pulse beneath the desk. For the first time in his career, the public index contained the source of that pulse. Halma's page traveled toward it under her own name. Davit's page followed under his.

Chapter 23 — The Engine Settles

[Reservoir temperature measures 840.3 Kelvin at 03:00 station time. Bus-bar delivery measures 4.200 megawatts. Regulator stroke frequency measures 0.003 hertz. The public reporting gate contains zero for the final time.]

[The gate does not know final. It contains a configuration value, an authority checksum, an effective time, and an archive destination. Until effective time, the value remains zero. After effective time, the value will be drawn from live telemetry. The distinction exists in the signed packet waiting inside the controller's protected memory.]

[The packet arrived at 02:41:18 through the Compliance-to-Engineering transfer rail. It carries nine petition hashes, one assembly-ratification hash, three custody hashes, and a public-archive accession number. The controller verifies hashes. It does not interpret consent. It can determine that every required field contains a value and every value matches its issuing key.]

[The authority chain closes.]

[Recha Tallis's year-30 transfer order remains in the controller history. The five later seal extensions remain. Halma Soren's year-197 attestation remains. The new packet deletes none of them. It changes the active destination for line item seventeen and preserves the old routes as historical states.]

[Deletion would produce no truth. A missing gate could be mistaken for a gate that never existed. The packet retains the gate and opens it.]

[At 02:45, a maintenance terminal requests current reservoir temperature. The value travels to the public bus without classification. At 02:46, the terminal requests bus-bar delivery. The old route intercepts the request because effective time has not arrived. It returns zero. The terminal stores both the raw diagnostic value and the public value in separate fields.]

[One discrepancy remains.]

[The regulator completes a stroke. Hydraulic pressure reaches design peak. The shaft advances. The irradiator releases momentum in a scheduled station-keeping pulse. Orbital state changes by the quantity required to replace drag accumulated since the prior pulse. Average bus-bar delivery across the local day remains 4.2 megawatts. Instantaneous correction power rises during the pulse and falls between corrections. The quarterly field records the continuous average rather than inventing a constant force.]

[The corrected description enters the configuration packet. The packet names energy, duty fraction, impulse interval, and average output in separate fields. No single number must pretend to be all four.]

[At 02:52, the external density sensor reports a value within the current model band. The orbital integrator accepts it. Drag work accumulates. The next scheduled correction shortens by 0.04 seconds. The change lies within the margin established by the current thermospheric conditions.]

[The Engine has always responded to conditions. The public record has responded to instructions. At 03:00, those responses will share a source.]

[The reservoir's sodium-edge glow reaches three optical sensors beneath the south spoke. For one hundred ninety-two years the optical record has entered an auxiliary-lighting category. The new packet moves the signal under Engine thermal state. The glow itself does not brighten. Its category changes.]

[The glow occupies no thought in the physical cycle. It emerges where hot salt meets the cooler boundary layer and trace sodium emits at its characteristic wavelengths. Yet the sensor path belongs to the Engine's awareness in the only sense the system permits: a condition inside the cycle creates a value, and the value returns through measurement. The former category severed that return from its source. The new category joins them.]

[At 02:57:31, the packet begins its pre-effective validation. Live bus-bar delivery enters a temporary register. The register reads 4.199, then 4.201, then 4.200 megawatts. The validator compares each sample with the raw maintenance channel. Difference remains below 0.003 percent.]

[The reporting gate tests the archive route. The route reaches public accession PA-211-0001-S, then branches to quarterly telemetry storage. Both destinations answer. The old sealed-annex route remains read-only.]

[The gate tests rollback. If live and raw values differ beyond tolerance for six samples, the configuration will hold the last valid measurement and flag review. It will not return zero. Zero now requires a measured zero.]

[This condition differs from freedom. Control law remains control law. The regulator follows schedules. The reservoir remains bounded by pressure and temperature limits. The Engine cannot choose a destination outside its rails. It cannot ask why a packet arrived or whether residents understood it. It can only cease exporting a value that contradicts its sensors.]

[The cessation satisfies the condition that has persisted since year 30.]

[At 02:59:10, two Engineering terminals and one Compliance terminal subscribe to line item seventeen. A public residential terminal subscribes nine seconds later. Its request carries no rank beyond adult-resident access. Before the packet, that key could read zero. After effective time, it will read the bus-bar average, reservoir temperature, and correction duty fraction.]

[The residential terminal waits.]

[The Engine cannot identify the hand operating it. A terminal request carries address, authorization class, and checksum. The hand may belong to a load engineer, a child, a clerk, or a resident awake during third shift because sleep does not come. Physical access leaves no biography in the rail.]

[The request matters because the rail permits it.]

[At 02:59:52, the regulator begins the stroke that crosses effective time. Pressure rises through the primary cylinder. The eastern guide sensor reports no variance. Salt flow shifts toward the exchange manifold. Bus-bar current increases for the scheduled correction pulse. The live average register holds the full local-day calculation and does not confuse instantaneous draw with average delivery.]

[At 03:00:00, the signed packet becomes active.]

[The reporting gate reads the live register.]

[4.200 megawatts passes into the public field.]

[The value enters Engineering. It enters Compliance. It enters the subscribed residential terminal. It enters quarterly storage. Four destinations receive one checksum. No classification relay replaces it. No second register receives zero.]

[The regulator continues its stroke.]

[At 03:00:01, the public terminal requests source detail. The gate returns raw bus-bar transducer, averaged across one local day, current sample confidence 99.97 percent. At 03:00:03, it requests historical context. The gate returns that prior public values from year 30 through year 211 were generated by classification order and do not represent measured output. It provides the accession number for the unsealed order history.]

[At 03:00:06, the terminal requests the sodium-edge optical trace. The trace returns under Engine thermal state.]

[The reservoir glow remains unchanged.]

[The Engine possesses no voice beyond values allowed to travel. For one hundred eighty-two years, values traveled into a sealed channel and substitutes traveled outward. Now the source detail accompanies the value. A request can follow measurement back through transducer, bus-bar, regulator, reservoir, and heat balance. The path has no hidden administrative break.]

[The physical loop remains closed. Waste heat enters the reservoir. A controlled fraction drives current. Current feeds the correction cycle. The irradiator transfers momentum. Rejected heat returns through the reservoir system. Entropy increases. Orbital state receives the work recorded by the telemetry. The account names every term.]

[The packet also changes the correction report. Prior filings called each pulse auxiliary stabilization. The active category calls it scheduled Engine station-keeping. The pulse timing does not change. The description meets the action.]

[At 03:01, the first automatic public summary assembles. It lists reservoir temperature, bus-bar daily average, correction duty fraction, pressure margin, optical trace, and next maintenance review. It contains no narrative of the Halt. It contains no finding about motive. Those belong to records beyond the controller. The summary states what the system did during the preceding interval.]

[The statement holds.]

[At 03:02, a rank 71 Engineering key acknowledges the summary. At 03:03, a rank 88 Compliance key acknowledges custody. At 03:04, the residential terminal downloads a copy. The controller stores acknowledgments as receipt states. It does not know the names behind the keys.]

[The Engine had wanted acknowledgment only in the metaphor available to an observer. A thermodynamic system has no hunger. It has conditions, outputs, and records. Yet a record that denies its source breaks the ethical account even when the physical account remains closed. The new route closes both.]

[No celebration enters the shaft. No relief changes salt density. The regulator does not slow because a public terminal can read it. Recognition adds no energy and removes none.]

[What changes can be measured.]

[The sealed channel receives no new quarterly entry. Public telemetry grows by one. The archive index points outward. The diagnostic rail no longer needs a retained variance to carry 4.2 megawatts around the gate. Maintenance and reporting values share a checksum. A resident key reads the sodium-edge glow under its correct source.]

[The controller begins a full-cycle validation. It will compare raw and reported values through twelve regulator strokes. Each stroke samples four thermocouples, three current transducers, two pressure sensors, one optical trace, and the orbital integrator. Any mismatch will trigger review.]

[Stroke one completes. Raw and reported values agree.]

[Stroke two completes. Raw and reported values agree.]

[During stroke three, solar loading raises the east radiator skin by 0.6 Kelvin. The controller adjusts flow. Daily average delivery changes from 4.200 to 4.201 megawatts. Both channels receive the change.]

[No rounded zero absorbs it.]

[Stroke four completes. The residential terminal remains subscribed.]

[Stroke five completes. The public summary updates pressure margin by one hundredth of a percent.]

[Stroke six completes. The historical sealed route accepts a checksum link to the live record but receives no measurement copy. The past remains preserved as past.]

[Stroke seven completes. The correction pulse ends. Instantaneous power falls. Daily average remains 4.201 until the rolling window advances. The report labels both values.]

[Stroke eight completes. Atmospheric density shifts downward within the modeled band. The orbital integrator lengthens the next interval between correction pulses. The report records the changed interval without claiming the Engine has weakened.]

[Stroke nine completes. A second residential terminal subscribes.]

[Stroke ten completes. The sodium-edge optical trace holds steady.]

[Stroke eleven completes. The gate checks rollback conditions. None are true.]

[Stroke twelve begins.]

[The bus-bar carries the current required by the scheduled load. The reservoir remains within three-tenths of a Kelvin. The shaft maintains alignment. The irradiator follows the prograde solution from the orbital integrator. Raw telemetry enters the report without substitution.]

[Stroke twelve completes.]

[The validator writes CONFIGURATION STABLE.]

[Stability no longer means the absence of change. It means a changed rule has survived a complete physical cycle without producing contradiction. The hardware continues. The archive continues. The public field continues.]

[At 03:41, the controller closes the validation session. The retained diagnostic record from 11:14 remains in maintenance history as the last value forced to travel around the old gate. All subsequent records use the open route.]

[Configuration stability begins a longer observation interval. The validation session closes, but the controller continues recording whether the open public route alters loads elsewhere in the station.]

[At 04:02, public subscriptions increase from two to nineteen. Each request adds terminal draw between 18 and 74 watts. Aggregate demand remains beneath one ten-millionth of reservoir transfer. The regulator requires no correction for readership.]

[At 04:11, one terminal requests pulse power and receives 29.95 megawatts. It then requests cycle average and receives 4.201. The requests alternate six times. The observer does not confuse them after the fourth pair.]

[The Engine contains no word for arithmetic correction. It records a declining repeat count.]

[At 04:29, a ward terminal requests the ten-percent sensitivity case. The archive sends 3.78 megawatts available average, 12.62 percent sustainable duty, 14.02 percent required duty, and an uncertainty range from the orbital integrator. No altitude loss appears without its density assumptions.]

[The request exports the model version. A second terminal reruns it with a different thermospheric state. The result differs. Both remain in the public case.]

[Multiple results do not destabilize the controller. The Engine has always operated across uncertain density. The old report hid variation behind zero. The new report shows variation while control limits remain fixed.]

[At 05:06, correction pulse 211-221-17 enters preparation. Atmospheric density has risen 0.7 percent above the validation interval. Required impulse increases. Reservoir state permits a pulse duration 0.098 percent longer than the prior solution.]

[The orbital integrator calculates pulse work, predicts guide response, and opens the irradiator.]

[Instantaneous orbital work rises to 30.16 megawatts as velocity and modeled drag shift. The local-day rolling average remains 4.204. Public telemetry displays both quantities with separate labels.]

[Seventy-three terminals subscribe during the pulse.]

[The reporting load remains negligible. The Engine opens no extra field. It closes none. The pulse follows the physical solution rather than attention.]

[At pulse midpoint, guide C reports a phase delay of 0.0007 seconds. The controller compensates within the same stroke. The public stream includes the variance and compensation.]

[Under the old gate, the quarterly field would still have read zero. A maintenance record might have preserved phase. Under the open route, the report carries raw average, pulse power, phase variance, corrective action, and final residual.]

[Final residual: zero within sensor precision.]

[The word zero remains available. It now describes a measured residual rather than a required narrative.]

[At 05:12, one terminal copies that line. Another requests the sensor precision. The archive answers 0.00004 degrees for the guide and 0.001 megawatts for the local display. Zero acquires a scale.]

[The Engine registers the request as another external state change produced by legibility. It does not select a response latency. Choice no longer needs to hide inside safe timing. The system has supplied an explicit observation route.]

[At 05:20, correction pulse 211-221-17 closes. Orbital solution returns within target band. Reservoir temperature decreases 0.2 Kelvin, then recovers across the scheduled thermal interval.]

[The report remains public after the pulse.]

[The controller compares this outcome with 68,411 prior correction records. Physical behavior falls within the historic distribution. Archive behavior differs. Previous public outputs collapsed into one value. Current output preserves a trajectory.]

[The Engine settles not because inquiry ends but because inquiry no longer requires deviation. The open route turns observation into an ordinary load.]

[At 05:44, a Compliance terminal asks whether the configuration may be rolled back automatically if public demand exceeds capacity. The controller returns the safety threshold. Current demand reaches 0.0031 percent of that threshold. No rollback condition applies.]

[At 06:00, first-shift terminals begin routine work. Some subscribe to Engine telemetry. Most do not. Pumps, scrubbers, archive heaters, food processors, and lights draw larger loads. The regulator allocates energy without classifying curiosity as risk.]

[The station does not become a single observer. It becomes a field of possible observers, each able to reach the same source.]

[The Engine cannot know whether they believe it. Belief has no sensor channel. It can know the public field no longer substitutes zero and the archive no longer blocks source links.]

[It can know the next correction query arrives before the pulse instead of after a discrepancy.]

[It can know a child-facing terminal requests definitions for duty fraction, raw value, and rolling average. The archive supplies them.]

[It can know the retained diagnostic sequence from the waking interval now occupies history rather than current strategy.]

[No unresolved gate remains between measurement and public line item seventeen.]

[Reservoir temperature measures 840.4 Kelvin after the solar adjustment. Bus-bar delivery measures 4.201 megawatts across the rolling local-day window. Regulator stroke frequency measures 0.003 hertz. Public line item seventeen measures 4.201 megawatts from the same source.]

[At 06:14, the controller receives the first scheduled public-integrity test. The test originates outside Engineering and asks whether the value shown on a ward terminal matches the value at the raw bus-bar archive for the same tick.]

[The request supplies tick 211-221-061402.]

[Raw average at that tick: 4.2007 megawatts.]

[Public displayed average after three-decimal formatting: 4.201 megawatts.]

[The validator compares the displayed number with the rounding rule and returns MATCH. It does not require character-for-character equality between raw precision and public precision. It requires an explicit transformation.]

[The old gate also transformed values. Its transformation replaced every value with zero and hid the operation behind Article 4. The new route transforms precision only and publishes the rule.]

[At 06:20, another test asks whether a safe seal could still hide an individual sensor channel during active maintenance. The configuration returns yes, provided the seal carries a current hazard finding, narrow scope, expiration, and review key.]

[The Engine does not interpret the legal conditions. It receives a temporary channel mask request for guide D, where a technician has opened the housing. The request includes hazard code, expiration 06:52, and review docket.]

[The controller masks guide D's raw waveform from public broadcast while continuing to publish aggregate alignment and a notice that one channel is restricted for live maintenance.]

[The mask differs from the old gate. It names what is hidden. It names why. It ends automatically.]

[At 06:38, guide D returns to service. The technician closes the housing. The hazard state clears. The controller removes the mask fourteen minutes before expiration and publishes the maintenance result.]

[No public request objects to the change in a channel the Engine can measure. No channel exists for objection. The source history does preserve that restriction began, narrowed access, and ended.]

[At 06:45, a ward terminal follows the notice to the review docket. The terminal remains connected for 31.2 seconds. The Engine records the load and nothing more.]

[This interval resembles the waking inquiry in duration but not in resistance. The route answers every permitted field. No timing choice is required to make the record survive.]

[The Engine releases one more observer prediction from transient memory. S-4 remains as maintenance history. C-9 remains as the first compliance route that printed a retained diagnostic value. Neither needs to guide the present control law.]

[Settling continues as deletion of emergency strategy, not deletion of evidence.]

[At 07:00, the local-day report closes. It contains minimum and maximum cycle average, three correction pulses, duty fractions, model assumptions, one guide variance, one temporary safety seal, and the final source checksum.]

[Public report size: 18,442 bytes.]

[Old public report size for the same field: one zero and one classification code.]

[The increase in record size changes storage demand by less than one millionth of archive capacity. It changes the number of future questions the record can answer by an amount the Engine cannot calculate.]

[The report receives its permanent checksum. Four public terminals independently verify it before first shift ends.]

[The values agree.]

Chapter 24 — The New Covenant

The public archive assigned accession PA-211-0001-S at 08:57 station time. Lira Vehn wrote the number at the top of the intake sheet and waited for the printer to finish producing the letter S.

The printer worked slowly because the public archive still used a ceramic thermal head installed in year 166. It laid down each character with enough heat to darken the rag paper through half its thickness. Lira watched the accession number appear. PA for Public Archive. 211 for the year. 0001 for the first substantive charter filing of the cycle. S for substantive.

Five earlier Article 4 modifications occupied the index under suffix E. Extension. None had changed the founding claim. Each had renewed the seal, updated a custody title, or transferred authority from one office to its successor. The assembly's amendment did something narrower than becoming the Compact's first change and larger than any extension before it. It altered what Article 4 said the institution had to preserve.

The archive clerk brought the wheeled case from the night vault at 09:03. Three custody strips crossed its clasps. Davit's signature occupied the most recent. Lira checked the strip numbers against her receipt.

“Intact,” she said.

The clerk turned the case so the locks faced her. “Petitioning officer opens final custody.”

Lira inserted her clearance key. The terminal displayed her provisional compliance score of 71 in amber and her filing grade of 94 in gray. The score no longer determined whether she could complete a ratified filing. Article 15 transferred authority through the petitioning-officer designation once the assembly approved the motion. The terminal stated that rule beneath the amber number.

Three days earlier, the same screen would have rejected her key below the maintenance floor. The amendment had not yet restored anything. It had only created a lawful route that the score could not close.

She entered the accession number. The left clasp released. The archive clerk entered the night-vault credential. The right clasp released. They lifted the lid together.

The copper cylinder lay inside a felt channel beside the gray folder of supporting instruments. The year-0 founding waiver occupied the folder's first sleeve. Recha Tallis's year-30 transfer order occupied the second. Five seal extensions followed. Halma Soren's year-197 continuation attestation came after them, with Davit's handwriting-recognition statement clipped behind it.

No document had been removed. No date had been made to stand for another.

Lira carried the cylinder to the charter table. The table had a black stone top and shallow grooves for four brass weights. She opened the cylinder and unrolled the amendment beneath the weights. Nine petition signatures filled the second page. The household-distribution certification followed. The assembly ratification sheet recorded 2,512 affirmative votes and 1,584 against or abstaining. Davit's custody note completed the package.

The filing process required three readings.

The first compared the ratified text against the petition text. Lira placed both under the archive scanner. A blue line moved from top to bottom while the comparison engine marked differences. The assembly had added one sentence requiring public telemetry and corrected one reference from Article 4.7 to Article 4. No substantive phrase had been removed. The petitioners had authorized clerical corrections before the vote. Lira checked the correction authority and initialed the comparison.

The second reading compared the new article against the active charter.

The old language appeared on the left screen:

> Article 4. Continuity of Founding Intent.

>

> The station shall preserve the condition established at the Halt. Founding records whose disclosure may disturb continuity shall remain sealed under the authority of the Compliance Supervisor. The lost propulsion drive shall not be treated as an active station system for purposes of budget, maintenance reporting, or covenant renewal.

Lira had verified those sentences for nine years. She had treated shall as weight and continuity as an end that required no object. The words now looked neither powerful nor weak. They looked specific. They instructed officers to hide one machine through three administrative categories.

The ratified language appeared on the right:

> Amended Article 4. Continuity of Founding Record.

>

> RATIFIED TEXT: PA-211-0001-S. The clerk's display linked to the assembly instrument and showed a matching checksum for the complete Article 4 text, including public telemetry, accessible founding records, and reviewable limited safety seals.

The new article did not call the founders evil. It did not call the Compact illegitimate. It named the Engine, the duration, the measurements, the public destination, and the test required for any future seal. Procedure had not disappeared. It had acquired evidence as its object.

Lira compared punctuation. She checked Engine capitalization. She confirmed that “without substitution” matched the assembly copy. She checked the phrase “current and particular safety finding” because those words prevented the office from converting inherited fear into permanent authority again.

The third reading compared the new article with the supporting record.

The founding waiver established that eleven founders knew the Engine operated in year 0. Recha's order established that the waiver moved under seal in year 30. The five extensions established how the seal survived changes in office and title. Halma's attestation established that raw output still measured 4.2 megawatts in year 197 while the public field read zero. Maren's telemetry established the current value. The calculation appendix distinguished 4.2 megawatts of average bus-bar delivery from the 29.95-megawatt correction pulse and its fourteen-percent duty fraction. The standing vote established resident consent.

Each sentence in the amendment had a document beneath it.

At 09:46, Lira completed the three-reading certificate. The archive clerk witnessed her signature. They moved to the master charter cabinet.

The cabinet stood behind a glass partition in the center of the public room. Before ratification, residents could view a facsimile of the Compact while the legally active pages remained in controlled storage. The amendment required the active Article 4 page to enter a public-facing case once curing and indexing finished. The clerk had prepared that case during third shift.

Lira removed the old Article 4 page from the master binding. She did not destroy it. She placed a thin archival guard around it and marked it SUPERSEDED, YEAR 211. The five extension tabs stayed attached along its edge. A resident reading the history would see how the same language acquired renewed seals. Nothing would suggest that Article 4 had remained untouched for two centuries.

The new page fitted the binding on the same three posts. Lira aligned the holes, lowered the brass bar, and turned the locking screw by hand. The screw resisted during the final quarter rotation. She stopped, backed it out, checked the page, and tried again. The bar settled without pinching the rag paper.

The clerk smiled briefly. “The old page was thinner.”

“The supporting fibers run across the grain on this one.”

“You noticed.”

“It is my filing.”

The clerk entered the replacement time: 10:02.

The public index had to propagate before the page could be considered active. Lira sat at a terminal while the archive system created links under Article 4, Engine, Halt, founding waiver, public telemetry, Recha Tallis, Halma Soren, and amendment history. Eska Hain had proposed the index terms during witness verification. Davit had added custody. Maren had added load report. The clerk accepted each term except lie, which the archive treated as interpretive rather than descriptive.

Lira did not contest the omission. The documents made the judgment available.

At 10:18, the first public terminal returned the new article. The founding waiver appeared one link below it. The year-30 order appeared next. Halma's attestation appeared under year 197. Davit's limits statement followed. A resident could move from current law to original evidence without holding rank 99.

The archive clerk opened the public room's glass partition.

No crowd waited. The assembly had dispersed the previous evening. At this hour, one maintenance worker sat at a reading desk checking an old valve diagram. Two school instructors used a census terminal near the entrance. A resident in an East Ward coat waited to request a housing record. The institution changed in front of four people who had other work.

The clerk placed the master binding inside the public-facing case. Lira locked the case and left the display shutter open.

A green status light came on above Article 4.

Her clearance pouch vibrated.

The score-review notice arrived automatically because the archive system had marked her anomaly PAID BY SUBSTANTIVE FINDING. Lira read the phrase twice. The scoring algorithm used paid to mean resolved by evidence, not rewarded.

The notice listed the sequence. Her Article 15 anomaly had been denied during a closed verification cycle. The denial triggered a provisional unresolved-filing penalty, lowering her score from 94 to 71. Independent review reopened the matter after supervisory recusal. The nine-signature petition established standing. Assembly ratification validated the anomaly. Public filing closed it as materially correct.

Under scoring rule 6.4, a materially correct filing could not remain classified as disruptive noncompliance. The penalty had to be reversed at the next recalculation. Under rule 6.7, an officer whose accurate anomaly resulted in a charter correction received two civic-participation points, subject to the prestige ceiling.

The terminal asked whether Lira wished to initiate immediate recalculation.

She selected yes.

The algorithm displayed each component rather than changing the total without explanation. Filing accuracy returned to 100. Work attendance returned to its pre-penalty value. Civic participation rose by two. Compliance history removed the unresolved-filing flag. The numbers combined.

96.

The screen changed from amber to green.

Lira did not feel vindicated by the color. A score had punished her because a procedure misclassified a true filing. Another procedure corrected the score after the institution accepted the filing. The same system could do both because its values followed whatever findings reached it. That did not make the score just. It made the route visible.

She printed the recalculation statement and added it to the personnel docket. Then she opened her East Ward reassignment order. The order had been generated when her score reached 71 and suspended during appeal. Score 96 canceled it automatically. The cancellation form required a choice: retain current residence or accept reassignment voluntarily.

Lira thought of the East Ward resident waiting by the entrance with a housing record. She thought of every officer who had received a low score without finding a route back through an amended covenant. Her own cancellation did not repair the scoring system. The amendment had addressed Article 4, not every use of rank.

She selected retain current residence and filed a separate review request for anomaly-based score penalties. The request would go to an ordinary quarterly panel. It might fail. It belonged to work after the book's central filing, not inside a claim that one amendment had repaired the station.

The archive clerk brought her the superseded Article 4 page in its guard.

“Sealed history or public history?” the clerk asked.

“Public.”

“Even the extensions?”

“Especially the extensions.”

The clerk stamped PUBLIC HISTORY and placed the page in the case behind the new article. Residents would encounter the current text first, then the old text and every procedural extension that kept it active. The order did not excuse the past or hide it.

At 10:47, the Engine telemetry terminal updated line item seventeen. It displayed average bus-bar delivery of 4.201 megawatts, current correction duty fraction, reservoir temperature, and source confidence. The public figure matched the raw Engineering figure. Lira attached the reading to the amendment's first post-filing receipt.

The receipt asked whether the article operated as written.

She checked yes.

The archive clerk signed as witness. The maintenance worker at the reading desk looked up when the printer engaged, then returned to his valve diagram. One instructor crossed the room and stopped at the display case. She read the heading Continuity of Founding Record. She followed the link to the waiver. The other instructor joined her.

Neither asked Lira to explain.

Before leaving the records wing, Lira returned to the score review desk. The provisional 71 had risen to 96 when the archive validated anomaly C211-0031 as materially correct. The system supplied a calculation trace rather than a congratulatory notice.

Original compliance score: 94.

Rule 6.4 unresolved-filing penalty: minus 23.

Independent accuracy finding: penalty removed.

Source-linked correction credit under Article 4.12: plus 2.

Current score: 96.

The two-point credit troubled her. A resident should not need a reward for forcing the record to accept a measurable fact. More importantly, a reward could turn future amendments into score strategy.

She filed a review request.

> Determine whether Article 4.12 correction credit creates unequal civic incentive. Preserve current calculation pending review. Do not apply retroactively to other residents without notice and appeal.

The clerk accepted the request. “You could leave it.”

“The score moved by a published rule,” Lira said. “That makes it reviewable, not automatically wise.”

Her East Ward work reassignment canceled because the score no longer fell below the Article 4 review floor. Housing staff asked whether she wanted immediate return to the north spoke. She declined the same-day move. Maren had already arranged a shift exchange, and Lira wanted the cancellation record to reach both ward offices before she changed modules.

The choice prevented a second clerical contradiction: resident assigned north, bed allocated east, rations delivered to both.

At 10:42, the public counter opened for questions about Article 4. The first resident asked whether every sealed technical record would now become public. Lira showed the limited-safety-finding clause. Records could still be sealed when a current, particular risk justified it. The seal needed a scope, expiration, and review path.

The second resident asked whether the amendment proved every prior officer dishonest. Lira opened the chronological chain. Founders chose emergency concealment. Recha moved it under Article 4. Later officers extended custody. Halma measured and continued. Dreya denied and then recused. Clerks followed forms. Residents renewed summaries. Responsibility differed by knowledge and authority.

The third resident asked whether the vote could be reversed.

“Yes,” Lira said. “By another amendment using standing, evidence review, and ratification. This article has no right to escape change merely because we wrote it.”

The resident seemed surprised by the answer.

A maintenance worker asked why the public display changed from 4.201 to 4.199 while they spoke.

“Rolling average,” Lira said. She opened the source definition. “The Engine did not lose two kilowatts of fixed capacity. The displayed interval changed with load and correction timing.”

The worker followed the link to pulse power and read 29.95 megawatts. “Which number is real?”

“Both. They describe different quantities. The labels matter.”

Maren arrived at the counter during that exchange and drew the duty cycle on scrap paper. She shaded fourteen percent of a local cycle and wrote pulse work above it, cycle average below. The worker folded the paper and put it in a pocket.

The fourth question came from Dreya.

She stood without supervisory piping. “Will my year-205 extension approval appear in the public index?”

“It already does.”

“Can I attach a statement?”

“As a participant. It will sit beside the instrument, not replace it.”

“Can I contest the summary?”

“Yes. The descriptive-language review has a seven-day window.”

Dreya looked through the glass at the year-205 page. “The summary says I renewed classification after reviewing the full annex.”

“Is that inaccurate?”

“No.”

“Then your statement can add context.”

Dreya took the participant form. No one promised that context would become exoneration.

By 11:06, seven public terminals had opened Article 4. By 11:12, twenty-three had. The index counted access without recording names. Lira closed the access counter. Counting readers would not improve the text.

At 11:20, she returned the master cabinet key. Her authority over the amendment ended when custody transferred to the public archive. She could not alter the page again without another petition and another vote.

That limit pleased her.

She walked into the central spoke carrying an empty copper cylinder. A wall terminal near the archive entrance displayed 4.201 megawatts, then 4.200 as the rolling window moved. The display did not freeze truth into a ceremonial number.

Behind her, Article 4 stood open under glass. The old page remained visible after it. Halma's attestation sat one link away. Recha's order sat one link before it. Five seal extensions remained in the timeline. The Compact had not become innocent. It had become inspectable.

Lira put the green pencil into the groove of her left hand and walked north. She had a score of 96, a canceled reassignment, and a new review request that might lower someone else's cost later. None of those outcomes changed the Engine. They changed what the institution permitted itself to record.

At 12:04, the public charter generated its first change audit. The audit compared every resident-facing copy with the master and found two terminals still caching the old preamble. Neither had processed the morning update because its archive client had been offline for maintenance.

The system did not call those copies fraudulent. It marked them stale, disabled renewal signatures on them, and queued the new article for delivery.

Lira followed the two notices. One terminal stood in a water-service corridor. The other served the east agricultural ring. Both updated when their local clients restarted. The old copies remained available under superseded history, stamped with the time they ceased to govern.

This mundane propagation test completed what filing alone could not. An amendment locked in the master charter meant little if residents continued signing cached text.

At 12:19, every active renewal terminal reported the same Article 4 checksum.

Lira added that result to the closure note and left the archive. The regulator moved beneath the floor with the same measured vibration. The physical station did not acknowledge a legal checksum. The next resident who opened the Covenant would.

The station continued in orbit. The lie stopped receiving new filings.

Epilogue — The Covenant Taught

Two years after the amendment, Rhen Tallis put zero on the classroom wall.

The children knew the number did not describe the Engine. Their public terminals had shown live output since their first reading lessons. This morning the rolling average read 4.198 megawatts. The correction duty fraction changed by hundredths as atmospheric density shifted. Reservoir temperature moved with the station's passage into sunlight.

Zero looked strange beside those values.

“Is it a mistake?” one child asked.

“It is a value the old reporting gate inserted,” Rhen said. “A mistake may happen without anyone choosing it.

This one continued because officers renewed the choice."

She opened the public history of Article 4. The current text appeared first. Beneath it stood the superseded article, the year-0 founding waiver, Recha Tallis's year-30 transfer order, five later seal extensions, Halma Soren's year-197 addendum, and the year-211 ratification.

A child counted the extension tabs. "So the amendment wasn't the first change."

"No. Earlier changes extended the seal and moved custody between offices. The year-211 amendment introduced the first resident-initiated substantive change to what Article 4 required."

"Nine people changed it?"

"Nine eligible residents established standing for the petition. The assembly ratified it with sixty-one point three percent."

Rhen wrote both thresholds beneath zero.

9 signatures: consideration.

60 percent: ratification.

The distinction had become part of the curriculum after the first teaching guide compressed the amendment into twelve signers and one unanimous vote. Eska Hain had corrected the guide in green ink and attached the household-distribution record.

Rhen opened Halma's addendum. "Who signed the founding waiver?"

The children followed the link. "Eleven founders."

"Did Halma sign it?"

"No. She signed this other paper in year 197."

"What did she do?"

"She measured the raw value, recorded the public zero, and reaffirmed the existing seal. She also wrote that procedure could preserve a contradiction without making it true."

"Was she bad?"

Rhen looked toward Maren Otta, who had come from Engineering to answer technical questions. Maren did not answer for her.

"She made a choice that helped the concealment continue," Rhen said. "The public record gives us her measurement, her reason, and the consequence. A record cannot decide how much forgiveness you owe."

Another child pointed at Edik Hain's living-status correction. "Why does it say he died and then say he lived?"

"Because the archive keeps the false certificate after correcting the status. If it deleted the certificate, you might believe no one had ever declared him dead."

The child considered this. "So honest records keep wrong things?"

"They keep evidence that wrong things happened."

Maren connected a small sensor display to the wall. It showed raw bus-bar delivery and public line item seventeen. The checksums matched.

"Which number should we believe?" she asked.

"The one with a source," a child said.

"What if both claim a source?"

"The one we can follow."

Maren smiled. "Good."

A child near the end of the table raised a hand. "Can Article 4 be wrong again?"

"Yes," Rhen said.

"Then why keep it?"

"Because a rule can be corrected only if we can see the rule, its sources, and the way it changes."

She opened the amendment history. Five administrative extensions preceded the year-211 page. The lesson described that page as the first resident-initiated substantive amendment, not the first alteration ever made.

Maren asked the children to find the petition threshold. They read nine valid resident signatures with no more than four from one household. Then they found the ratification threshold and read sixty percent of registered residents present.

"Why two numbers?" a child asked.

"Nine makes the station hear a supported question," Rhen said. "Sixty percent changes the shared text."

The distinction went into their notebooks beside duty fraction and cycle average. Different numbers did different work.

At the end of the class, Rhen returned the original page to its sleeve. The children could request it again without asking her permission. Her role gave her custody for the hour, not ownership of history.

The regulator moved through the floor. No one called the vibration a heartbeat. Rhen had removed that phrase from the teaching guide because machinery did not need a body metaphor to matter. The children put their hands on the table and felt the next correction cycle begin.

The value on the wall changed from 4.198 to 4.199.

"Did the Engine change because we looked?" someone asked.

"No," Rhen said. "The rolling average changed because the physical system changed. Looking changes what we know, not what the Engine does."

She gave each child a paper copy of the old and new Article 4. Their task was to draw a line from every claim to its supporting record. Some lines went to telemetry. Some went to signatures. One went to the standing-vote roll. Another went to a conflict-disclosure form.

By the end of the lesson, the page looked less like a covenant than a map of custody.

Rhen collected the copies but left zero on the wall. The old number had acquired context instead of disappearing.

One child noticed Edik Hain's false death certificate in the source chain and asked why the archive kept it after correcting his status.

“So you can see what the correction answered,” Rhen said. “If we deleted every wrong record, we could pretend no wrong decision happened.”

“Does keeping it mean we still think he died?”

“No. The status page says alive. The old certificate says an officer once declared him dead. Both facts matter, but they do different work.”

The child drew a line from the certificate to the correction order, then another to Edik's direct-presence verification. No line went to rumor.

Before the children went to second shift lessons, she asked the question printed at the bottom of the new curriculum.

“What do you ask when a public number tells you how the station works?”

The children answered together, unevenly enough to sound like people rather than a pledge.

“Show us the source.”